WHO Guideline on fortification of edible oils and fats with vitamins A and D for public health

World Health Organization

WHO Guideline on fortification of edible oils and fats with vitamins A and D for public health

World Health Organization

WHO guideline on fortification of edible oils and fats with vitamins A and D for public health

ISBN 978-92-4-011510-1 (electronic version)
ISBN 978-92-4-011511-8 (print version)

Suggested citation. WHO guideline on fortification of edible oils and fats with vitamins A and D for public health. Geneva: World Health Organization; 2025. Licence: **CC BY-NC-SA 3.0 IGO**.

Cataloguing-in-Publication (CIP) data. CIP data are available at **https://iris.who.int/**.

Contents

Acknowledgements

This guideline was led by Dr Luz Maria De-Regil, Unit Head, Multisectoral Actions in Food Systems Unit, Department of Nutrition and Food Safety, with the technical support of Ms Manasi Shukla Trivedi and technical contributions from the following World Health Organization (WHO) staff (in alphabetical order): Dr Katrin Engelhardt, Dr Maria Nieves Garcia-Casal, Dr Jason Montez, Dr Hector Pardo-Hernandez, Dr Juan Pablo Peña-Rosas and Dr Lisa Rogers.

WHO gratefully acknowledges the technical input of the members of the WHO guideline development group on food fortification with micronutrients, who were involved throughout the process, Dr Ayed Amr (University of Jordan), Dr Salima Al Maamari (Ministry of Health, Sultanate of Oman), Dr Rukhsana Haider (Training and Assistance for Health and Nutrition Foundation, Bangladesh), Dr Maria Elena del Socorro Jefferds (U.S. Centers for Disease Control and Prevention), Mr Phillip Joseph Makhumula-Nkhoma (independent consultant), Ms Andrea S Papan (Institute of Development Studies), Dr Dalip Ragoobirsingh (University of the West Indies), Ms Rusidah Selamat (Ministry of Health, Malaysia), Professor Clark Murray Skeaff (University of Otago, New Zealand) and Ms Ayodele Elizabeth Tella (TechnoServe). Dr Erika Ota (St. Luke's International University) provided methodological support in line with the requirements of the *WHO handbook for guideline development.* We thank the peer-reviewers Dr Andreas Blüthner (Dr. Blüthner & Partner), Dr Karen Codling (independent consultant), Dr Reina E Stone (UC Davis Department of Nutrition), Mr Penjani Mkambula (Global Alliance for Improved Nutrition), Dr N Sreekumaran Nair (Jawaharlal Institute of Postgraduate Medical Education and Research) and Dr Sirimavo Nair (The Maharaja Sayajirao University of Baroda) for their thoughtful feedback on a preliminary version of this guideline. Special thanks are also extended to the authors of the systematic review and supporting literature reviews that were used to inform this guideline.

We would like to express our gratitude to Ms Marion Blacker and Ms Rebekah Thomas from the WHO Guidelines Review Committee Secretariat for their technical support throughout the process. Thanks are also due to the Office of Compliance, Risk Management and Ethics for their support in the management of conflicts of interest procedures.

WHO gratefully acknowledges the financial contribution of the Bill & Melinda Gates Foundation for the completion of this work. The donor did not participate in any decision related to the guideline development process, including the composition of research questions, membership of the guideline development group, conduct and interpretation of systematic reviews, or formulation of the recommendations.

Executive summary

Vitamins A and D are two fat-soluble vitamins critical to maintaining the health and well-being of individuals. Signs and symptoms of deficiency appear when these vitamins are consumed insufficiently or when individuals face conditions or situations that preclude them from absorbing or synthesizing these vitamins or that exacerbate their metabolism, such as infection.

Vitamin A deficiency remains a leading cause of preventable blindness in children and also increases the risks of severe illness or death from infections. As of 2005 *(1)*, one third of the global pre-school aged children were estimated to have vitamin A deficiency with the highest prevalence in Africa (44%) and South-East Asia (50%). A pooled analysis of population-based surveys *(2)* from 138 low-income and middle-income countries reported varying trends in the prevalence of vitamin A deficiency among children by region and a decline in global prevalence from 39% in 1991 to 29% in 2013. The global prevalence of vitamin A deficiency is estimated to be 15% among pregnant women (1995–2005) with the highest risk of deficiency in the regions of Africa and South-East Asia *(1)*.

Vitamin D deficiency has become a public health concern, particularly in regions with insufficient sunshine or with sun avoidance behaviours that prevent or impede dermal synthesis, and infrequent access to or intake of dietary sources of vitamin D. Vitamin D deficiency may cause rickets in children and osteomalacia in adults and is associated with increased risks of respiratory illness and small for gestational age births. Although data on the global prevalence of vitamin D deficiency are unavailable, it is estimated that up to 1 billion people worldwide may be at risk of deficiency *(3, 4)*.

Fortification of edible oils and fats with vitamins A and D is widely implemented as a feasible public health intervention to improve intake of these vitamins, given these foods are consumed by most age groups. Fortification of margarine with vitamin A and D was introduced in several countries as early as the 1920s. Thirty-nine countries mandate or allow voluntary fortification of edible oils with vitamin A alone, and 13 countries mandate or allow voluntary co-fortification with both vitamins A and D.

Purpose of the guideline

The main objective of this guideline is to provide locally adaptable, clear, evidence-informed global recommendations on the fortification of edible oils and fats with vitamins A and D as a strategy to improve the vitamin A and D and health status of populations. The focus of this document is on the use of fortification as a public health strategy and not on market-driven fortification of edible oils and fats.[1]

This guideline aims to help Member States and their partners to make informed decisions on the appropriate nutrition actions to achieve the 2030 Sustainable Development Goals *(5)* and the global targets set in the World Health Organization (WHO) *Comprehensive implementation plan on maternal, infant and young child nutrition (6)*. The development of these recommendations considered gender equality, equity in access to fortified foods, and human rights approaches with the aim of leaving no one behind.

The recommendations in this guideline are intended for a wide audience, including policy-makers, expert advisers, and technical and programme staff in ministries and organizations involved in the design, implementation and scale-up of nutrition actions for public health. The recommendations are particularly relevant to the design and implementation of food fortification programmes as part of a comprehensive food-based strategy for combating micronutrient inadequacies and deficiencies.

[1] Market-driven fortification refers to situations in which food manufacturers take the initiative to add one or more micronutrients to processed foods, usually within regulatory limits, to increase sales and profitability. Fortification as a public health strategy refers to the practice of deliberately increasing the content of essential micronutrients (i.e. vitamins and minerals, including trace elements) in a food to improve the nutritional quality of the food supply and provide a public health benefit with minimal risk to health.

The guideline complements other WHO guidelines on vitamin A fortification of multiple staple foods, including maize flour and corn meal *(7)*, rice *(8)* and wheat flour *(9)*. Its implementation requires consideration of WHO guidelines on healthy diets *(10)*, including those on saturated fatty acids, *trans*-fatty acids and polyunsaturated fatty acids *(11, 12)*, which contain recommendations that aim to reduce the risk of unhealthy diets and diet-related noncommunicable diseases.

Guideline development methodology

WHO developed the evidence-informed recommendations in this guideline using the procedures outlined in the *WHO handbook for guideline development (13).* The steps in this process included identification of the priority questions and outcomes; retrieval of the evidence; assessment and synthesis of the evidence; formulation of recommendations, including research priorities; planning for dissemination; outlining considerations related to health equity, gender equality, human rights, implementation, ethics and regulation; and proposing alternatives for impact evaluation and updating of the guideline. The Grading of Recommendations Assessment, Development and Evaluation (GRADE) methodology[2]*(16)* was followed to prepare evidence-to-decision profiles related to preselected topics, based on a commissioned systematic review and other narrative syntheses of the evidence.

The guideline development group comprised multidisciplinary experts, methodologists and end users, with consideration given to geographical and gender diversity. The guideline development group participated in two meetings. The first meeting was held virtually on 10 and 11 October 2022 and had an objective to define the priority questions and outcomes. The second meeting was held in Geneva, Switzerland on 21 and 22 June 2023 to discuss the evidence that would inform the development of draft recommendations. A follow-up virtual meeting was held on 7 September 2023 to discuss draft remarks and research needs.

The members of the systematic review team participated in the guideline development process as resource persons by presenting evidence and helping to identify research priorities. Six experts peer reviewed the guideline.

Summary of the evidence

The evidence-to-decision summaries presented to the guideline development group to inform the recommendations were based on a systematic review *(17)* addressing the guideline questions in population, intervention, control, outcomes (PICO) format, a summary of available pre- and post-surveys from national or subnational fortification programmes in Indonesia, Malawi and Oman, and a desk review of implementation considerations.

Fortification with vitamin A

A systematic review *(17)* assessed the benefits and harms of edible oils and fats fortified with vitamin A as a public health intervention in the general population. Four out of five included studies used edible oil as the vehicle for fortification and one used margarine. The evidence showed fortification with vitamin A had no significant effect on serum retinol levels. This was the case for evidence from both randomized controlled trials (RCTs) (mean difference (MD) 0.35 µmol/L, 95% confidence interval (CI) –0.43 to 1.12; two trials, 514 participants; low-certainty evidence) and controlled clinical trials (CCTs) (MD 0.31 µmol/L, 95% CI –0.18 to

[2] The GRADE handbook *(14)* describes the process of rating the quality of the best available evidence and developing health care recommendations following the approach proposed by the GRADE Working Group *(15)*. The GRADE approach defines the overall rating of confidence in the body of evidence from systematic reviews as the extent to which one can be confident of an estimate of effect across all outcomes considered critical to the recommendation. Each critical outcome is given a confidence rating based on the certainty of evidence – high, moderate, low or very low. High-certainty evidence indicates that we are very confident that the true effect lies close to the estimate of the effect. Moderate-certainty evidence indicates that we are moderately confident in the effect estimate and that the true estimate is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low-certainty evidence indicates that our confidence in the effect estimate is limited, and the true effect may be substantially different from the estimate of the effect. Very low-certainty evidence indicates that we have very little confidence in the effect estimate and the true effect is likely to be substantially different from the estimate of effect.

0.80; two trials, 205 participants; very low-certainty evidence). Vitamin A fortification had no significant effect on subclinical vitamin A deficiency (serum retinol equal to or less than 0.70 µmol/L) in RCTs (none of 268 participants in the intervention group versus none of 144 participants in the control group, risk ratio (RR) not estimable; one trial, 412 participants, low-certainty evidence) and CCTs (RR 0.21, 95% CI 0.01 to 4.10; one trial, 31 participants; very low-certainty evidence). One CCT reported morbidity scores (defined as frequency of illness multiplied by duration of illness) and found no significant differences between study groups. No studies reported data on clinical vitamin A deficiency, all-cause mortality or adverse effects.

The findings from the review were complemented by the findings of representative pre- and post-surveys of edible oils and fats fortification programmes in Oman *(18, 19)*, Indonesia *(20)* and Malawi *(21)*.

In Oman, a national survey conducted in 2004 *(19)* showed that among children with normal C-reactive protein values the overall prevalence of vitamin A deficiency (serum retinol less than 0.70 µmol/L) was approximately 6.4% among children aged 6–59 months. The prevalence of vitamin A deficiency among non-pregnant women of reproductive age was 0.4%. When women with high C-reactive protein values were excluded, there were no cases of vitamin A deficiency. However, a limitation reported by the survey was that the number of participants with C-reactive protein analysed was lower because of non-response to blood withdrawal. The survey also showed that oil was available in most households (95%), and Omani products were the most consumed. Over half of households reported consuming at least 1 L of oil per person per month. These survey findings led to the introduction of mandatory fortification of oils and ghee with vitamin A. A subsequent national survey conducted in 2017 found that the prevalence of vitamin A deficiency (measured using ad hoc retinol-binding protein levels, adjusted for inflammation) among the Omani population was 0.2% among non-pregnant women and 9.5% among children aged 6–59 months. However, it was recognized that the comparability of the surveys has limitations given that the biomarkers and adjustments for inflammation were not consistent. Also, data on coverage is not reported.

In West Java, Indonesia *(20),* pre- and post-surveys were conducted in 24 villages in the Tasikmalaya and Ciamis districts, where fortified oil had a coverage of 50% or higher. The sampling frame targeted the poor population suspected to be at highest risk of low serum retinol. Mean serum retinol concentrations at baseline ranged from 30.7 µg/dL (standard deviation (SD) 12.6) among children aged 6–11 months to 34.3 µg/dL (SD 12.6) among children aged 5–9 years. In lactating mothers, average serum retinol was 30.7 µg/dL (SD 12.6) while it was 42.7 µg/dL (SD 19.2) among non-lactating women. At endline, mean retinol was higher among all groups, with the largest increase among non-lactating women of 8.1 µg/dL (19.0%). Vitamin A deficiency (serum retinol <20 µg/dL, adjusted for inflammation) was significantly lower at endline for all groups. The largest proportional difference was among children aged 24–59 months (9.9% at baseline, 0.4% at endline; $P < 0.001$).

In Malawi *(21)*, there have been consistent declines in the prevalence of vitamin A deficiency after introducing fortification of edible oil and sugar with vitamin A. However, it is impossible to isolate the effect of fortification of edible oils with vitamin A in Malawi.

Fortification with vitamin D

A systematic review *(17)* assessed the benefits and harms of edible oils and fats fortified with vitamin D as a public health intervention in the general population. Out of three included studies, one was with edible oil while two used margarine as a vehicle for fortification. There was no significant effect of vitamin D fortification on serum 25-hydroxyvitamin D (25(OH)D) concentration (MD 6.59 nmol/L, 95% CI –6.89 to 20.07; one trial; 62 participants, low-certainty evidence).

To our knowledge, there are no national or subnational surveys that have evaluated the effectiveness of fortification of oils and/or fats on vitamin D status. Finland reported improvements in vitamin D concentrations after implementing a national fortification programme, but it is impossible to ascertain whether the change is due to consumption of fortified fats or other fortified foodstuffs such as milk *(22)*. Similarly, Oman's National Nutrition Survey *(18, 19)* measured vitamin D concentrations after fortification of oil with vitamin D, but a lack of baseline data impedes drawing conclusions.

Recommendations

The overarching principle underlying the following recommendations is that fortification of edible oils and fats may be considered when there is an indication of vitamin A or D deficiency in multiple age groups; edible oils and fats are regularly and sufficiently consumed by large population groups; and the industry is reasonably consolidated to permit reaching large segments of the population with fortification.

Oils and fats are energy-dense macronutrients and their selection as food vehicles for fortification requires a balanced consideration of both quantity (how much is consumed) and type (fatty acid composition) to protect consumers from the risk of diet-related noncommunicable diseases. This means limiting – as much as possible – the fortification of oils and fats that contain trans fat or high amounts of saturated fats. Countries should make efforts to ensure the availability and affordability of healthier oils, enabling populations to transition to healthier dietary patterns.

Fortification programmes also need to consider the sensory and physical effects of the added fortificants, the stability of the fortificants under different storage and cooking conditions, the type of oil or fat, oxidative stability of the oil or fat, consumption of other fortified foods, coexisting micronutrient interventions, and the costs, feasibility and acceptability of fortification.

The following recommendations are based on the available evidence, balance of potential benefits and harms, and consideration of equity, gender equality, human rights and cost.

- Fortification of edible oils and/or fats with vitamin A, is recommended as a public health intervention to prevent subclinical vitamin A deficiency in populations at risk of vitamin A deficiency *(strong recommendation,[3] low and very low certainty of evidence)*.
- Fortification of edible oils and/or fats with vitamin D may be considered as a public health intervention to prevent vitamin D deficiency *(conditional recommendation,[4] low certainty of evidence)*.

Table 1 provides the amount of vitamins A and D to consider adding to oils and fats as part of a fortification intervention.

It is important that countries adapt the amount based on evidence about inadequacy of micronutrient intakes; per capita consumption of oils and fats; the prevalence of deficiency; coexisting interventions; policies for compliance with WHO guidelines on total fats, saturated fatty acids, *trans*-fatty acids and polyunsaturated fatty acids; and technological and cost considerations and other relevant contextual factors.

[3] A strong recommendation is one for which the guideline development group is confident that the desirable effects of adherence outweigh the undesirable effects. Implications of a strong recommendation are that most people in these settings would desire the recommended fortification of edible oils and fats. For policy-makers, a strong recommendation indicates that the recommendation can be adopted as policy in most situations.

[4] A conditional recommendation is one for which the guideline development group concludes that the desirable effects of adherence probably outweigh the undesirable effects, although the trade-offs are uncertain. Implications of a conditional recommendation for populations are that while many people would desire the recommended fortification of edible oils and fats, a considerable proportion would not. With regard to policy-makers, a conditional recommendation means that there is a need for substantial debate and involvement from stakeholders before considering the adoption of fortification of edible oils and fats with these vitamins in each setting.

[5] See Annex D – A procedure for estimating feasible fortification levels for a mass fortification programme, *Guidelines on food fortification with micronutrients (23).*

Table 1. Amounts of vitamins A and D to consider adding to oils and fats, based on safety limits[5]

Nutrient [a]	Chemical form of the compound	Amount of nutrient to be added to oil or fat [b]	
		Minimum (mg/kg[c] oil or fat)	Maximum (mg/kg oil or fat)
Vitamin A	Retinyl palmitate Retinyl acetate	12	26
Vitamin D	Vitamin D3 (cholecalciferol) Vitamin D2 (ergocalciferol)	0.20	0.43

Note: This table is for general guidance. The amounts of nutrients should be adapted according to the needs of a country. The estimates assume only edible oils and fats are being fortified. The suggested range will need to be adjusted as per the dietary intake from all sources including other large-scale food fortification programmes and/or micronutrient interventions with vitamins A and D being implemented at the same time.

[a] Antioxidants permitted by the Codex Alimentarius General Standard for Food Additives (CXS 192-1995) in food category 02.1.2 (vegetable oils and fats) or national regulatory authorities may also be added to the base oils and fats to prevent oxidative damage to the oils and fats and improve the stability of vitamins A and D.

[b] Countries are encouraged to adapt the suggested amounts of vitamins A and D through a modelling approach that is guided by country-specific information on prevalence of inadequate intake of vitamin A and/or D in different population subgroups and consumption estimate of selected oil or fat in different population subgroups. Calculation of reduction in the prevalence of inadequate intakes (i.e. the proportion below the EAR) and the risk of excessive intakes (i.e. the proportion above the UL) for different levels of fortification is important to determine the appropriate fortification level using a risk-based approach. Detailed calculation methods and alternative approaches are available in Chapter 7 and Annex D, *Guidelines on food fortification with micronutrients (23).*

[c] Refer to the following unit conversion as applicable: vitamin A (retinol): 1 IU = 0.3 µg RE (retinol equivalent); vitamin D (cholecalciferol or ergocalciferol): 1 IU = 0.025 µg.

Key remarks

Any fortification guideline must be considered within the context that the goal of food-related policies, and most desirable health outcome, is for everyone to be able to follow a dietary pattern that meets dietary guidelines by consuming a range of foods that have a natural or inherent composition that provides desirable amounts of nutrients for health.

The guideline development group recognized tensions associated with the selection of oils and fats as vehicles for food fortification. It acknowledged the need to fortify foods that are widely consumed by populations at risk of micronutrient deficiencies. However, it also acknowledged that the rise of diet-related noncommunicable diseases associated with the consumption of foods high in salt, sugar and unhealthy fats raises concerns about whether fortifying oils and fats – particularly those made up of a high proportion of saturated or trans-fatty acids – overshadows or contradicts efforts to promote the consumption and affordability of healthier diets among the most vulnerable.

The following remarks are suggestions intended to provide some considerations for implementation of the recommendations and are based on the discussions of the guideline development group.

- When vitamin A and/or D deficiency is a public health problem in a country, establishing a food fortification programme with edible oils and fats as food vehicles can be considered to address the problem. Dietary intake of edible oils and fats among different population groups and coexisting interventions such as supplementation should be taken into consideration. Competent authorities should determine whether fortification of edible oils and/or fats with vitamin A and/or D should be mandatory or voluntary, depending on the severity and extent of the public health problem.

- Oils and fats are energy-dense macronutrients. The implications of fortifying edible oils and fats with vitamins A and D for the noncommunicable disease burden must be evaluated with regard to each country's context. Governments may also wish to be satisfied that market promotion of fortified foods does not conflict with, or compromise, any national food and nutrition policies on healthy eating. The objective of fortification is to shift consumption from non-fortified oils or fats to fortified oils or fats, and not to increase total consumption of oils and fats.

- The choice of oils and fats may be influenced by availability (supply), consumer preference (demand), regulatory requirements, cost considerations and industry capacity to innovate. Decision-making needs to consider ongoing efforts to both replace saturated fatty acids with polyunsaturated and monounsaturated fatty acids from plant sources, and comply with WHO guidelines on total fats, saturated fatty acids, trans-fatty acids and polyunsaturated fatty acids. Adoption of multisectoral policies to support production of healthier oils might help to increase their use by industry and consumers.

- Feasibility of a fortification programme should be evaluated in each context prior to its initiation while considering the implications of whether the programme is envisaged as voluntary or mandatory. It is important to assess the following key aspects of the selected edible oil or fat and nutrient(s) – coverage, cost, consumption, coexisting micronutrient interventions, industry capacity and consolidation, compatibility of nutrients, bioavailability, and the stability of the nutrients under prevailing conditions of storage, transportation, packaging and cooking. Dietary assessment and modelling, sensory studies, stability studies, consumer acceptability surveys, qualitative research and consumption monitoring can provide vital inputs for fortification programme design. Input from all involved sectors to the design and delivery of fortification programmes is pivotal to ensure a safe, practical and sustainable approach.

- The regulatory framework, including regulations, coordination and oversight mechanisms, compliance monitoring and enforcement measures, plays a key role in the success of any fortification programme. Fortification requirements should be integrated into the existing regulatory framework rather than implemented as a stand-alone intervention.
  Regulatory authorities and relevant national bodies should review scientific evidence and data on nutrient gaps, consumption pattern of selected edible oils or fats, acceptability, feasibility, appropriate fortificant forms and methods to establish realistic target levels/range for industry compliance. Global and regional guidelines, including general principles and guidelines by the Codex Alimentarius Commission, WHO guidelines and any relevant regional standards can provide vital inputs for formulation of fortification standards and regulations as well as operational aspects of monitoring and compliance.

- For effective implementation, regulatory monitoring is essential to ensure nutrient, quality and safety standards as outlined in regulations and standards are followed. Regulatory monitoring encompasses all monitoring activities conducted at the production level, and monitoring at customs, warehouses and retail stores. These activities are carried out by the relevant regulatory authorities and by producers as part of self-regulation programmes.
  Adopting a risk-based approach enables regulatory authorities to prioritize and direct resources in addressing key problems. One such example is prioritizing the regulatory monitoring of bulk/unbranded oil to ensure traceability and check compliance against standards including fortification standards, as applicable.

- Nutrition labelling has a dual purpose of protecting the health of consumers and ensuring fair trade practices. When instituting fortification programmes, governments have a duty to ensure that consumers are not misled or deceived about the true nutritional properties of the fortified oil or fat. Irrespective of the nature of the fortification programme – that is, voluntary or mandatory – labelling requirements should apply to all fortified foods.

- The effect of fortification programmes on nutrient status and health outcomes should be monitored regularly through a monitoring and evaluation plan with appropriate indicators, including indicators that consider equity, gender equality and human rights. Monitoring of multiple micronutrient interventions in the same population groups is critical to prevent excessive intake of nutrients.

- Aspects of equity, gender and human rights should be integrated across the different stages of the fortification programme lifecycle, that is, programme design, monitoring and evaluation.

Research needs

Diet evolves over time, being influenced by many social and economic factors that interact in a complex manner. Additionally, other factors, such as advances in research and new guidance and regulations at a country or global level, may also need to be considered. Programme design should consider a periodic review of available data related to micronutrient needs, consumption patterns of fortified foods, other sources of the micronutrient in the diet, contextual factors and the validity of relevant design assumptions.

The following research needs were identified during the guideline's development.

- There is limited information regarding actual consumption of oils and fats, particularly in regions where vitamin A or vitamin D deficiency is prevalent, which limits understanding of the dietary gap to be addressed through fortification, the use of fortified oils or fats when a programme is implemented and the subsequent impact on nutrient adequacy.
- Further studies are needed to ensure more effective use of these recommendations, including those that:
    - study efficacy and effectiveness, especially for vitamin D fortification and co-fortification with both vitamins A and D;
    - assess the prevalence of vitamin deficiencies in populations using different biochemical indicators to identify populations that are most at risk of deficiency or excessive intake, or in need of intervention;
    - contribute towards international consensus on assessment of vitamin D deficiency in populations;
    - further determine the fortificant concentration that can lead to adequate vitamin A and D intake from edible oils and fats in different population subgroups;
    - assess the coverage of fortified oils and fats across different population subgroups; and
    - assess the contribution of fortification to nutrient adequacy, and its potential health impact and adverse effects.
- The stability of different vitamin A and D compounds needs to be established for their use in oils and fats, including in relation to packaging, storage, distribution, consumption and cooking methods in different settings. This includes assessing the stability of vitamins:
    - with different packaging materials and exposure to light
    - with different cooking methods
    - in relation to storage time and oxidative status of the oil or fat
- National studies are encouraged to demonstrate that large-scale fortification reaches those that may be at higher risk of micronutrient deficiencies such as women, children, certain socioeconomic groups, people with low dietary diversity or living in households that are food insecure, and those living in specific geographical regions.
- Data collection, including coverage studies, should capture the needs of various groups through an intersectional lens. Further research is needed to integrate aspects of health equity, gender equality and human rights into programme design, monitoring and evaluation.

Plans for updating the guideline

The WHO Secretariat will continue to follow research developments in edible oils and fats fortification, particularly those related to questions for which the certainty of evidence was found to be low or very low. If the guideline merits an update, or if there are concerns about the validity of the guideline, the Department of Nutrition and Food Safety, in collaboration with other WHO departments or programmes, will coordinate the guideline update, following the formal procedures of the *WHO handbook for guideline development*.

Scope and purpose

WHO is committed to increasing the impact of public health measures in every country, ensuring healthy lives and promoting well-being for all people at all ages. As part of its unique normative function in public health, WHO aims to provide global, evidence-informed recommendations on the fortification of edible oils and fats with vitamins A and D to improve health outcomes associated with micronutrient status. This guideline will support the work of WHO regional and country offices and will help Member States and their partners to make evidence-informed decisions on appropriate actions in their efforts to improve access to quality essential health services, support countries to be prepared for health emergencies and address the determinants of health. The guideline will also help to increase countries' capacity to respond to their owns needs related to improving micronutrient status and to prioritize essential nutrition actions in national health policies, strategies and plans.

The guideline aims to help Member States and their partners in their efforts to make informed decisions on the appropriate nutrition actions to achieve the 2030 Sustainable Development Goals *(5)*, in particular, goals 2 (end hunger, achieve food security and improved nutrition, and promote sustainable agriculture) and 3 (ensure healthy lives and promote well-being for all at all ages). The guideline will also support Member States in their efforts to achieve the global targets of the *Comprehensive implementation plan on maternal, infant and young child nutrition (6), The global strategy for women's, children's and adolescents' health (2016–2030) (24)* and the World Health Assembly resolution to accelerate efforts for preventing micronutrient deficiencies and their consequences through safe and effective food fortification *(25).*

The recommendations in this guideline are intended for a wide audience, including policy-makers, their expert advisers, and technical and programme staff at ministries and organizations involved in the design, implementation and scale-up of nutrition actions for public health. This guideline is intended to contribute to discussions among stakeholders when selecting or prioritizing interventions to be undertaken in their specific context. It presents the key recommendations and a summary of the supporting evidence. Further details of the evidence base are provided in Annex 1 and in the reference.

[6] This publication is a WHO guideline. A WHO guideline is any document, whatever its title, containing WHO recommendations about health interventions, whether they be clinical, public health or policy interventions. A recommendation provides information about what policy-makers, health care providers or patients should do. It implies a choice between different interventions that have an impact on health and that have ramifications for the use of resources. All publications containing WHO recommendations are approved by the WHO Guidelines Review Committee.

Background

Globally, more than 2 billion people suffer from vitamin or mineral deficiencies, with serious long-term consequences for children, families, and societies *(26)*. Even though they are only required in tiny amounts, vitamins, and minerals (known collectively as micronutrients) are vital for health. The most common micronutrient deficiencies – caused by insufficient intake or absorption – relate to iron, vitamin A, folate and iodine. Micronutrient deficiencies can cause serious health issues and can impair intellectual capacity and educational attainment, but they are correctable and preventable.

A lack of iron, folate, and vitamins B_{12} and A, for example, can lead to anaemia, which can cause fatigue, weakness, shortage of breath and dizziness, and lead to problems functioning in work or at school. It is estimated that 39.8% of children aged under 5 years (2019) and 35.5% of pregnant women (2023) worldwide are anaemic *(27)*.

Vitamin A deficiency

Vitamin A deficiency may cause anaemia by impairing iron metabolism, but the mechanism by which this occurs is unclear. In studies of vitamin A-deficient populations, improving vitamin A status reduces the prevalence of anaemia in most but not all studies *(28)*. A systematic review, which included 19 studies involving over 310 000 women, concluded that vitamin A supplementation during pregnancy can reduce the risk of maternal anaemia *(29)*.

Vitamin A deficiency remains the leading cause of preventable blindness in children and also increases the risks of severe illness or death from infections. As of 2005 *(1)*, one third of the global pre-school aged children were estimated to have vitamin A deficiency with the highest prevalence in Africa (44%) and South-East Asia (50%). A pooled analysis of population-based surveys *(2)* from 138 low-income and middle-income countries reported varying trends in the prevalence of vitamin A deficiency among children by region and a decline in global prevalence from 39 (1991) to 29% (2013). Globally, approximately 15% of pregnant women (1995–2005) are estimated to have vitamin A deficiency which may increase the risk of maternal mortality *(1)*. Vitamin A deficiency is a public health problem in more than half of all countries. It is especially a problem in Africa and South-East Asia and affects young children and pregnant women in low-income countries the most. Usually, vitamin A deficiency develops in an environment of ecological, social and economic deprivation, in which the key risk factors for vitamin A deficiency are a diet low in sources of vitamin A (such as dairy products, eggs, fruits and vegetables), poor nutritional status and a high rate of infections, in particular, measles and diarrhoeal diseases *(23)*.

Supplementation with high doses of preformed vitamin A is currently one of the most widely used interventions to deliver vitamin A. More than 80 countries worldwide are implementing universal vitamin A supplementation programmes targeted to children aged 6–59 months through semi-annual national campaigns. The aim of supplementation with a high dose of vitamin A is to reduce mortality associated with measles, diarrhoea and other illnesses, and not to sustainably improve the vitamin A status of populations. For this reason, although vitamin A supplementation provides a protective dose against vitamin A deficiency, complementary interventions are needed to ensure continued vitamin A intake, such as biofortification, fortification, point-of-use fortification with micronutrient powders, promotion of dietary diversity, nutrition education, and prevention and control of infectious disease *(30)*.

Vitamin D deficiency

Vitamin D is an essential nutrient for bone health and may influence the risks of respiratory illness, adverse pregnancy outcomes and chronic diseases of adulthood. Because many countries have a relatively low supply of foods rich in vitamin D and inadequate exposure to natural ultraviolet B radiation from sunlight, a proportion of the global population is at risk of vitamin D deficiency. Bone and musculoskeletal effects are the most studied and substantial health outcomes of vitamin D deficiency. However, an increasing number of studies have suggested that vitamin D may influence other important health outcomes in infants, children, and pregnant and lactating women *(31)*.

Milk and other dairy products, including dried milk powder and evaporated milk, are often fortified with vitamin D. In many countries, margarine is also fortified with vitamin D. Vitamin D deficiency has become an issue of public health concern, particularly in regions with insufficient sunshine or with sun avoidance behaviours that prevent or impede dermal synthesis, and infrequent access to or intake of dietary sources of vitamin D. In such situations, fortification of milk and margarine with vitamin D have proved to be useful strategies.

Food fortification to address micronutrient deficiency

More than 3 billion people worldwide cannot afford a healthy diet, made up of the diverse mix of nutrient-rich foods needed to promote health and provide all the essential nutrients *(32)*. The world urgently needs to improve access to affordable healthy diets.

Emerging, unexpected situations that affect health, food safety or trade have a deep impact on micronutrient status, especially that of vulnerable populations such as children, women and older persons. The COVID-19 pandemic has likely worsened the already high prevalence of micronutrient deficiencies worldwide.

Lockdowns and physical distancing may have led to decreased family income and reduced access to crops, food, services, health care and social protection programmes that may have resulted in increased rates of malnutrition and micronutrient deficiency. Modelling predicts that the COVID-19 pandemic will have a significant impact on maternal and child undernutrition and child mortality in the current generation, leading to large long-term negative consequences for productivity *(33)*.

Fortification is an evidence-informed intervention that contributes to the prevention, reduction and control of micronutrient deficiencies. It can be used to correct a demonstrated micronutrient deficiency in the general population (through mass or large-scale fortification) or in specific population groups (through targeted fortification) such as children, pregnant women and the beneficiaries of social protection programmes *(25)*. By adding essential vitamins and minerals to staple foods and condiments, such as wheat and maize flours, rice, cooking oil and salt, in accordance with national consumption patterns and deficiencies, countries can correct and further prevent a demonstrated micronutrient deficiency.

Edible oils are important candidate vehicles for fortification in many low- and middle-income countries, as they are centrally processed, broadly distributed and widely consumed by all age groups. 39 countries mandate or allow voluntary fortification of edible oils with vitamin A alone, and 13 countries mandate or allow voluntary co-fortification with both vitamins A and D *(34)*.

Properties of edible oils and fats

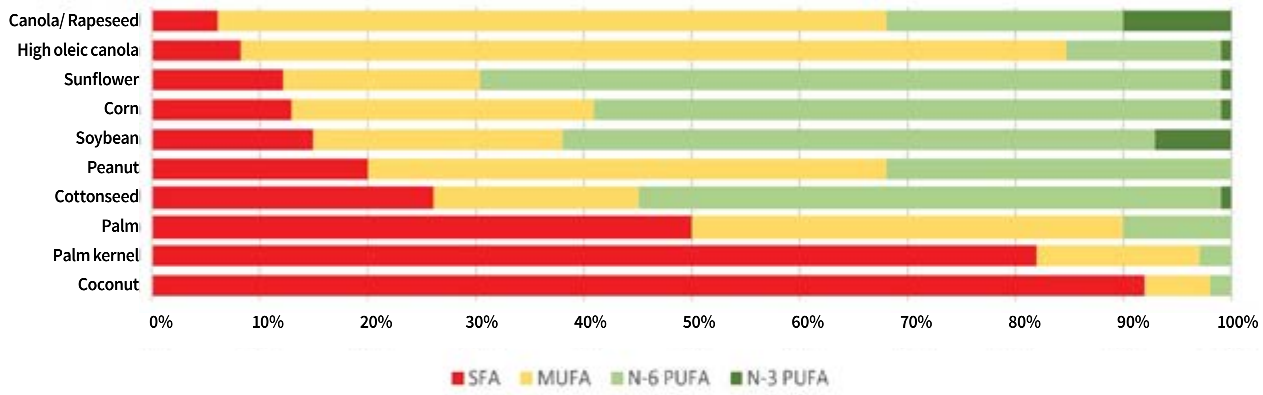
Oils and fats are made up of triglycerides, which consist of a glycerol molecule attached to three chain-like molecules that mostly consist of carbon and hydrogen, called fatty acids. Fatty acids can be grouped into four main types: saturated, monounsaturated, polyunsaturated and *trans*-. Oils and fats are mixtures of different types of fatty acids as components of triglycerides (Fig. 1). Both the type of fatty acid and the placement can affect the functional properties of oils and fats, including their melting temperature (the temperature at which the mixture is liquid rather than a solid – higher melting temperatures mean that the fat is more likely to be solid at room temperature), crystallization characteristics and resistance to oxidation (a reaction with oxygen that leads to off-flavours and dark colours) *(35)*. Although the terms oils and fats are often used interchangeably, they are often used to distinguish triglycerides in the liquid state at ambient temperatures (oils) from those in the solid state (fats). Saturated fatty acids give solidity and texture, and high resistance to oxidation. Monounsaturated fatty acid oil is liquid and has good resistance to oxidation, while polyunsaturated fatty acid oil is liquid and more susceptible to oxidation.

The major chemical reactions affecting the shelf stability in oils and fats are oxidation and hydrolysis. Oxidation progresses at different rates depending on factors such as temperature, light, availability of oxygen, and the presence of moisture and metals (such as iron). The type of oil also influences the rate of oxidation. Edible oils and fats can be tested to measure the degree of deterioration as well as stability against such changes over time.

The degree of oxidation can be measured using the peroxide value (PV). It is one of the most used methods for determining the freshness and quality of oils and fats. Anisidine value (AV) test is used to assess the secondary oxidation products of oils and fats and is an indicator of excessive oil deterioration especially in deep frying process. The Totox value, calculated by the formula AV + 2PV, indicates the overall oxidation state of oils and fats.

Hydrolytic or lipolytic rancidity is the breakdown of triglycerides into constituent free fatty acids because of a reaction with water (usually in the presence of an active catalyst). This reaction is favoured by presence of moisture, high temperature and natural lipolytic enzymes. Acid value measures the free fatty acids resulting from hydrolytic rancidity of oils and fats.

Fig. 1. Approximate fatty acid composition of selected vegetable oils



SFA: saturated fatty acid; MUFA: monounsaturated fatty acid; N-6 PUFA: omega-6 polyunsaturated fatty acid; N-3 PUFA: omega-3 polyunsaturated fatty acid.

Source: WHO *(35)*.

Edible oils and fats come from both plant and animal sources and have important functional and nutritional properties in foods. Over the last five years, WHO has published several guidelines on the consumption of oils and fats. WHO recommends that adults should limit total fat intake to 30% of total energy intake or less. Fat consumed by everyone aged 2 years and older should be primarily unsaturated fatty acids, with no more than 10% of total energy intake coming from saturated fatty acids and no more than 1% of total energy intake from *trans*-fatty acids from both industrially produced and ruminant animal sources. Saturated fatty acids can be found in fatty meat, dairy foods, and hard fats and oils such as butter, ghee, lard, palm oil and coconut oil and *trans*-fatty acids in baked and fried foods, pre-packaged snacks, and meat and dairy foods from ruminant animals, such as cows or sheep. Saturated and *trans*-fatty acids in the diet can be replaced with other nutrients such as polyunsaturated fatty acids and monounsaturated fatty acids from plant sources *(11, 12, 36).*

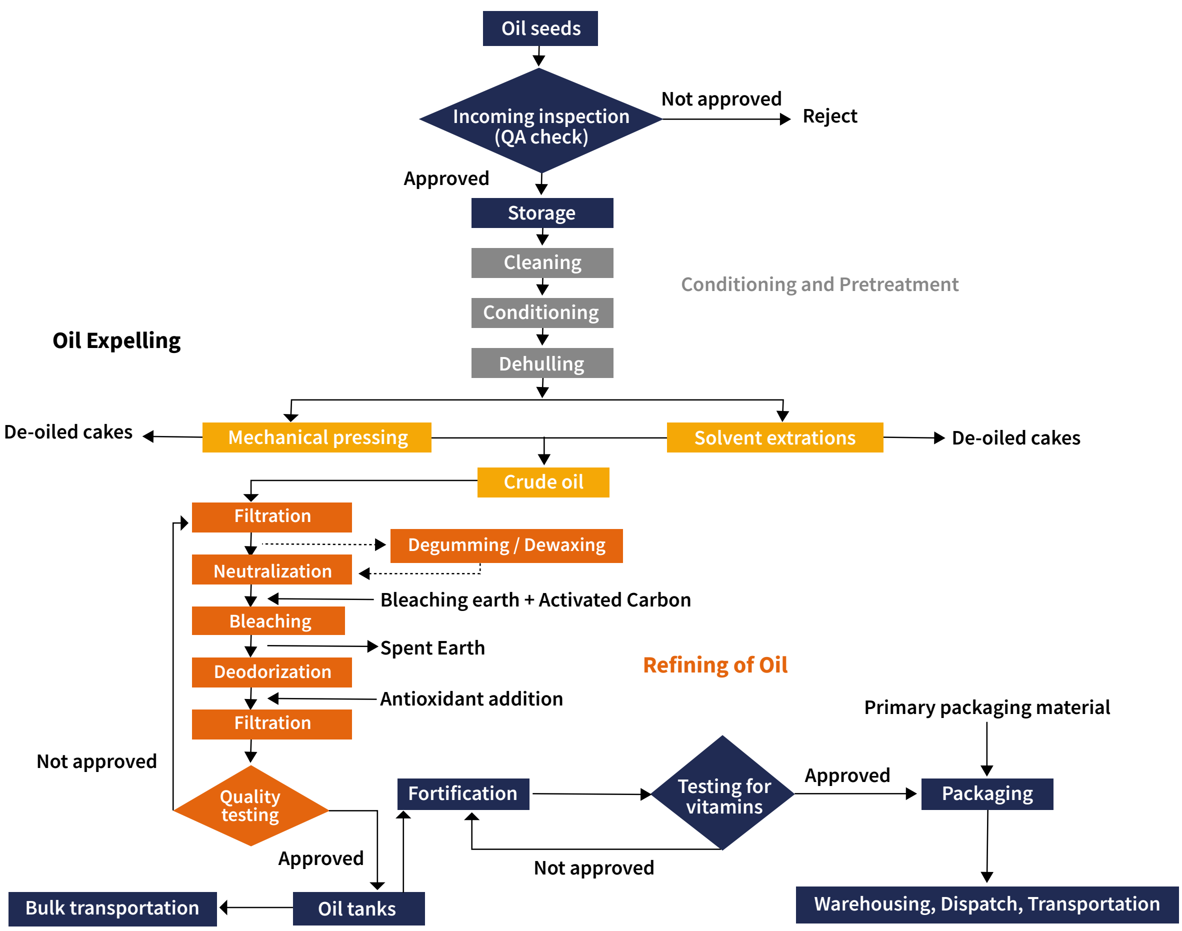
Trans fats, or *trans*-fatty acids, are unsaturated fatty acids that come from either industrial or natural sources. Eliminating trans fat is a powerful way to prevent heart disease and reduce the high costs incurred for individuals and the economic burden, including medical costs and lost productivity. WHO remains committed in supporting countries in their efforts to eliminate industrial trans fats. The REPLACE action package provides a strategic approach to eliminating industrially produced trans fat from national food supplies. The package comprises an overarching technical document that provides a rationale and framework for this integrated approach to trans fat elimination, along with six modules and additional web resources to facilitate implementation.

Fortification of edible oils and fats

Oil fortification consists of adding appropriate amounts of concentrate of vitamins A and D to refined, bleached and deodorised oil at 45–50 °C. The solubility of commercially available blends of vitamins A and D in vegetable oils is excellent. Common blends contain 1 million (300 000 µg retinol activity equivalents) and 1.7 million (510 000 µg retinol activity equivalents) international units (IU) of vitamin A palmitate and 100 000 IU of vitamin D (2500 µg vitamin D) in liquid form, stabilized with vitamin E (α-tocopherol) or a mixture of butylated hydroxyanisole and butylated hydroxytoluene. To ensure that the vitamins are uniformly distributed, mixing takes place in vertical tanks that contain turbines or propeller agitators. Mixing can also be done through circulation of the oil using pumps, from bottom of tank to the top. Edible antioxidants (butylated hydroxyanisole and butylated hydroxytoluene) or natural antioxidants (e.g. α-tocopherol, ascorbyl palmitate) may be added to protect both the vitamin A and the oil. Vitamin A is sensitive to oxidation by air. Loss of activity is accelerated by heat and exposure to light. Oxidation of fats and oils can destroy fat-soluble vitamins, including vitamin A. The stability of vitamin A in oil therefore depends greatly on the stability of the oil itself. Vitamin A stability is enhanced in high quality oils with low PV as these oils are less prone to oxidation. To maintain vitamin A activity, fortified oil can be packaged in light-protected, sealed containers. Replacing the container headspace with inert gas helps to retain the stability of both the oil and vitamin A until the container is opened *(37).*

The fortification process and supply chain (Fig. 2) in any given country involves an intricate network of public and private entities that link suppliers (of oilseeds, crude oil and refined oil), wholesalers, premix suppliers, oil refineries, packers, retailers and food processors to the final consumers. Other stakeholders include transporters; companies that supply oilseeds, agrochemicals and agricultural equipment; irrigation companies; industry associations; inspection agencies; government departments of commerce, tax and agriculture; and other state agencies that control flow through the supply chain through governmental policies *(38).*

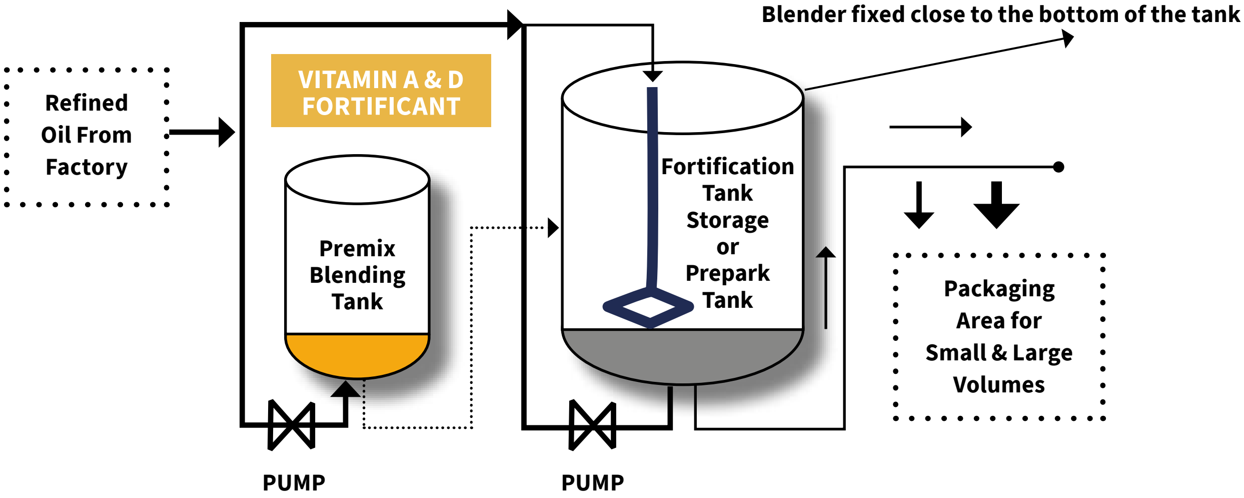
Fig. 2. Process flow diagram for edible oil processing and fortification



Source: Global Alliance for Improved Nutrition *(39).*

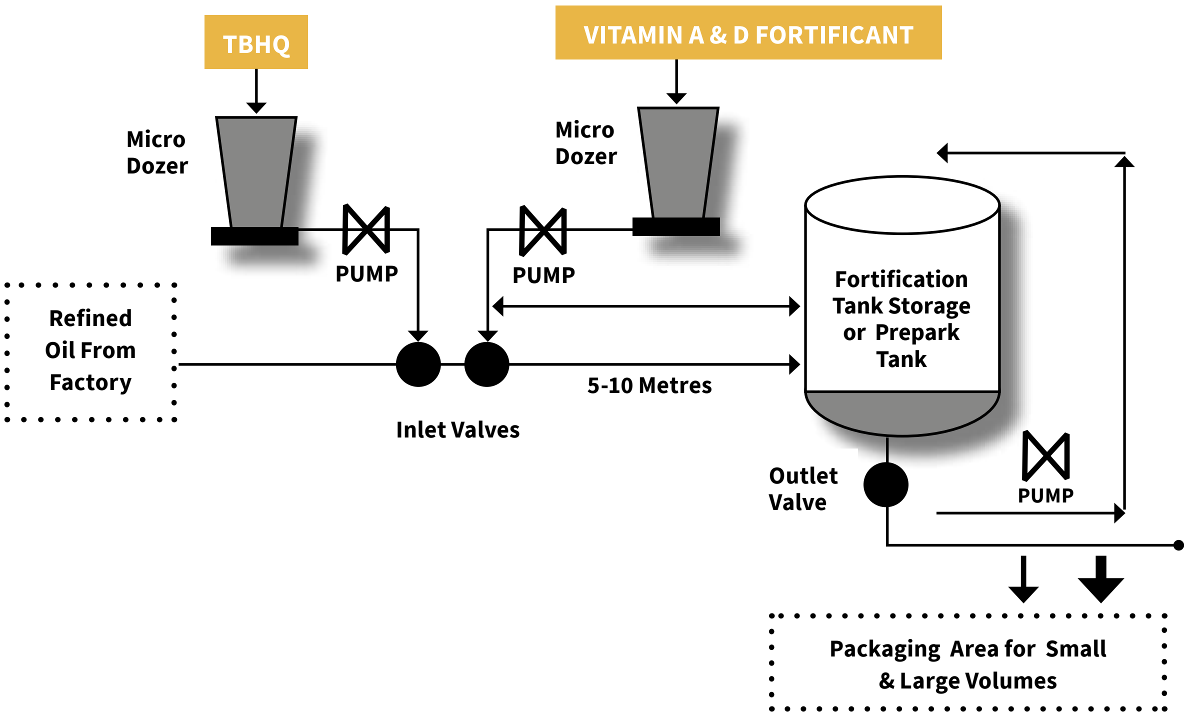
The production and fortification of oil or margarine can be carried out in either a batch-mixing or a continuous mixing process (Fig. 3 and Fig. 4). The vitamin blend is premeasured according to the batch size and mixed with warm oil, in a ratio as high as 1 : 5, until a uniform solution is obtained. The premix is then incorporated into the oil or margarine before the emulsifying process *(37).*

Fig. 3. Batch-mixing process for mixing of vitamin pre-blend with oil



Source: Global Alliance for Improved Nutrition *(39).*

Fig. 4. Continuous mixing process for mixing of vitamin pre-blend with oil



Source: Global Alliance for Improved Nutrition *(39).*

To date (2024), thirty-nine countries mandate or allow voluntary fortification of edible oils with vitamin A alone, and 13 countries mandate or allow voluntary co-fortification with both vitamins A and D *(34)*. Table 2 shows the amount and chemical forms of vitamins A and D included in regulations and standards.

Table 2. Amount and chemical forms of vitamins A and D included in oil fortification regulations and standards

Micronutrient	Range of amount used (mg/kg oil)	Most common amount used (mg/kg oil)	Number of countries with regulations or standards	Most common chemical form of compound used (number of countries)
Vitamin A	3.5–41.25	20	39	Retinyl palmitate (31) Retinyl acetate (4) Retinol (4) Beta-carotene (2) Unspecified (5)
Vitamin D	0.01–0.75	0.14	13	D_3 cholecalciferol (9) D_2 ergocalciferol (3) Unspecified (4)

Source: Based on data compiled by the Global Fortification Data Exchange (2024).

Objectives

The main objective of this guideline is to provide locally adaptable, clear, evidence-informed global recommendations on the fortification of edible oils and fats with vitamins A and D as a public health strategy to improve the micronutrient status of populations. The recommendations are grounded in gender equality, equity and human rights approaches with the aim of leaving no one behind.

The guideline complements the *Guidelines on food fortification with micronutrients* published by WHO and the Food and Agricultural Organization of the United Nations (FAO) *(23)*, and other WHO guidelines on fortification of rice *(8)*, maize flour and corn meal *(7)* and wheat flour *(9).* The recommendations in this guideline should be considered in the context of WHO guidelines on healthy diets *(10)*, including those on saturated fatty acids, *trans*-fatty acids and polyunsaturated fatty acids. WHO has updated its guidance on total fat, saturated fatty acids and *trans*-fatty acids, based on the latest scientific evidence. Two new guidelines, on saturated fatty acid and *trans*-fatty acid intake for adults and children *(11)* and total fat intake for the prevention of unhealthy weight gain in adults and children *(12),* contain recommendations that aim to reduce the risk of unhealthy weight gain and diet-related noncommunicable diseases.

Summary of the evidence

Edible oils and fats fortification and population micronutrient status and health-related outcomes

The evidence-to-decision summaries presented to the guideline development group to inform the recommendations were based on a systematic review addressing the guideline questions in PICO format, a summary of available pre- and post-surveys from national or subnational fortification programmes in Indonesia, Malawi and Oman, and a desk review of implementation considerations.

Fortification with vitamin A

A systematic review assessed the benefits and harms of edible oils and fats fortified with vitamin A as a public health intervention in the general population. The review included five studies comparing vitamin A fortification with no fortification with vitamin A, including two RCTs *(40-42)*, two CCTs *(43, 44)* and one birth cohort study *(45)*. The duration of the interventions varied from eight weeks to six months. In four studies, oil was the vehicle for fortification; in the other study, margarine was. The vitamin A compound added to the oil was not specified in one study. Retinyl palmitate was added in two studies (30 IU/g in one study and 38.06–84.06 IU/g in the other), provitamin A carotenoids were added in another (8.5 IU/g), and a combination of retinol and beta-carotene was added in the remaining study (862 µg retinol equivalents (RE) per 30 g of margarine, made up of 108 µg RE from beta-carotene and 754 µg RE added as retinyl palmitate). The age at outcome measurement was 4–40 years.

Serum retinol concentrations were measured in two RCTs, which had an intervention duration of six months, and two CCTs, which had an intervention duration ranging from two to five months. Two studies, one RCT and one CCT, evaluated subclinical vitamin A deficiency. All-cause morbidity was measured in two RCTs and one CCT; however, only one RCT and the CCT reported results. No studies reported data on clinical vitamin A deficiency, all-cause mortality or adverse effects.

Dietary vitamin A intake and the retinol content of breast milk were each measured in one trial. None of the studies reported data on serum ferritin concentrations or anaemia prevalence.

The evidence from RCTs showed no significant effect of fortification with vitamin A on serum retinol concentrations (MD 0.35 µmol/L, 95% CI –0.43 to 1.12; two trials, 514 participants; low-certainty evidence). This was consistent with the evidence from non-randomized studies (MD 0.31 µmol/L, 95% CI –0.18 to 0.80; two trials, 205 participants; very low-certainty evidence). Similarly, the evidence from one RCT showed no significant effect of fortification on subclinical vitamin A deficiency, measured as serum retinol equal to or less than 0.70 µmol/L (none of 268 participants in the intervention group and none of 144 participants in the control group, RR not estimable; one trial; low-certainty evidence), as did the evidence from one CCT (RR 0.21, 95% CI 0.01 to 4.10; one trial, 31 participants; very low-certainty evidence). The evidence from one RCT (low-certainty evidence) and one CCT (very low-certainty evidence) showed no significant effect of fortification on all-cause morbidity.

Low-certainty evidence from one RCT showed no significant difference in vitamin A intake between participants receiving the vitamin A fortification intervention and the controls (MD 15.7 µg RE/day, 95% CI –105.82 to 74.42; one trial, 412 participants). Very low-certainty evidence from one CCT showed a significant effect of fortification (MD 240.6 µg RE/day, 95% CI –175.1 to 306.1; one trial, 31 participants).

Consumption of oil fortified with vitamin A resulted in better vitamin A status, measured as retinol concentrations in breast milk (MD 0.79 µmol/L, 95% CI 0.72 to 0.86; one trial, 63 participants; very low-certainty evidence), and lower risk of low retinol concentrations in breast milk, defined as less than 1.05 µmol/L (RR 0.04, 95% CI 0.01 to 0.14; one trial, 101 participants; very low-certainty evidence). Other markers of vitamin A status were not measured in the included trials.

The authors of the review concluded that fortification of oils and fats with vitamin A may result in little to no difference in serum retinol concentrations in general populations. They highlighted that further studies are needed to determine the fortification dose that can lead to adequate vitamin A intakes, and to assess potential adverse effects.

The WHO guideline development group considered the following outcomes to be critical for decision-making: clinical vitamin A deficiency (xerophthalmia and night blindness), serum or plasma retinol concentration, subclinical vitamin A deficiency (defined as serum retinol concentration less than 0.70 µmol/L), adverse effects (hypervitaminosis and liver toxicity), all-cause morbidity and all-cause mortality. The following secondary outcomes were included: vitamin A status (as defined by trialists – liver stores of vitamin A, concentration of retinol in breast milk, modified relative dose response (MRDR) test results, retinol-binding protein concentration, serum or plasma beta-carotene concentration), intake of dietary vitamin A, and iron status (ferritin) and anaemia.

On application of the GRADE methodology, the certainty of the direct evidence for the only critical outcome for which evidence was available ranged from low to very low. The GRADE summary of findings table for the fortification of edible oils and fats with vitamin A is provided in Annex 1.

The findings from the review were complemented by the findings of representative pre- and post-surveys of edible oils and fats fortification programmes in Oman *(18, 19)*, Indonesia *(20)* and Malawi *(21)*.

In Oman, a national survey conducted in 2004 *(19)* showed that among children with normal C-reactive protein values the overall prevalence of vitamin A deficiency (serum retinol less than 0.70 µmol/L) was approximately 6.4% among children aged 6–59 months. The prevalence of vitamin A deficiency among non-pregnant women of reproductive age was 0.4%. When women with high C-reactive protein values were excluded, there were no cases of vitamin A deficiency. However, a limitation reported by the survey was that the number of participants with C-reactive protein analysed was lower because of non-response to blood withdrawal. The survey also showed that oil was available in most households (95%), and Omani products were the most consumed. Over half of households reported consuming at least 1 L of oil per person per month. These survey findings led to the introduction of mandatory fortification of oils and ghee with vitamin A. A subsequent national survey conducted in 2017 found that the prevalence of vitamin A deficiency (measured using ad hoc retinol-binding protein levels, adjusted for inflammation) among the Omani population was 0.2% among non-pregnant women and 9.5% among children aged 6–59 months. However, it was recognized that the comparability of the surveys has limitations given that the biomarkers and adjustments for inflammation were not consistent. Also data on coverage is not reported.

In West Java, Indonesia *(20)*, pre- and post-surveys were conducted in 24 villages in the Tasikmalaya and Ciamis districts, where fortified oil had a coverage of 50% or higher. The sampling frame targeted the poor population suspected to be at highest risk of low serum retinol. Mean serum retinol concentrations at baseline ranged from 30.7 µg/dL (standard deviation (SD) 12.6) among children aged 6–11 months to 34.3 µg/dL (SD 12.6) among children aged 5–9 years. In lactating mothers, average serum retinol was 30.7 µg/dL (SD 12.6) while it was 42.7 µg/dL (SD 19.2) among non-lactating women. At endline, mean retinol was higher among all groups, with the largest increase among non-lactating women of 8.1 µg/dL (19.0%). Vitamin A deficiency (serum retinol <20 µg/dL, adjusted for inflammation) was significantly lower at endline for all groups. The largest proportional difference was among children aged 24–59 months (9.9% at baseline, 0.4% at endline; $P < 0.001$).

In Malawi *(21)*, there have been consistent declines in the prevalence of vitamin A deficiency after introducing fortification of edible oil and sugar with vitamin A. However, it is impossible to isolate the effect of fortification of edible oils with vitamin A in Malawi.

Fortification with vitamin D

A systematic review *(17)* assessed the benefits and harms of edible vegetable oils and fats fortified with vitamin D as a public health intervention in the general population. The review included three studies comparing vitamin D fortification with no fortification with vitamin D, including one RCT (46, 47), and two birth cohort studies *(48-61)*.

In the birth cohort studies, women were exposed or not exposed to vitamin D fortification during pregnancy, and outcomes were investigated during the exposure period or nine to 27 years after birth. Of the three studies investigating the effect of vitamin D fortification, one investigated fortified canola oil (1000 IU vitamin D/25 g) and two investigated mandatory margarine fortification (1.25 µg vitamin D/100 g).

Serum 25(OH)D concentrations were measured in a single RCT that included 62 participants. The RCT found no significant difference between groups (MD 6.59 nmol/L, 95% CI –6.89 to 20.07; low-certainty evidence). No studies reported data on vitamin D deficiency, osteomalacia in older persons, nutritional rickets in children, adverse effects, morbidity or mortality.

Based on one RCT, vitamin D fortification had no effect on vitamin D intake (MD 22.35 µg/day, 95% CI –52.88 to 8.18; one trial, 62 participants; low-certainty evidence). In terms of maternal and infant outcomes, based on one birth cohort study, there was no significant difference in the incidence of gestational diabetes mellitus between the group exposed to vitamin D fortification and those not exposed (RR 0.87, 95% CI 0.75 to 1.01; one trial, 28 871 participants; very low-certainty evidence). A birth cohort study showed that the rate of pre-eclampsia also did not significantly differ between the exposed and non-exposed groups (RR 1.04, 95% CI 0.96 to 1.12; one trial, 73 237 participants; very low-certainty evidence).

Based on one birth cohort study, at age 7 years, children of mothers exposed to vitamin D-fortified margarine during pregnancy had a significantly lower body mass index (BMI) (MD –0.1 kg/m^2, 95% CI –0.17 to –0.03; one trial, 10 832 participants; low-certainty evidence), lower risk of overweight (RR 0.92, 95% CI 0.86 to 0.98; one trial, 10 832 participants; low-certainty evidence) and lower risk of obesity (RR 0.85, 95% CI 0.77 to 0.95; one trial, 10 832 participants; low-certainty evidence) compared with children of mothers not exposed to vitamin D-fortified margarine during pregnancy.

Consumption of vitamin D-fortified oil had no significant effect on serum parathyroid hormone concentrations (MD 0.10 pmol/L, 95% CI –0.99 to 1.99; one trial, 36 participants; low-certainty evidence) or serum bone alkaline phosphatase concentrations (MD 5.76 IU/L, 95% CI –0.12 to 11.64; one trial, 36 participants; low-certainty evidence).

One birth cohort study investigated fracture events among children whose mothers were exposed or not exposed to vitamin D-fortified margarine and found a lower risk of fracture events in the exposed group compared with the non-exposed group (RR 0.84, 95% CI 0.82 to 0.85; one trial, 217 983 participants; low-certainty evidence). In the same birth cohort study, children of mothers exposed to vitamin D-fortified margarine had a lower risk of type 1 diabetes mellitus than children of those not exposed (RR 0.78, 95% CI 0.67 to 0.91; one trial, 261 956 participants; low-certainty evidence). The risk of childhood asthma did not differ significantly between the two groups (RR 0.96, 95% CI 0.90 to 1.03; one trial, 222 247 participants; low-certainty evidence).

The authors concluded that fortification of edible oils and fats with vitamin D results in little to no difference in serum 25(OH)D concentration. Several aspects of providing fortified oils and fats to the general population as a public health intervention should be further investigated, including the optimal fortification dose, effects on vitamin D deficiency and its clinical symptoms, and potential adverse effects.

The WHO guideline development group considered the following outcomes to be critical for decision-making: vitamin D deficiency (serum 25(OH)D (as defined by trialists)), vitamin D serum 25(OH)D concentrations, osteomalacia in older persons, nutritional rickets in children, adverse effects (hypervitaminosis, hypercalcaemia and kidney stones), all-cause morbidity and all-cause mortality. The following secondary outcomes were included: intake of dietary vitamin D; maternal and infant outcomes (preterm birth, low birth weight and small for gestational age, gestational diabetes, weight gain during pregnancy, pre-eclampsia, postpartum depression and postpartum haemorrhage); concentration of vitamin D in breast milk; growth, BMI and weight change (overweight and obesity); parathyroid hormone concentration (serum), alkaline phosphatase concentration (serum); longer-term outcomes (bone health, diabetes, hypertension, other cardiovascular diseases, respiratory infections, asthma and menopause onset (as defined by trialists)).

On application of the GRADE methodology, the certainty of the direct evidence for the only critical outcome for which evidence was available was graded as low. The GRADE summary of findings table for the fortification of edible oils and fats with vitamin D is provided in Annex 1.

Recommendations

The overarching principle underlying the following recommendations is that fortification of edible oils and fats may be considered when there is an indication of vitamin A or D deficiency in multiple age groups; edible oils and fats are regularly and sufficiently consumed by large population groups; and the industry is reasonably consolidated to permit reaching large segments of the population with fortification.

Oils and fats are energy-dense macronutrients and their selection as food vehicles for fortification requires a balanced consideration of both quantity (how much is consumed) and type (fatty acid composition) to protect consumers from the risk of diet-related noncommunicable diseases. This means limiting – as much as possible – the fortification of oils and fats that contain trans fat or high amounts of saturated fats. Countries should make efforts to ensure the availability and affordability of healthier oils, enabling populations to transition to healthier dietary patterns.

Fortification programmes also need to consider the sensory and physical effects of the added fortificants, the stability of the fortificants under different storage and cooking conditions, the type of oil or fat, oxidative stability of the oil or fat, consumption of other fortified foods, coexisting micronutrient interventions, and the costs, feasibility and acceptability of fortification.

The following recommendations are based on the available evidence, balance of potential benefits and harms, and consideration of equity, gender equality, human rights and cost.

- Fortification of edible oils and/or fats with vitamin A is recommended as a public health intervention to prevent subclinical vitamin A deficiency in populations at risk of vitamin A deficiency *(strong recommendation,*[7] *low and very low certainty of evidence).*
- Fortification of edible oils and/or fats with vitamin D may be considered as a public health intervention to prevent vitamin D deficiency *(conditional recommendation,*[8] *low certainty of evidence).*

Table 3 provides the amount of vitamins A and D to consider adding to oils and fats as part of a fortification intervention.

It is important that countries adapt the amount based on evidence about inadequacy of micronutrient intakes; per capita consumption of oils and fats; the prevalence of deficiency; coexisting interventions; policies for compliance with WHO guidelines on total fats, saturated fatty acids, *trans*-fatty acids and polyunsaturated fatty acids; and technological and cost considerations and other relevant contextual factors.

[7] A strong recommendation is one for which the guideline development group is confident that the desirable effects of adherence outweigh the undesirable effects. Implications of a strong recommendation are that most people in these settings would desire the recommended fortification of edible oils and fats. For policy-makers, a strong recommendation indicates that the recommendation can be adopted as policy in most situations.

[8] A conditional recommendation is one for which the guideline development group concludes that the desirable effects of adherence probably outweigh the undesirable effects, although the trade-offs are uncertain. Implications of a conditional recommendation for populations are that while many people would desire the recommended fortification of edible oils and fats, a considerable proportion would not. With regard to policy-makers, a conditional recommendation means that there is a need for substantial debate and involvement from stakeholders before considering the adoption of fortification of edible oils and fats with these vitamins in each setting.

Table 3. Amounts of vitamins A and D to consider adding to oils and fats, based on safety limits[9]

Nutrient [a]	Chemical form of the compound	Amount of nutrient to be added to oil or fat [b]	
		Minimum (mg/kg[c] oil or fat)	Maximum (mg/kg oil or fat)
Vitamin A	Retinyl palmitate Retinyl acetate	12	26
Vitamin D	Vitamin D3 (cholecalciferol) Vitamin D2 (ergocalciferol)	0.2	0.43

Note: This table is for general guidance. The amounts of nutrients should be adapted according to the needs of a country. The estimates assume only edible oils and fats are being fortified. The suggested range will need to be adjusted as per the dietary intake from all sources including if other large-scale food fortification programmes and/or micronutrient interventions related to vitamins A and D are being implemented at the same time.

[a] Antioxidants permitted by the Codex Alimentarius General Standard for Food Additives (CXS 192-1995) in food category 02.1.2 (vegetable oils and fats) or national regulatory authorities may also be added to the base oils and fats to prevent oxidative damage to the oils and fats and improve the stability of vitamins A and D.

[b] Countries are encouraged to adapt the suggested amounts of vitamins A and D through a modelling approach, guided by country-specific data on prevalence of inadequate intake of vitamin A and/or D in different population subgroups and consumption estimate of selected oil or fat in different population subgroups. Calculation of reduction in the prevalence of inadequate intakes (i.e. the proportion below the EAR) and the risk of excessive intakes (i.e. the proportion above the UL) for different levels of fortification is important to determine the appropriate fortification level using a risk-based approach. Detailed calculation methods and alternative approaches are available in Chapter 7 and Annex D, *Guidelines on food fortification with micronutrients (24).*

[c] Refer to the following unit conversion as applicable: vitamin A (retinol): 1 IU = 0.3 µg RE (retinol equivalent); vitamin D (cholecalciferol or ergocalciferol): 1 IU = 0.025 µg.

[9] See Annex D – A procedure for estimating feasible fortification levels for a mass fortification programme, *Guidelines on food fortification with micronutrients (23).*

Remarks

Any fortification guideline must be considered within the context that the goal of food-related policies, and most desirable health outcome, is for everyone to be able to follow a dietary pattern that meets dietary guidelines by consuming a range of foods that have a natural or inherent composition that provides desirable amounts of nutrients for health.

The guideline development group recognized tensions associated with the selection of oils and fats as vehicles for food fortification. It acknowledged the need to fortify foods that are widely consumed by populations at risk of micronutrient deficiencies. However, it also acknowledged that the rise of diet-related noncommunicable diseases associated with the consumption of foods high in salt, sugar and unhealthy fats raises concerns about whether fortifying oils and fats – particularly those made up of a high proportion of saturated or *trans*-fatty acids – overshadows or contradicts efforts to promote the consumption and affordability of healthier diets among the most vulnerable.

The following remarks are suggestions intended to provide some considerations for implementation of the recommendations, and are based on the discussions of the guideline development group.

- When vitamin A and/or D deficiency is a public health problem in a country, establishing a food fortification programme with edible oils and fats as food vehicles can be considered to address the problem. Dietary intake of edible oils and fats among different population groups and coexisting interventions such as supplementation should be taken into consideration. Competent authorities should determine whether fortification of edible oils and/or fats with vitamin A and/or D should be mandatory or voluntary, depending on the severity and extent of the public health problem.
- Oils and fats are energy-dense macronutrients. The implications of fortifying edible oils and fats with vitamins A and D for the noncommunicable disease burden must be evaluated with regard to each country's context. Governments may also wish to be satisfied that market promotion of fortified foods does not conflict with, or compromise, any national food and nutrition policies on healthy eating. The objective of fortification is to shift consumption from non-fortified oils or fats to fortified oils or fats, and not to increase total consumption of oils and fats.
- The choice of oils and fats might be influenced by availability (supply), consumer choice (demand), regulatory requirements, cost considerations and industry capacity to innovate. Decision-making needs to consider ongoing efforts to both replace saturated fatty acids with polyunsaturated and monounsaturated fatty acids from plant sources, and comply with WHO guidelines on total fats, saturated fatty acids, trans-fatty acids and polyunsaturated fatty acids. Adoption of multisectoral policies to support production of healthier oils might help to increase their use by industry and consumers.
- Feasibility of a fortification programme should be evaluated in each context prior to its initiation while considering the implications of whether the programme is envisaged as voluntary or mandatory. It is important to understand the following aspects of the selected edible oil or fat and nutrient(s) – coverage, cost, consumption, coexisting micronutrient interventions, industry capacity and consolidation, compatibility of nutrients, bioavailability, and the stability of the nutrients under prevailing conditions of storage, transportation, packaging and cooking. Dietary assessment and modelling, sensory studies, stability studies, consumer acceptability surveys, qualitative research and consumption monitoring can provide vital inputs for fortification programme design. Input from all involved sectors to the design and delivery of fortification programmes is pivotal to ensure a safe, practical and sustainable approach.
- The regulatory framework, including regulations, coordination and oversight mechanisms, compliance monitoring and enforcement measures, plays a key role in the success of any fortification programme. Fortification requirements should be integrated into the existing regulatory framework rather than as a stand-alone intervention.
  Regulatory authorities and relevant national bodies should review scientific evidence and data on nutrient gaps, consumption of selected edible oils or fats, acceptability, feasibility, appropriate fortificant forms and methods to establish realistic target levels/range for industry compliance. Global and regional guidelines, including general principles and guidelines by the Codex Alimentarius Commission, WHO guidelines and any relevant regional standards can provide vital inputs for formulation of fortification standards and regulations as well as operational aspects of monitoring and compliance.

- For effective implementation, regulatory monitoring is essential to ensure nutrient, quality and safety standards set out in regulations and standards are followed. Regulatory monitoring encompasses all monitoring activities conducted at the production level, and monitoring at customs, warehouses and retail stores, by the relevant regulatory authorities and by producers as part of self-regulation programmes.
  Adopting a risk-based approach enables regulatory authorities to prioritize and direct resources in addressing key problems. One such example is prioritizing the regulatory monitoring of bulk/unbranded oil helps to ensure traceability and check compliance against standards including fortification standards, as applicable.
- Nutrition labelling has a dual purpose of protecting the health of consumers and ensuring fair trade practices. When instituting fortification programmes, governments have a duty to ensure that consumers are not misled or deceived about the true nutritional properties of the fortified oil or fat. Irrespective of the nature of the fortification programme – that is, voluntary or mandatory – labelling requirements should apply to all fortified foods.
- The effect of fortification programmes on nutrient status and health outcomes should be monitored regularly through a monitoring and evaluation plan with appropriate indicators, including indicators that consider equity, gender equality and human rights. Monitoring of multiple micronutrient interventions in the same population groups is critical to prevent excessive intake of nutrients.
- Aspects of equity, gender and human rights should be integrated across the different stages of the fortification programme lifecycle, that is, programme design, monitoring and evaluation.

Summary of considerations to determine the direction and strength of the recommendations

In addition to considering the certainty of the evidence described in the reviews, the guideline development group considered the following factors in determining the strength and direction of the recommendations: the priority of the problem; the balance of benefits and harms; equity, gender equality and human rights; acceptability; values and preferences; feasibility; and cost.

The group was presented with an evidence-to-decision framework that considered these factors. This approach is consistent with the WHO guideline development procedures.

Further details of the group's considerations are provided in Annex 2.

Certainty of evidence

For the evidence related to vitamin A:

- low and very low certainty was the most common GRADE rating for primary (critical) outcomes;
- of the six primary (critical) outcomes for vitamin A, three were assessed as low certainty (serum retinol (RCT), subclinical vitamin A deficiency (RCT) and all-cause morbidity (RCT));
- of the six primary (critical) outcomes for vitamin A, three were assessed as very low certainty (serum retinol (CCT), subclinical vitamin A deficiency (CCT) and all-cause morbidity (CCT)); and
- of the six primary (critical) outcomes for vitamin A, three outcomes were not measured (vitamin A deficiency (xerophthalmia, night blindness), all-cause mortality and adverse effects).

For the evidence related to vitamin D:

- low certainty was the most common GRADE rating for critical outcomes;
- of the seven primary (critical) outcomes for vitamin D, one was assessed as low certainty (vitamin D serum 25(OH)D concentrations); and
- of the seven primary (critical) outcomes for vitamin D, six outcomes were not measured (vitamin D deficiency, osteomalacia in elderly people, nutritional rickets, adverse effects, all-cause morbidity and all-cause mortality).

Priority of the problem

The group concluded that vitamin A deficiency is an important public health problem and health care providers, policy-makers and communities in all settings are likely to place a high value on food fortification to combat vitamin A deficiency.

The group agreed that vitamin D deficiency has a negative impact on bone health and that there is increasing interest in vitamin D and its possible positive effects on health. However, the group acknowledged that there is uncertainty in estimating the global prevalence of vitamin D deficiency.

Balance of benefits and harms

Based on the available evidence, the group concluded that there are benefits of fortification of edible oils and fats with vitamins A and D.

The evidence for undesirable effects was uncertain, because studies were limited in number and quality. The group acknowledged that countries that fortify edible oils and fats may also fortify other food items with the same nutrient and that monitoring of multiple interventions in the same population groups is critical to prevent excessive intakes of nutrients.

The group concluded that the overall benefits of fortification of edible oils and fats outweighed the harms.

Equity, gender equality and human rights

The group discussed equity from the perspective of coverage, potential to reach the most vulnerable population groups with the fortification vehicle (oils and fats), potential to reach the most vulnerable population groups with the intervention (fortification), and affordability.

One of the major challenges in the fortification of oil is reaching the unorganized market where oil is sold in in bulk or unbranded packs. Proper regulations and a normative framework for fortification are necessary to ensure the sustainable reach of the fortification programme to the most vulnerable population groups.

There is often a difference in access to fortified oils and fats between urban and rural areas, between certain socioeconomic groups and between population groups residing in border areas and those in other areas, emphasizing the importance of regulatory monitoring.

Following discussion, the group concluded that fortification of edible oils and fats with vitamins A and D could probably increase health equity. The group recommended that equitable access to and consumption of fortified oils and fats must be evaluated with regard to each country's context. In general, effective nutrition interventions are more likely to decrease health inequities if they are accompanied by concurrent interventions that address the root cause of the problem. Strategies for fortification therefore need to be aligned with relevant programmes, especially poverty reduction and other social interventions.

The group acknowledged that gender inequality is a cross-cutting determinant of health that intersects with other determinants. There are multiple opportunities to ensure gender equality is considered across the fortification value chain. These include working with technical institutions to increase supply chain actors' awareness of the benefits of food fortification for women and children and their recognition of women as caregivers, consumers of fortified foods and as entrepreneurs or staff participating in fortified food production. They also include building men's awareness of nutrition topics, communicating with vulnerable families and marginalized groups, and engaging with women in programme design and implementation to contextualize interventions.

Following discussion, the group recommended that gender equality must be integrated into all stages of programme design and contextualized to the needs of different settings.

Nutrition is one of many underlying socioeconomic determinants of the right to the highest attainable standard of health. The group reflected on the different human rights instruments that relate to nutrition including Article 25 of the Universal Declaration of Human Rights *(62)*, Article 1 of the Universal Declaration on the Eradication of Hunger and Malnutrition *(63)*, Article 11 of the International Covenant on Economic, Social and Cultural Rights *(64)* , Article 12 of the Convention on the Elimination of All Forms of Discrimination Against Women *(65)* and Article 24 of the Convention on the Rights of the Child *(66)*.

Adequate consumption of essential nutrients through consumption of fortified foods has been linked to a reduction in the disease burden (measured as disability-adjusted life-years) resulting in people leading healthier and more productive lives and thereby contributing towards the fulfilment of human rights.

The group recommended that programmes for fortification of edible oils and fats should be designed using a human rights-based approach. Recognizing that access to a diverse diet may not be possible in all contexts at all times, food fortification can help to fill gaps in nutrient intake. A human rights-based approach to fortification could relate to:

- the physical availability of fortified foods
- economic access to fortified foods
- food habits and cultural norms
- links with gender equality and equity
- the regulatory environment for fortification
- the role of different stakeholders in the fortification value chain.

The group also acknowledged that proper regulations and a normative framework for fortification are necessary to ensure the sustainable reach of the fortification programme to the most vulnerable population groups.

Competent authorities should determine whether fortification of edible oils and/or fats with vitamins A and/or D should be mandatory or voluntary, depending on the severity and extent of the public health problem caused by deficiency. Input from all involved sectors, including oil processors and producers, industry and trade organizations, public institutions, academia, research organizations, and consumer associations and protection groups, when developing regulations and standards will help to ensure a realistic approach.

Acceptability

The group discussed acceptability through the perspective of different stakeholders in the fortification value chain, including consumers, industry and government.

Oils and fats are widely consumed. Based on data from 176 countries, edible oil availability ranges from 0 to 71.81 g per capita per day *(67)*. Margarine has a long history of use, particularly in the European countries where it was originally introduced as a replacement for butter.

The group noted that as of 2024, 35 countries have legislation for mandatory fortification of edible oils and eight countries have legislation for voluntary fortification. Of these 43 countries, 39 countries have standards that include vitamin A, 13 countries have standards that include vitamin D and two countries have standards that include vitamin E. The amount of vitamin A permitted to be added in the 39 countries with mandatory and voluntary standards for vitamin A ranges from 3.5 to 41.25 mg/kg oil. For the 13 countries with mandatory and voluntary standards for vitamin D, the amount permitted to be added ranges from 0.01 to 0.75 mg/kg oil *(68)*.

The group noted that, for fats such as margarine, fortification was widely practised and accepted by consumers, especially in European countries *(69)*. In some countries such as Belgium, Poland and Sweden, fortification of margarine, fat spreads and related products is mandatory. The Netherlands has a national covenant to fortify margarine and other spreadable fats with vitamin D. Most countries allow voluntary fortification, including Finland, Germany and Norway.

The group also discussed changes to the organoleptic properties (such as the appearance, aroma, taste and texture) of oils and fats, the retention of vitamins during cooking and acceptability by industry as factors that could affect the overall acceptability of fortification. The group considered that fortification of oils and fats is a well-accepted intervention, although it recognized that some groups oppose adding any substance to foods.

Values and preferences regarding health outcomes

The group discussed values and preferences regarding the health outcomes related to vitamin A and D, and centred its discussion on end users, that is, those who buy fortified oils and fats. All guideline development group members considered that micronutrient deficiencies are important public health problems and that health care providers, policy-makers and family members in all settings are likely to place a high value on food fortification to combat micronutrient deficiencies. The group concluded there is no important uncertainty or variability in how much the general population value the main health outcomes related to vitamins A and D deficiencies.

Values and preferences regarding fortification as an intervention

The group also discussed values and preferences regarding fortification of edible fats and oils as an intervention. The group noted that fortification requires less behaviour change than other interventions, meaning it is likely to be acceptable and feasible. However, the group recognized that some groups oppose adding any substance to foods.

The group mentioned that there might be reservations from industry for various reasons including:

- insufficient business incentives
- limited awareness, internal capacity and technical knowledge
- varying costs of premix and equipment
- varying demands of consumers.

The group concluded there is possibly important uncertainty or variability in how much the general population value fortification of edible oils and fats as a way to receive vitamins A and D in their diets.

Feasibility

The group concluded that fortification of edible oils and fats with vitamins A and D can be considered a feasible intervention as it is currently implemented in multiple countries. However, implementation by centralized industry producers and regular monitoring by governments are critical, preferably alongside other agricultural, health, education and social intervention programmes that promote the consumption of adequate quantities of good quality nutritious foods among the nutritionally vulnerable.

Feasibility of a fortification programme should be evaluated in each context prior to its initiation while considering the implications of whether the programme is envisaged as voluntary or mandatory. It is important to understand the following aspects of the selected edible oil or fat and nutrient(s) – coverage, cost, consumption, coexisting micronutrient interventions, industry capacity and consolidation, compatibility of nutrients, bioavailability, and the stability of the nutrients under prevailing conditions of storage, transportation, packaging and cooking. Dietary assessment and modelling, sensory studies, stability studies, consumer acceptability surveys, qualitative research and consumption monitoring data can provide vital inputs for fortification programme design. These inputs help to ensure meaningful quantities of nutrients reach those at the lower end of the consumption spectrum while not exceeding unacceptable levels for those at the higher end.

Cost and cost effectiveness

The cost increase caused by fortification of oils with vitamin A amounts to only 0.1–0.3% of the retail price *(37)*. Since January 2000, the cost and cost-efficiency of food fortification programmes has been estimated for eight food vehicles across 19 LMICs. Most of this evidence has been generated for fortified refined oil and wheat flour. The median per capita fortification programme cost (2021 US dollars) for refined oil is estimated at US$ 0.03. The median cost per DALY averted (2021 US dollars) is estimated to be US$ 23 for refined oil.

The group discussed examples of costs and cost–effectiveness evaluations.

- In Bangladesh, fortifying vegetable oil with vitamin A would cost US$ 1.27 million annually, and save 406 877 disability-adjusted life-years at an average cost of US$ 3.25 per disability-adjusted life-year *(70)*.

- In the United Republic of Tanzania, the incremental cost of fortifying sunflower oil with vitamin A could be as low as US$ 0.13 per litre for small-scale producers, US$ 0.06 per litre for medium-scale producers and US$ 0.02 per litre for large-scale producers. The estimated cost per disability-adjusted life-year averted for fortified sunflower oil was US$ 281 for large-scale producers and could be as low as US$ 626 for medium-scale producers and US$ 1507 for small‑scale producers under ideal conditions *(71)*.
- In Ethiopia, a programme to fortify imported oils (approximately 98% of all oils consumed) with vitamin A, including start-up investments and recurring monitoring and evaluation and border inspection activities, would cost approximately US$ 3.6 million over 10 years. Over a 10-year period, the cost per life saved among non-breastfed children would range from US$ 706 to US$ 973 *(72)*.

Based on the available information, fortification of edible oils and fats with vitamin A, supported by appropriate legislation and enforcement of regulations, can be a cost-effective strategy to address vitamin A deficiency.

Research needs

Diet evolves over time, being influenced by many social and economic factors that interact in a complex manner. Additionally, other factors, such as advances in research and new guidance and regulations at a country or global level, may also need to be considered. Programme design should consider a periodic review of available data related to micronutrient needs, consumption patterns of fortified foods, other sources of the micronutrient in the diet, contextual factors and the validity of relevant design assumptions.

The following research needs were identified during the guideline's development.

- There is limited information regarding actual consumption of oils and fats, which limits understanding of the dietary gap to be addressed through fortification, the use of fortified oils or fats when a programme is implemented, and the subsequent impact on nutrient adequacy.
- Further studies are needed to ensure more effective use of these recommendations, including those that:
    - study efficacy and effectiveness, especially for vitamin D fortification and co-fortification with both vitamins A and D;
    - assess the prevalence of vitamin deficiencies in populations using different biochemical indicators to identify populations that are most at risk of deficiency or excessive intake, or in need of intervention;
    - contribute towards international consensus on assessment of vitamin D deficiency in population;
    - further determine the fortificant concentration that can lead to adequate vitamin A and D intake from edible oils and fats in different population subgroups;
    - assess the coverage of fortified oils and fats across different population subgroups; and
    - assess the contribution of fortification to nutrient adequacy, and its potential health impact and adverse effects.
- The stability of different vitamin A and D compounds needs to be established for their use in oils and fats, including in relation to packaging, storage, distribution, consumption and cooking methods in different settings. This includes assessing the stability of vitamins:
    - with different packaging materials and exposure to light
    - with different cooking methods
    - in relation to storage time and oxidative status of the oil or fat
- National studies are encouraged to demonstrate that large-scale fortification reaches those that may be at higher risk of micronutrient deficiencies such as women, children, certain socioeconomic groups, people with low dietary diversity or living in households that are food insecure, and those living in specific geographical regions.
- Data collection, including coverage studies, should capture the needs of various groups through an intersectional lens. Further research is needed to integrate aspects of equity, gender equality and human rights into programme design, monitoring and evaluation.

Dissemination, implementation and ethical considerations

Dissemination

This guideline will be disseminated through Department of Nutrition and Food Safety mailing lists, the Department of Nutrition and Food Safety website *(73)*, conferences and panels.

In addition, the guideline will be disseminated through a broad network of international partners, including WHO country and regional offices, ministries of health, WHO collaborating centres, universities, other United Nations agencies and nongovernmental organizations. Derivative products that are useful for end users, such as summaries and a collation of recommendations related to food fortification, may be developed.

Particular attention will be given to improving access to these guidelines for stakeholders that face high, or specific, barriers to accessing information, or to those that play a crucial role in the implementation of the guideline's recommendations, for example, policy-makers and decision-makers at the subnational level that disseminate the contents of the guideline. Disseminated information may emphasize the benefits of food fortification programmes in populations or regions where micronutrient deficiencies and their consequences are of public health significance. This is particularly important in rural communities or highly isolated settings where access to fortified foods is often limited or difficult.

The guideline's executive summary may be translated into the other five United Nations languages and disseminated through the WHO regional offices. Specialized technical assistance will be provided to any WHO regional office willing to translate the full guideline into any of these languages and to support countries in implementing it.

Human rights, gender equality and equity considerations

A situational analysis is a starting point to understand underlying biases and structures that affect fortification programme design. To illustrate, the most severe effects of vitamin A deficiency are seen in young children and pregnant women in low-income countries. Some population groups tend to be more susceptible to vitamin D deficiency, including adolescents, adults, pregnant women, and people living in regions with insufficient sunshine or with sun avoidance behaviours that prevent or impede dermal synthesis. Dietary intake data and disaggregated data on the prevalence of micronutrient inadequacies and deficiencies across population groups highlight the differential impact of policies and programmes on vulnerable population groups.

A useful strategy to disaggregate data are included in the acronym PROGRESS-Plus: place of residence; race, ethnicity, culture and language; occupation; gender and sex; religion; socioeconomic status; social capital; and other relevant social determinants (e.g. age, disability status, migration status, health-system configuration, political environment) *(74)*. Disaggregated data are also useful for monitoring and evaluating the programme. WHO has developed guidance on health equity to support Member States, which can be found in the *Handbook on health inequality monitoring with a special focus on low- and middle-income countries (75)* and the Health Equity Assessment Toolkit (HEAT) *(76)*. These resources aim to assist Member States to assess within-country health inequalities and can inform Member States adaptation of this guideline.

Although many factors affect the success of fortification programmes, high population coverage is needed to have a public health impact. A recent review aimed to provide recent global coverage estimates of salt, wheat flour, vegetable oil, maize flour, rice and sugar among countries with mandatory fortification legislation. The review found that 87.0% of people consumed vegetable oil (based on data from 10 countries), 86.7% of people consumed fortifiable (that is, industrially processed) oil (based on data from seven countries), and 40.1% of people consumed oil that was fortified to any extent (based on data from five countries). Data on adequately fortified vegetable oil (according to national or international standards) were limited *(77)*.

Equitable access to and consumption of fortified oils and fats must be evaluated for each context from the perspective of coverage, potential to reach the most vulnerable population groups with the fortification vehicle (oils and fats), potential to reach the most vulnerable population groups with the intervention (fortification), and affordability. In general, effective nutrition interventions are more likely to decrease health inequities if they are accompanied by concurrent interventions that address the root cause of the problem. Strategies for fortification therefore need to be aligned with relevant programmes, especially poverty reduction and other social interventions. One of the major challenges faced in the fortification of oil is reaching the unorganized market where oil is sold in bulk or unbranded packs. Proper regulations and a normative framework for fortification are necessary to ensure the sustainable reach of the fortification programme to the most vulnerable population groups.

It is also recommended the pathways and distribution channels required to reach and benefit hard-to-reach population groups should be carefully identified. Policy-makers and programme managers may consider appropriate measures to guarantee that the intervention is implemented as it was designed.

Gender refers to socially constructed norms that impose and determine power, roles and relationships between women, men, boys, girls and trans and gender-diverse people, including people with non-binary gender identities *(78)*. Gender norms, roles and relations vary from society to society and evolve over time. Gender inequality disproportionately affects women and girls, posing barriers to access health information and critical services, including restrictions on mobility, lack of decision-making autonomy, limited access to finances, lower literacy rates and discriminatory attitudes of health care providers.

The increasing disparity in health and nutrition outcomes of women, girls and other vulnerable groups highlights the need for a nuanced gender-based inclusive approach towards programme design and implementation. Gender mainstreaming is an internationally accepted strategy that aims to institutionalize gender equality across sectors. Gender mainstreaming is the process of assessing the implications for women, men and gender-diverse people of any planned action within a health system, including legislation, policies, programmes or service delivery, in all technical areas and at all levels.

WHO has developed guidance on gender mainstreaming to raise awareness and develop skills on gender analysis and gender-responsive planning in health sector activities. The first edition of the manual dates from 2011, and WHO is now updating it in light of new scientific evidence and conceptual progress on gender, health and development *(76, 79)*.

WHO is developing a guideline on the health of trans and gender diverse people. The new guideline will provide evidence and implementation guidance on health sector interventions aimed at increasing access and utilization of quality and respectful health services by trans and gender diverse people *(80)*.

Based on the information from the gender analysis, strategies can be developed to plan for gender-based inequalities in policies, programmes and services. Gender analysis for a fortification intervention would typically analyse:

- equal access to fortified foods for women and girls compared with men and boys;
- equal access to information and awareness about the benefits of fortified foods;
- differences in food purchasing power;
- gender social norms around dietary restrictions and food consumption;
- role of women in household decision-making and their influence on adopting fortified foods; and
- gender dimensions of fortification supply and distribution, such as women as workers, distributors, smallholders and consumers.

Policy-makers and programme managers may consider gender norms and roles to ensure that fortification programmes are designed to reduce gender disparities.

Regulatory considerations

A regulatory framework is key for the success of food fortification programmes. A regulatory framework includes the framework of regulations, coordination and oversight mechanisms, compliance monitoring and enforcement measures.

Fortification regulations and standards typically include:

- recommended nutrients, fortification target levels, minimum and/or maximum levels and chemical forms of nutrients to be used for fortification;
- labelling, claims and advertising; and
- regulatory monitoring, sampling procedures and enforcement measures to assure compliance.

Other specifications might include limits on physical and microbiological contaminants *(81)*. Several Codex Alimentarius standards provide general guidance that may be helpful to regulators, such as on the addition of essential nutrients; labelling and claims; and composition, quality and food safety factors in edible oils and fats *(82-85)*.

Competent authorities need to determine whether the standards for fortification of edible oils and fats should be mandatory or voluntary. This decision may be based on the severity and extent of public health need. Input from all involved sectors, including producers, public institutions, academia, research organizations and consumer protection groups, when developing the regulations and standards will help to ensure a safe and sustainable approach.

The choices of oils and fats used in a country might be influenced by availability (supply), consumer choice (demand), regulatory requirements, cost considerations and industry capacity to innovate. Country-specific regulations on standards of identity (definitions and descriptions) and permitted additives (including nutrients) should be referred to. Oils and fats are highly susceptible to food fraud, including economically motivated adulteration. Appropriate measures and controls such as supply chain traceability systems and vulnerability assessment *(86)* can be incorporated as a preventive approach against fraud.

The choice of vitamin A and D compounds used in fortification is a balance between cost; bioavailability; stability; micronutrient interactions; acceptability of texture, taste, smell and colour; and evolving regulatory requirements and consumer needs and preferences, including those related to the origin (source) of vitamins.

The *Code of practice for food premix operations* was created by the Pan American Health Organization as a first step to assure premix quality for fortification programmes. It considers quality not only in terms of adequate types and levels of nutrients added, but also in relation to hygiene, food safety and good manufacturing practices, thereby assuring that premix meets requirements for human consumption *(87)*. Purity criteria for premix compounds will also need to be stipulated, and reference made to texts such as the *Food Chemicals Codex (88)*.

A well-designed regulatory monitoring system is essential to implementing a fortification programme to ensure nutrient, quality and safety standards as set out in regulations and standards are followed. WHO and FAO have developed guidelines on fortification that describe the key functions of regulatory monitoring and identify criteria for evaluating monitoring systems. The criteria consider the role of national authorities in establishing procedures, methodologies and reporting requirements to evaluate the fortification programme; allocation of responsibilities between different actors; and monitoring mechanisms *(23)*. Regulatory monitoring encompasses all monitoring activities conducted at the production level, and monitoring at customs, warehouses and retail stores, by the relevant regulatory authorities and by producers as part of self-regulation programmes. Production-level regulatory monitoring comprises both internal and external monitoring. Internal monitoring refers to the quality control and quality assurance practices conducted by producers, importers and packers. External monitoring refers to the inspection and auditing activities carried out by governmental authorities *(23)*. Adopting a risk-based approach enables regulatory authorities to prioritize and direct resources in addressing key problems. One such example is prioritizing the regulatory monitoring of oil sold in bulk containers or unbranded packs to ensure traceability and check compliance against standards including fortification standards, as applicable.

Governments should also consider trade regulations. Mandatory fortification may impose trade restrictions on imported products based on health criteria, because either they are unfortified or they have been fortified differently. On the other hand, nations with similar needs may benefit from a common agreement on fortification policies and regulations that could be regionally adopted *(23, 89).*

Depending on the stage of planning and implementation of a food fortification programme, regulatory considerations may include the following.

- Initial phase:

  - Establish a coordination and communication mechanism between the different bodies responsible for implementation of food fortification.
  - Evaluate safety, technological, regulatory and economic constraints when defining thresholds for nutrient addition.
  - Consider alignment with existing global and regional benchmarks while setting standards and regulations.
  - Integrated approach: Include fortified food and premix production, importation and exportation within existing licensing systems to reduce the need for multiple concurrent systems.
  - Include supervision of fortified foods and their producers within the existing food safety regulatory framework, controls and monitoring protocols.
  - Adopt a risk-based approach for all food-related inspections and enforcement activities to optimize the use of limited resources and direct the resources to high-risk businesses.
  - Plan the following: which food samples to test, which micronutrients to test, where to collect the food samples (point of sampling), sampling protocols, handling and transportation of samples, and analytical methods per nutrient per food vehicle.
  - Establish sufficient testing capacity for effective monitoring compliance with food fortification regulations.
  - Consider use of rapid test kits and portable devices for self-checks and initial screening to detect the presence of micronutrients.
  - Define the conditions under which rapid test kits or portable devices can be used and how they should be interpreted for regulatory purposes.
  - Ensure appropriate labelling regulations are applicable irrespective of voluntary or mandatory food fortification.
  - Develop comprehensive tools, checklists, and guidelines to aid in food fortification inspections.
  - Provide adequate training to laboratory staff and inspectors collecting samples, to ensure their proficiency, minimize errors and improve reliability of results.

- Implementation and sustenance phase:

  - Establish online systems to provide easy access to regulations and guidelines, increase transparency and accessibility of information and streamline regulatory processes.
  - Promote use of risk ranking approaches to target resources for risk management.
  - Develop a toolbox of differentiated enforcement measures against non-compliance with an objective to raise awareness and encourage self-regulation.
  - Use information integration and sharing to enhance industry compliance, decision-making, preventive and corrective actions from inspections.
  - Develop tools to incentivize competition and promote industry compliance (e.g. food safety index, fortification index or score cards).
  - Expand the network of accredited laboratories to facilitate timely testing and reporting.
  - Invest in public–private partnerships, if possible, to strengthen surveillance systems and food testing infrastructure to assure adequately fortified and safe foods for consumers.
  - Partner with international or national bodies to update food sampling and testing methods and laboratory manuals in line with the latest analytical advancements.
  - Update the training and resources to cover relevant issues, including the latest scientific evidence, standards, other regulatory requirements, and emerging practices in the field.
  - Level the playing field for small businesses by providing them with clear, practical advice that is easy to understand and implement.
  - Build a culture of continuous improvement – periodically provide technical capacity training and resources for government agencies responsible for monitoring and enforcing food fortification regulations.

- Continuously review the regulatory performance indicators to assess compliance, impact and identify areas for improvement.
- Promote active collaborations with centres of excellence or reference centres on fortification, nutrition, food safety, noncommunicable diseases and other allied sectors.
- Promote a data culture to facilitate the interpretation and use of data to influence and inform decision-making.
- Adopt foresight approach to address issues that will, or could, emerge in the medium-to-long term due to climate change, changing food consumption patterns, technological advancements, and food safety considerations, to name a few.
- Ensure that laboratory systems are internationally accredited, effectively networked, and financially sustainable.

Implementation considerations

This guideline provides Member States with evidence-informed recommendations on the effects and safety of fortifying edible oils and fats with vitamins A and D as a strategy to improve the health status of populations. This guideline is intended to help Member States and their partners make informed decisions about what interventions are best suited to their context, needs, resources and ongoing programmes, observing existing human rights standards and pursuing health equity. As of 2024, 35 countries have legislation for mandatory fortification of edible oils and eight countries have legislation for voluntary fortification. Of these 43 countries, 39 countries have standards that include vitamin A, 13 countries have standards that include vitamin D and two countries have standards that include vitamin E.

If Member States decide to adopt the recommendations in this guideline at the national or subnational level, a thorough assessment of the policy implications is needed. The following illustrative considerations seek to support Member States that are considering fortification of edible oils and fats with vitamins A and/or D.

The adoption and adaptation of this recommendation should be framed under a country's existing national strategy on the prevention and control of micronutrient deficiencies. The choice of intervention to prevent micronutrient deficiencies should be considered in the context of the strategy, and include consideration of the costs, feasibility, accessibility and acceptability among different stakeholders (e.g. decision-makers, lawmakers, programme managers, farmers, manufacturers, industry organizations, importers, exporters, retailers, consumer organizations, organizations with opposing views). Mapping the different stakeholders and their interests and involvement in the intervention is a useful practice *(90).*

Food fortification programmes should not be considered as a replacement for adequate and diverse diets. Food fortification efforts should hence coincide with initiatives to improve diets, especially in population groups with less diverse diets, and with other dietary counselling programmes in place. To coordinate this, policy-makers need to determine which multisectoral approaches represent the most appropriate allocation of resources, produce greater benefits and align with the programme objectives.

The programme should have well-defined objectives that consider available resources, existing policies, suitable delivery platforms and suppliers, communication channels and potential stakeholders. Ideally, a programme for fortification of edible oils and fats should be implemented as part of a coordinated and comprehensive programme aiming to address micronutrient deficiencies. A comprehensive fortification programme may include several fortification vehicles. The selection of the foods to be fortified and the levels of nutrients to be added to the foods must be coordinated.

The decision to implement the recommendations in this guideline must be made in the context of achieving or maintaining nutritional adequacy and avoiding excess energy intake. Before implementation, analysis is recommended of capacity for edible oils and fats production, importation and industrialization; the nutrients needed to be added to (vitamin A or D or both); and consumption of edible oils and fats as culinary ingredients and ingredients in food products.

A detailed discussion of how the recommendations might be implemented is beyond the scope of this guideline. However, considerations for policy-makers and programme managers when discussing possible measures include:

- assessing the prevalence of micronutrient deficiencies in population groups to decide on the fortification vehicle, premix formulation, cost estimates and other implementation requirements;
- assessing current intake (quantity and quality) of total oils and fats in the population relative to the recommendations in *Total fat intake for the prevention of unhealthy weight gain in adults and children: WHO guideline and Saturated fatty acid and trans-fatty acid intake for adults and children: WHO guideline*;
- establishing the following parameters when designing a programme:
    - the use gap, which is the proportion of households in a population not consuming edible oils and/or fats; and
    - the feasibility gap, which is the difference between the proportion of households consuming edible oils and/or fats in any form and those consuming them in a fortifiable form;
- assessing the effectiveness of fortifying edible oils and fats compared with other fortification vehicles and micronutrient interventions;
- assessing nutrient intakes and potential adverse effects due to overlapping interventions in the same population groups;
- setting the appropriate fortification levels considering:
    - country-specific information on the prevalence of inadequate intake of vitamin A and/or D in different population subgroups and consumption estimates of selected oil or fat in different population subgroups;
    - potential reduction in the prevalence of inadequate intakes (i.e. the proportion below the EAR) for different levels of fortification;
    - potential risk of excessive intakes (i.e. the proportion above the UL) for different levels of fortification;
    - stability of the nutrients (both in the preblend and in the fortified edible oil or fat) under prevailing conditions of packaging, storage, distribution, transportation, and commonly used cooking methods;
    - safety, technological and cost constraints; and
    - relevant global and regional guidelines;
- identifying fit-for-purpose distribution channels, that is, channels to reach and benefit the most hard-to-reach population groups;
- identifying barriers to access and use of fortified edible oils and fats in a population, such as:
    - not all oil being industrially processed or sold in packaged format (some is sold loose or in bulk);
    - unpackaged oils tending to have a higher percentage of rancidity compared with packaged oils, which besides posing health concerns, poses challenges for the stability of added vitamins and for traceability and compliance for regulatory purposes – some countries offer suitable case studies, such as Bangladesh *(91)*; and
    - households in some low-income and rural settings not having access to packaged edible oils and fats – in these cases, concurrent measures (including awareness-raising campaigns, food subsidies, cash transfer programmes, direct distribution of culturally appropriate fortified foods, and point-of-use fortification or supplementation) can help to prevent and mitigate health inequities;
- integrating aspects of equity, gender and human rights across the different stages of the fortification programme lifecycle, that is, programme design, monitoring and evaluation;
- developing policy measures to reduce intake of total fat, where necessary, through a range of public health interventions, many of which are already being implemented by countries, including:

- nutrition labelling (i.e. mandatory nutrient declaration) and front-of-pack labelling systems;
- consumer education;
- product reformulation;
- regulation of marketing;
- fiscal measures; and
- adoption of culturally and contextually specific food-based dietary guidelines that consider locally available food and dietary customs;

- monitoring the effect of fortification programmes on nutrient status and health outcomes through an adequately designed monitoring and evaluation plan with appropriate indicators, including indicators that consider equity, gender equality and human rights;
- strengthening regulatory monitoring to ensure the quality and food safety of fortified foods through;
    - effective internal and external quality assurance;
    - a risk-based approach for all food-related inspections and enforcement activities;
    - proportionate and gradual enforcement measures in response to non-compliance;
    - use of digital/technology-enabled tools to facilitate information sharing, traceability, analysis, reporting and timely decision-marking by stakeholders;
    - a network of accredited and approved laboratories to conduct routine micronutrient testing;
    - clear and precise guidance on sampling protocols and analytical methods for different nutrients and fortified foods;
    - use of rapid test methods for self-checks and initial screening to detect the presence of micronutrients as per the context;
    - a toolkit of comprehensive checklists, manuals and guidelines to aid in food fortification testing and inspections;
    - periodic technical capacity training for regulatory officials responsible for monitoring and enforcing food fortification regulations; and
    - promoting a food safety culture;
- periodically assessing the following parameters during the programme life cycle:
    - demographic, technological and urbanization shifts;
    - evolving food choices and dietary patterns;
    - patterns of consumption of fortifiable edible oils and/or fats, nutrient intake and nutrient needs;
    - the fortification gap, which is the difference between the proportion of households consuming fortifiable edible oils and/or fats and those consuming fortified edible oils and/or fats; and
    - the quality gap, which is the difference between the proportion of households consuming edible oils and/or fats that are fortified to any extent and those consuming oils and/or fats fortified in accordance with the relevant national fortification standards *(92)*.

Monitoring and evaluation of guideline implementation

A monitoring and evaluation plan appropriate indicators, including those that consider equity, gender equality and human rights, is encouraged at all stages of a fortification programme *(75).* The impact of this guideline can be evaluated within countries (i.e. monitoring and evaluation of the programmes implemented at national or regional scale) and across countries (i.e. the adoption and adaptation of the guideline globally).

The Department of Nutrition and Food Safety has developed a logic model for fortification of maize flour and corn meal with vitamins and minerals for public health. The logic model depicts the plausible relationships between inputs and expected contributions to the Sustainable Development Goals, especially goals 2 and 3, by applying the micronutrient programme evaluation theory (Annex 3). It is adapted from the WHO–Centers for Disease Control and Prevention logic model for micronutrient interventions and could be adapted and applied to fortification of edible oils and fats *(93).*

Member States can adjust the model and use it with appropriate indicators to design, implement, monitor and evaluate the escalation of nutrition actions in public health programmes. The WHO–Centers for Disease Control and Prevention eCatalogue of indicators for micronutrient programmes *(94)* uses this logic model and can be customized for programmes of fortification of edible oils and fats with vitamins A and D as a public health strategy. The eCatalogue is a user-friendly, non-comprehensive web resource for those providing technical assistance in monitoring, evaluation and surveillance of micronutrient-related public health programmes. It provides potential indicators with standard definitions that can be selected, downloaded and adapted to a local programme context.

Since 1991, WHO has hosted the Micronutrients Database as part of the Vitamin and Mineral Nutrition Information System *(95).* Part of WHO's mandate is to assess the micronutrient status of populations, monitor and evaluate the impact of strategies for the prevention and control of micronutrient malnutrition, and track related trends over time. The Department of Nutrition and Food Safety manages the Micronutrients Database through a network of regional and country offices, and in close collaboration with national health authorities.

For evaluation at the global level, the Department of Nutrition and Food Safety has developed the web-based Global Nutrition Targets Tracking Tool *(96)*, which allows users to explore different scenarios to achieve the rates of progress required to meet the 2025 global nutrition targets. The Department of Nutrition and Food Safety has also developed a centralized platform for sharing information on nutrition actions in public health practice implemented around the world, called the Global database on the Implementation of Food and Nutrition Actions *(97).* The database provides a repository of policies, programmes and actions, mechanisms and country commitments related to improving nutrition. It is an interactive platform that presents standardized information on nutrition policies and action, enabling greater access to data, which can be used to assess and monitor policies, learn lessons to improve policies, support healthy diets and reduce the burden of diet-related diseases. By sharing programmatic details, specific country adaptations and lessons learned, the database will provide examples of how guidelines are being translated into actions.

Guideline development process

This guideline was developed in accordance with the WHO evidence-informed guideline development procedures, as outlined in the *WHO handbook for guideline development (13).*

The development of this guideline started in February 2010. The project included the development of guidelines on fortification of five staple foods (wheat, maize, rice, condiments, sugar and oil) with a group of micronutrients involved in maintaining health and development and for which deficiencies have severe consequences at the individual and the public health level (iron, folate, vitamin A, zinc and iodine). This fortification guideline is the fifth guideline of the project. Earlier in the project, the Department of Nutrition for Health and Development (now the Department of Nutrition and Food Safety) published guidelines on fortification of salt, maize, rice and wheat flour.

Initially, the guideline questions related to edible oils were defined and scoped for vitamin A only. Since the questions were discussed in 2010, vitamin D deficiency has emerged as an important public health topic and several countries have fortified edible oils and fats with vitamin D, which merited the review of its efficacy, safety and cost–effectiveness. Because of this, a new question was added to assess the benefits and harms of fortification with vitamin D, alone or with vitamin A and a new guideline development process was started de novo.

A WHO steering committee (Annex 4), led by the Department of Nutrition and Food Safety, was established with representatives from relevant WHO departments with an interest in the provision of scientific nutrition advice. The steering committee guided and provided overall supervision to the guideline development process. Two additional groups were formed: a guideline development group and a systematic review team.

The WHO guideline development group on food fortification with micronutrients was established for the biennium 2022–2024 (Annex 5). Its role was to advise WHO on the choice of critical outcomes for decision-making within the scope of this guideline. It included experts from the WHO guideline development group on nutrition actions (2013–2014) who were available to continue as guideline development group members *(98, 99)* and those identified through open calls for specialists *(100)*, taking into consideration a balanced gender mix, multiple disciplinary areas of expertise and representation from all WHO regions. Efforts were made to include content experts, methodologists, representatives of potential stakeholders (such as managers and other health professionals involved in the health care process) and technical staff from WHO and ministries of health from Member States. Representatives of commercial organizations cannot be members of a WHO guideline development group.

The systematic review team (Annex 6) participated only in open meetings and were not allowed to participate in the decision-making process.

The final draft guideline was peer reviewed by six content experts (Annex 7), who provided technical feedback. The peer-reviewers were identified through various expert panels within and outside WHO. Peer-reviewers are not involved in the guideline development process *(13)* and are only asked to provide comments on the final draft guideline. Their role is to identify any errors or missing data and to comment on clarity, setting-specific issues and implications for implementation; they may not change the recommendations formulated by the guideline development group. Reviews from such individuals or organizations on a draft guideline may be helpful in anticipating and dealing with controversy, improving the clarity of the final document and promoting engagement with all stakeholders.

This document is a WHO guideline and, after executive clearance, represents the decisions, policy or views of WHO.

Scope of the guideline, evidence appraisal and decision-making

An initial set of questions (and the components of the questions) to be addressed in the guideline was the starting point for formulating the recommendations. The questions were drafted by technical staff at the Department of Nutrition and Food Safety, based on the policy and programme guidance needs of Member States and their partners. The PICO format was used. The questions were discussed and reviewed by the WHO steering committee and the guideline development group and were modified as needed. The guideline development group discussed the relative importance of outcomes as critical or important. The final key questions, along with the outcomes that were identified as critical for decision-making, are provided in PICO format in Annex 8.

To inform this guideline, and as detailed in *Summary of the evidence*, WHO commissioned a systematic review of the evidence to determine the benefits and harms of fortification of edible oils and fats with vitamin A and D for health-related outcomes in the general population. The Cochrane methodology for systematic reviews of interventions *(101)* was followed. No language restrictions were applied in the search. Evidence summaries were prepared according to the GRADE approach to assess the overall certainty of the evidence *(102, 103)*. GRADE considers the study design; the limitations of the studies in terms of their conduct and analysis; the consistency of the results across the available studies; the directness (or applicability and external validity) of the evidence with respect to the populations, interventions and settings where the proposed intervention may be used; and the precision of the summary estimate of the effect.

WHO also commissioned a desk review to:

- review the enabling environment from the perspective of trade, technology, feasibility, economic impact and legislation;
- systematically analyse information to define needs and priorities (looking for additional data on actual practice), potential risks, emerging issues and opportunities;
- assess barriers and facilitators that can hinder or enhance the uptake and/or adaptation of implementation knowledge products;
- provide input to the guideline development process and document technical considerations for the fortification of edible oils and fat with vitamins A and D; and
- inform field testing and rollout strategies to promote the uptake of recommendations and adaptation into local contexts.

The desk review also aimed to provide evidence to inform the domains of the GRADE evidence-to-decision framework related to implementation considerations, namely cost, acceptability, feasibility, and equity, gender equality and human rights.

As part of the desk review, key stakeholders were interviewed about implementation considerations. The objectives of the interview were explained to the stakeholders, who were also informed that no attributions would be made to individuals, and they were asked to provide oral consent before they were interviewed.

To identify additional evidence not captured by the systematic review or desk review, more than 10 international organizations working on micronutrient interventions were contacted to identify pre- and post-surveys or studies still in progress.

Both the systematic review and the GRADE summary of findings tables for each of the critical outcomes (Annex 1) were used for drafting this guideline.

The findings of systematic review, desk review and pre- and post-surveys and the draft recommendations were discussed by the WHO steering committee and the guideline development group at a second guideline development group meeting, held in Geneva on 20 and 22 June 2023. During the meeting, the guideline development group members determined the strength of the recommendation, taking into account the desirable and undesirable effects of the intervention; the certainty of the available evidence; values and preferences related to the intervention in different settings; and the cost and feasibility of the intervention in different settings. These aspects were discussed openly during the meeting. The WHO Secretariat (Annex 9) gathered and disclosed a summary of the consensus decision-making results to the guideline development group. If there was no unanimous consensus (primary decision rule), more time was given for deliberations and a second round of voting took place. If unanimous agreement was still not reached, two thirds of the votes of the guideline development group were required for approval of the proposed recommendation (secondary decision rule). Divergent opinions could be recorded in the guideline. The results from voting are kept on file by WHO for up to five years. WHO staff present at the meeting, and the systematic review team involved in collecting and grading the evidence, did not participate in the consensus-building process. A chair who had expertise in managing group processes and interpreting evidence was nominated at the opening of the first meeting, and the guideline development group approved the nomination unanimously. Members of the WHO Secretariat were always available to help guide the overall meeting process but did not vote and did not have veto powers.

Management of competing interests

According to the processes recommended in the *WHO handbook for guideline development (13)* and the declaration of interests for WHO experts *(104)*, all experts participating in WHO meetings must declare any interest relevant to the meeting, before their participation. The responsible technical officer and the relevant departments reviewed the declaration-of-interest statements for all guideline development group members, before finalization of the group composition and invitation to attend a guideline development group meeting. All participants at guideline development meetings, including members of the guideline development group and systematic review team, submitted declaration-of-interest and confidentiality forms and their curriculum vitae before each meeting. Meeting participants participated in their individual capacity and not as institutional representatives. In addition to the forms, participants verbally declared any interests that could be perceived to affect their objectivity and independence in providing advice to WHO at the beginning of each meeting. The procedures for management of competing interests strictly followed the WHO guidelines for declaration of interests *(97).* The management of perceived or real conflicts of interest declared by members of the guideline development group that are relevant to this guideline is summarized below.

Professor Mette Berger an honorary professor at Lausanne University – Faculty of Biology & Medicine, Switzerland. She was the Head of the European Society for Clinical Nutrition and Metabolism (ESPEN) micronutrient guideline group 2019–2022. She declared receiving honoraria for lectures on parenteral nutrition, micronutrient guidelines and indirect calorimetry at medical congresses (ESPEN, The European Society of Intensive Care Medicine (ESICM)) from private sector organizations including Fresenius Kabi, Aguettant, Nestle, Baxter and DSM.

Dr Berger participated in the virtual sessions of 10 and 11 October 2023. No individual voting was held in that meeting and thereafter upon WHO seeking additional clarification on the declared honoraria from the private sector she decided to withdraw herself from the guideline development group. Dr Berger's participation in the first virtual guideline development group meeting is recorded as an external expert.

Ms Ayodelle Tella worked as Country Lead – Large Scale Food Fortification Programmes at TechnoServe Nigeria, a non-profit organization. Her role involved oversight of the activities of the country team, working closely with industry leaders for different food vehicles (edible oil, sugar and wheat flour), and operation managers to improve the competitiveness and compliance of food processors in Nigeria, to achieve the programme's objective of increasing fortification compliance. She also represented TechnoServe nationally, engaging with potential partners and stakeholders: processors, government institutions, regulators, technical industry experts, service providers and related agencies.

TechnoServe is a non-profit organization that has an entrepreneurial approach to development. Food and beverage companies are listed among their partners as well as non-profit organizations. Several board members work for the private sector but do not seem to have direct engagement with fortification or oil production.

For the second guideline development group meeting, she confirmed that TechnoServe did not receive funds specifically to scale up edible oil fortification in the past year. She declared that the fortification programme funded in the year previous to the meeting was to set up a technical assistance mechanism for producers in eight countries, for wheat flour, maize flour, edible oil, sugar and rice. She confirmed her role as a programme manager for Nigeria and not the direct grant manager.

Ms Tella fully participated in decision-making and was asked to verbally disclose to the other meeting participants her employer's involvement in projects related to food fortification.

Mr Phillip Joseph Makhumula-Nkhoma worked as an independent consultant with experience in development of standards, quality assurance and quality control for fortified foods. He has experience in developing guidelines on internal and external monitoring for the fortification of edible oils and laboratory testing methods for vitamin A in fortified oil. He declared having worked in the past for WFP, GAIN, FAO, FFI, TechnoServe, NI, UNICEF, USAID and HKI.

In May 2023, Phillip took up a full-time assignment in Nairobi, Kenya as a Senior Technical Advisor for the USAID AFFORD project through TechnoServe. This information was received after the second guideline development group meeting. After discussion with the WHO Office of Compliance, Risk Management and Ethics (CRE), it was deemed that the job change did not compromise his participation in the guideline development group.

The members of the systematic review team declared that they had no conflicts of interest.

Plans for updating the guideline

The WHO Secretariat will continue to follow research developments in edible oils and fats fortification, particularly those related to questions for which the certainty of evidence was found to be low or very low. If the guideline merits an update, or if there are concerns about the validity of the guideline, the Department of Nutrition and Food Safety, in collaboration with other WHO departments or programmes, will coordinate the guideline update, following the formal procedures of the *WHO handbook for guideline development*.

References

1. Global prevalence of vitamin A deficiency in populations at risk 1995–2005: WHO Global Database on Vitamin A Deficiency. Geneva: World Health Organization; 2009 (***https://iris.who.int/handle/10665/44110***).
2. Stevens GA, Bennet JE, Hennocq Q, Lu Y, De-Regil LM, Rogers L et al. Trends and mortality effects of vitamin A deficiency in children in 138 low-income and middle-income countries between 1991 and 2013: a pooled analysis of population-based surveys. Lancet Glob Health. 2015;3(9):E528–36 (***https://doi.org/10.1016/S2214-109X(15)00039-X***).
3. Amrein K, Scherkl M, Hoffmann M, Neuwersch-Sommeregger S, Köstenberger M, Tmava Berisha A et al. Vitamin D deficiency 2.0: an update on the current status worldwide. Eur J Clin Nutr. 2020;74(11):1498–513 (***https://doi.org/10.1038/s41430-020-0558-y***).
4. Nair R, Maseeh A. Vitamin D: the "sunshine" vitamin. J Pharmacol Pharmacother. 2012;3(2):118–26 (***https://doi.org/10.4103/0976-500X.95506***).
5. The 17 goals [website]. United Nations Department of Economic and Social Affairs; 2021 (***https://sdgs.un.org/goals***).
6. Comprehensive implementation plan on maternal, infant and young child nutrition. Geneva: World Health Organization; 2014 (WHO/NMH/NHD/14.1; ***https://iris.who.int/handle/10665/113048***).
7. WHO guideline: fortification of maize flour and corn meal with vitamins and minerals. Geneva: World Health Organization; 2016 (***https://iris.who.int/handle/10665/251902***). Licence: CC BY-NC-SA 3.0 IGO.
8. Guideline: fortification of rice with vitamins and minerals as a public health strategy. Geneva: World Health Organization; 2018 (***https://iris.who.int/handle/10665/272535***). Licence: CC BY-NC-SA 3.0 IGO.
9. Guideline: fortification of wheat flour with vitamins and minerals as a public health strategy. Geneva: World Health Organization; 2022 (***https://iris.who.int/handle/10665/354783***). Licence: CC BY-NC-SA 3.0 IGO.
10. Healthy diet [website]. World Health Organization; 2025 (***https://www.who.int/initiatives/behealthy/healthy-diet***).
11. Saturated fatty acid and trans-fatty acid intake for adults and children: WHO guideline. Geneva: World Health Organization; 2023 (***https://iris.who.int/handle/10665/370419***). Licence: CC BY-NC-SA 3.0 IGO.
12. Total fat intake for the prevention of unhealthy weight gain in adults and children: WHO guideline. Geneva: World Health Organization; 2023 (***https://iris.who.int/handle/10665/370421***). Licence: CC BY-NC-SA 3.0 IGO.
13. WHO handbook for guideline development, second edition. Geneva: World Health Organization; 2014 (***https://iris.who.int/handle/10665/145714***).
14. Schünemann H, Brożek J, Guyatt G, Oxman A. GRADE handbook [website]. McMaster University; 2013 (***https://gdt.gradepro.org/app/handbook/handbook.html***).
15. GRADE: welcome to the GRADE working group [website]. Grade Working Group; 2024 (***https://www.gradeworkinggroup.org/***).
16. Schünemann HJ, Brennan S, Akl EA, Hultcrantz M, Alonso-Coello P, Xia J et al. The development methods of official GRADE articles and requirements for claiming the use of GRADE – statement by the GRADE guidance group. J Clin Epidemiol. 2023;159:79–84 (***https://doi.org/10.1016/j.jclinepi.2023.05.010***).
17. Szabó É, Csölle I, Felső R, Kuellenberg de Gaudry D, Nyakundi PN, Ibrahim K et al. Benefits and harms of edible vegetable oils and fats fortified with vitamins A and D as a public health intervention in the general population: a systematic review of interventions. Nutrients. 2023;15(24):5135 (***https://doi.org/10.3390/nu15245135***).
18. Petry N, Al-Maamary SA, Woodruff BA, Alghannami S, Al-Shammakhi SM, Al-Ghammari IK et al. National prevalence of micronutrient deficiencies, anaemia, genetic blood disorders and over- and undernutrition in Omani women of reproductive age and preschool children. Sultan Qaboos Univ Med J. 2020;20(2):e151–64 (***https://doi.org/10.18295/squmj.2020.20.02.005***).
19. Oman Ministry of Health, International Micronutrient Malnutrition Prevention and Control (IMMPaCt) Program, Centers for Disease Control and Prevention, WHO Regional Office for the Eastern Mediterranean, United Nations Children's Fund. National micronutrient status and fortified food coverage survey: survey report. Muscat; 2009 (***https://groundworkhealth.org/wp-content/uploads/2020/04/MOH-2009_National-Food-fortification-coverage-survey.pdf***).
20. Sandjaja, Jus'at I, Jahari AB, Ifrad, Htet MK, Tilden RL et al. Vitamin A-fortified cooking oil reduces vitamin A deficiency in infants, young children and women: results from a programme evaluation in Indonesia. Public Health Nutr. 2015;18(14):2511–22 (***https://doi.org/10.1017/S136898001400322X***).
21. Williams AM, Tanumihardjo SA, Rhodes EC, Mapango C, Kazembe B, Phiri F et al. Vitamin A deficiency has declined in Malawi, but with evidence of elevated vitamin A in children. Am J Clin Nutr. 2021;113(4):854–64 (***https://doi.org/10.1093/ajcn/nqab004***).
22. Jääskeläinen T, Itkonen ST, Lundqvist A, Erkkola M, Koskela T, Lakkala K, Dowling KG, Hull GL, Kröger H, Karppinen J, Kyllönen E, Härkänen T, Cashman KD, Männistö S, Lamberg-Allardt C. The positive impact of general vitamin D food fortification policy on vitamin D status in a representative adult Finnish population: evidence from an 11-y follow-up based on standardized 25-hydroxyvitamin D data. Am J Clin Nutr.

2017;105(6):1512–1520 (https://doi.org/10.3945/ajcn.116.151415).
23. Allen L, de Benoist B, Dery O, Hurrell R, editors. Guidelines on food fortification with micronutrients. Geneva: World Health Organization and Food and Agriculture Organization of the United Nations; 2006 (https://iris.who.int/handle/10665/43412).
24. The global strategy for women's, children's and adolescents' health (2016–2030). Geneva: World Health Organization; 2014 (https://platform.who.int/docs/default-source/mca-documents/rmncah/global-strategy/ewec-globalstrategyreport-200915.pdf).
25. New WHA resolution to accelerate efforts on food micronutrient fortification [news release]. World Health Organization; 29 May 2023: (https://www.who.int/news/item/29-05-2023-new-wha-resolution-to-accelerate-efforts-on-food-micronutrient-fortification).
26. Conference outcome document: Rome declaration on nutrition. Rome: Food and Agriculture Organization of the United Nations and World Health Organization; 2014 (https://openknowledge.fao.org/server/api/core/bitstreams/3992f1da-6392-4050-ad09-431e489eacfb/content).
27. Anaemia in women and children [website]. World Health Organization; 2025 (https://www.who.int/data/gho/data/themes/topics/anaemia_in_women_and_children).
28. IVACG statement: vitamin A and iron interactions. Washington, DC: International Vitamin A Consultative Group; 1998 (https://pdf.usaid.gov/pdf_docs/Pnacp993.pdf).
29. McCauley ME, van den Broek N, Dou L, Othman M. Vitamin A supplementation during pregnancy for maternal and newborn health outcomes [website]. The Cochrane Collaboration; 2025 (https://www.cochrane.org/CD008666/PREG_vitamin-supplementation-during-pregnancy-maternal-and-newborn-health-outcomes).
30. Wirth JP, Petry N, Tanumihardjo SA, Rogers LM, McLean E, Greig A et al. Vitamin A supplementation programs and country-level evidence of vitamin A deficiency. Nutrients. 2017;9(3):190 (https://doi.org/10.3390/nu9030190).
31. Roth DE, Abrams SA, Aloia J, Bergeron G, Bourassa MW, Brown KH et al. Global prevalence and disease burden of vitamin D deficiency: a roadmap for action in low- and middle-income countries. Ann N Y Acad Sci. 2018;1430(1):44–79 (https://doi.org/10.1111/nyas.13968).
32. Food and Agriculture Organization of the United Nations, International Fund for Agricultural Development, United Nations Children's Fund, World Food Programme, World Health Organization. The state of food security and nutrition in the world 2023. Rome: Food and Agriculture Organization of the United Nations; 2023 (https://doi.org/10.4060/cc3017en).
33. Osendarp S, Akuoku JK, Black RE, Headey D, Ruel M, Scott N et al. The COVID-19 crisis will exacerbate maternal and child undernutrition and child mortality in low- and middle-income countries. Nat Food. 2021;2:476–84 (https://doi.org/https://doi.org/10.1038/s43016-021-00319-4).
34. Mandatory fortification: interactive map [website]. Global Fortification Data Exchange; 2025 (https://fortificationdata.org/interactive-map-fortification-legislation/).
35. REPLACE trans fat: an action package to eliminate industrially produced trans-fatty acids. Module 2: promote. How-to guide for determining the best replacement oils and interventions to promote their use. Geneva: World Health Organization; 2020 (https://iris.who.int/handle/10665/324821). Licence: CC BY-NC-SA 3.0 IGO.
36. WHO updates guidelines on fats and carbohydrates [news release]. World Health Organization; 17 July 2023: (https://www.who.int/news/item/17-07-2023-who-updates-guidelines-on-fats-and-carbohydrates).
37. Diosady LL, Krishnaswamy K. Micronutrient fortification of edible oils. In: Mannar MGV, Hurrell RF, editors. Food fortification in a globalized world: Academic Press; 2018:167–74 (https://doi.org/10.1016/C2014-0-03835-X).
38. Mbuya MNN, Thorpe J, Carpio A, Islam A, Saha A, Balagamwala M et al. Why do companies fortify? Drivers of compliance with edible oil fortification in Bangladesh. Geneva: Global Alliance for Improved Nutrition; 2020 (https://www.gainhealth.org/sites/default/files/publications/documents/gain-working-paper-series-8-why-do-companies-fortify-drivers-of-compliance-with-edible-oil-fortification-in-bangladesh.pdf).
39. Training course on edible oil fortification by Global Alliance for Improved Nutrition (GAIN) [website]. GAIN India; 2023 (https://staplefoodfortification.org/).
40. Candelaria LV, Magsadia CR, Velasco RE, Pedro MRA, Barba CVC, Tanchoco CC. The effect of vitamin A-fortified coconut cooking oil on the serum retinol concentration of Filipino children 4–7 years old. Asia Pac J Clin Nutr. 2005;14(1):43–53 (https://apjcn.qdu.edu.cn/14_1_12.pdf).
41. Atalhi N, El Hamdouchi A, Barkat A, Elkari K, Hamrani A, El Mzibri M et al. Combined consumption of a single high-dose vitamin A supplement with provision of vitamin A fortified oil to households maintains adequate milk retinol concentrations for 6 months in lactating Moroccan women. Appl Physiol Nutr Metab [Physiologie appliquée, nutrition et métabolisme]. 2020;45(3):275–82 (https://doi.org/10.1139/apnm-2019-0116).
42. Atalhi N, Choua G, Elhamdouchi A, Elhaloui N, Elmzibri M, Haskell M et al. Impact of daily consumption of vitamin A fortified oil on human milk vitamin A concentration in lactating Moroccan women. In: 11th European Nutrition Conference (FENS), Madrid, Spain, 26–29 October 2011. Ann Nutr Metab; 2011 (https://doi.org/https://doi.org/10.1159/000334393).
43. Marliyati SA, Martianto D, Andarwulan N, Fauzi S. Efficacy of non-branded cooking oil fortified with carotene from RPO on blood retinol and IgG of children aged 7–9 years. PJN. 2016;15(5):419–26 (https://doi.org/10.3923/pjn.2016.419.426).
44. Donglan W, Qingmin X, Yan H, Shutian LI, Weiqiang C, Yingyong C. Effects of vitamin A-fortified edible oil on

improving the immune function of children. Acta Nutr Sin. 2006;28(1):40–6 (***https://search.bvsalud.org/gim/resource/es/wpr-558269***).
45. Keller A, Ängquist L, Jacobsen R, Vaag A, Heitmann BL. A retrospective analysis of a societal experiment among the Danish population suggests that exposure to extra doses of vitamin A during fetal development may lower type 2 diabetes mellitus (T2DM) risk later in life. Br J Nutr. 2017;117(5) (***https://doi.org/10.1017/S000711451700037X***).
46. Ghasemifard N, Hassanzadeh-Rostami Z, Abbasi A, Naghavi AM, Faghih S. Effects of vitamin D-fortified oil intake versus vitamin D supplementation on vitamin D status and bone turnover factors: A double blind randomized clinical trial. Clin Nutr ESPEN. 2022;47:28–35 (***https://doi.org/10.1016/j.clnesp.2021.12.025***).
47. Ghasemifard N, Nasimi N, Hassanzadeh-Rostami Z, Abbasi A, Faghih S. Effects of vitamin D fortified canola oil on vitamin D and lipid profiles in healthy adults: a double-blind randomized trial. Iran J Nutr Sci Food Technol. 2020;15(2):fa1–9 (***https://www.cabidigitallibrary.org/doi/full/10.5555/20203305091***).
48. Stougaard M, Damm P, Frederiksen P, Jacobsen R, Heitmann BL. Exposure to vitamin D from fortified margarine during fetal life and later risk of pre-eclampsia: the D-tect Study. Public Health Nutr. 2018;21(4):721–31 (***https://doi.org/10.1017/S1368980017003135***).
49. Stougaard M, Damm P, Frederiksen P, Jacobsen R, Heitmann BL. Extra vitamin D from fortification and the risk of preeclampsia: the D-tect study. PLoS ONE. 2018;13(1):e0191288 (***https://doi.org/10.1371/journal.pone.0191288***).
50. Moos C, Duus KS, Frederiksen P, Heitmann BL, Andersen V. Exposure to the Danish mandatory vitamin D fortification policy in prenatal life and the risk of developing coeliac disease – the importance of season: a semi ecological study. Nutrients. 2020;12(5):1243 (***https://doi.org/10.3390/nu12051243***).
51. Duus KS, Moos C, Frederiksen P, Andersen V, Heitmann BL. Prenatal and early life exposure to the Danish mandatory vitamin D fortification policy might prevent inflammatory bowel disease later in life: a societal experiment. Nutrients. 2021;13(4):1367 (***https://doi.org/10.3390/nu13041367***).
52. Heitmann BL. D-tecting disease – from exposure to vitamin D during critical periods of life (D-tect) [website]. National Library of Medicine; 2017 (***https://ClinicalTrials.gov/show/NCT03330301***).
53. Keller A, Stougård M, Frederiksen P, Thorsteinsdottir F, Vaag A, Damm P et al. In utero exposure to extra vitamin D from food fortification and the risk of subsequent development of gestational diabetes: the D-tect study. Nutr J. 2018;17:1–9 (***https://doi.org/10.1186/s12937-018-0403-5***).
54. Thorsteinsdottir F, Maslova E, Jacobsen R, Frederiksen P, Keller A, Backer V et al. Exposure to vitamin D fortification policy in prenatal life and the risk of childhood asthma: results from the D-Tect study. Nutrients. 2019;11(4):924 (***https://doi.org/10.3390/nu11040924***).
55. Jacobsen R, Abrahamsen B, Bauerek M, Holst C, Jensen CB, Knop J et al. The influence of early exposure to vitamin D for development of diseases later in life. BMC Public Health. 2013;13:1–8 (***https://doi.org/10.1186/1471-2458-13-515***).
56. Jacobsen R, Hypponen E, Sørensen TIA, Vaag AA, Heitmann BL. Gestational and early infancy exposure to margarine fortified with vitamin D through a national Danish programme and the risk of type 1 diabetes: The D-tect study. PLoS ONE. 2015;10(6):e0128631 (***https://doi.org/10.1371/journal.pone.0128631***).
57. Jacobsen R, Moldovan M, Vaag AA, Hypponen E, Heitmann BL. Vitamin D fortification and seasonality of birth in type 1 diabetic cases: D-tect study. J Dev Orig Health Dis. 2016;7:114–9 (***https://doi.org/10.1017/S2040174415007849***).
58. Jensen CB, Berentzen TL, Gamborg M, Sørensen TIA, Heitmann BL. Does prenatal exposure to vitamin D-fortified margarine and milk alter birth weight? A societal experiment. Br J Nutr. 2014;112(5):785–93 (***https://doi.org/10.1017/S0007114514001330***).
59. Händel MN, Frederiksen P, Osmond C, Cooper C, Abrahamsen B, Heitmann BL. Prenatal exposure to vitamin D from fortified margarine and risk of fractures in late childhood: period and cohort results from 222 000 subjects in the D-tect observational study. Br J Nutr. 2017;117(6):872–81 (***https://doi.org/10.1017/S000711451700071X***).
60. Jensen C, Stougård M, Sørensen T, Heitmann B. Does prenatal exposure to vitamin D-fortified margarine and milk alter birth weight? A societal experiment – Corrigendum. Br J Nutr. 2016;116(2):377–9 (***https://doi.org/10.1017/S0007114516002014***).
61. Jensen CB, Gamborg M, Berentzen TL, Sørensen TIA, Heitmann BL. Prenatal exposure to vitamin-D from fortified margarine and milk and body size at age 7 years. Eur J Clin Nutr. 2015;69(10):1169–75 (***https://doi.org/10.1038/ejcn.2015.55***).
62. Universal declaration of human rights. New York: United Nations General Assembly; 1948 (***https://www.un.org/en/about-us/universal-declaration-of-human-rights***).
63. Universal declaration on the eradication of hunger and malnutrition. United Nations Human Rights Office of the High Commissioner; 1974 (***https://www.ohchr.org/en/instruments-mechanisms/instruments/universal-declaration-eradication-hunger-and-malnutrition***).
64. International covenant on economic, social and cultural rights. United Nations Human Rights Office of the High Commissioner; 1966 (***https://www.ohchr.org/en/instruments-mechanisms/instruments/international-covenant-economic-social-and-cultural-rights***).
65. Convention on the Elimination of All Forms of Discrimination against Women. New York: United Nations General

Assembly; 1979 (*https://www.ohchr.org/en/instruments-mechanisms/instruments/convention-elimination-all-forms-discrimination-against-women*).
66. Convention on the Rights of the Child. New York: United Nations General Assembly; 1989 (*https://www.ohchr.org/en/instruments-mechanisms/instruments/convention-rights-child*).
67. Daily food intake/availability: interactive map [website]. Global Fortification Data Exchange; 2025 (*https://fortificationdata.org/map-availability/*).
68. Vitamin A level, Vitamin D level: interactive map [website]. Global Fortification Data Exchange; 2025 (*https://fortificationdata.org/map-nutrient-levels-in-fortification-standards/*).
69. Turck D, Bohn T, Castenmiller J, de Henauw S, Hirsch-Ernst K-I, Knutsen HK et al. Scientific opinion on the tolerable upper intake level for vitamin D, including the derivation of a conversion factor for calcidiol monohydrate. EFSA Journal. 2023;21(8):e08145 (*https://doi.org/10.2903/j.efsa.2023.8145*).
70. Fiedler JL, Lividini K, Guyondet C, Bermudez OI. Assessing alternative industrial fortification portfolios: a Bangladesh case study. Food Nutr Bull. 2015;31(1):57–74 (*https://doi.org/10.1177/156482651503600106*).
71. Walters D, Ndau E, Saleh N, Mosha T, Horton S. Cost-effectiveness of sunflower oil fortification with vitamin A in Tanzania by scale. Matern Child Nutr. 2019;15(S3):e12720 (*https://doi.org/10.1111/mcn.12720*).
72. Vosti S, Engle-Stone R, Woldegebreal D, Luo H, Asfaw E, Kagin J et al. Estimated effectiveness and cost-effectiveness of a fortified edible oils program in Ethiopia. Curr Dev Nutr. 2020;4 (*https://doi.org/10.1093/cdn/nzaa064_025*).
73. Nutrition and food safety [website]. World Health Organization; 2025 (*https://www.who.int/teams/nutrition-and-food-safety/overview*).
74. O'Neill J, Tabish H, Welch V, Petticrew M, Pottie K, Clarke M et al. Applying an equity lens to interventions: using PROGRESS ensures consideration of socially stratifying factors to illuminate inequities in health. J Clin Epidemiol. 2014;67(1):56–64 (*https://doi.org/10.1016/j.jclinepi.2013.08.005*).
75. Handbook on health inequality monitoring with a special focus on low- and middle-income countries. Geneva: World Health Organization; 2013 (*https://iris.who.int/handle/10665/85345*).
76. Health inequality monitor [website]. World Health Organization; 2024 (*https://www.who.int/data/inequality-monitor*).
77. Rohner F, Wirth JP, Zeng W, Petry N, Donkor WES, Neufeld LM et al. Global coverage of mandatory large-scale food fortification programs: a systematic review and meta-analysis. Adv Nutr. 2023;14(5):1197–210 (*https://doi.org/10.1016/j.advnut.2023.07.004*).
78. Frequently asked questions on sexual and gender diversity, health and human rights: an introduction to key concepts. Geneva: World Health Organization; 2024 (*https://www.who.int/publications/m/item/frequently-asked-questions-on-sexual-and-gender-diversity--health-and-human-rights--an-introduction-to-key-concepts*).
79. Gender mainstreaming for health managers: a practical approach. Geneva: World Health Organization; 2011 (*https://iris.who.int/handle/10665/44516*).
80. WHO announces the development of a guideline on the health of trans and gender diverse people [news release].World Health Organization; 18 December 2023 (*https://www.who.int/news/item/18-12-2023-who-announces-the-development-of-a-guideline-on-the-health-of-trans-and-gender-diverse-people*).
81. Codex Alimentarius Commission. General principles for the addition of essential nutrients to foods. Food and Agriculture Organization of the United Nations, World Health Organization; 2015 (CAC/GL 9-1987; *https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?lnk=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXG%2B9-1987%252FCXG_009e_2015.pdf*).
82. Codex Alimentarius Commission. Guidelines on nutrition labelling. Food and Agriculture Organization of the United Nations, World Health Organization; 2021 (CXG 2-1985; *https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?lnk=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXG%2B2-1985%252FCXG_002e.pdf*).
83. Codex Alimentarius Commission. Standard for olive oils and olive pomace oils. 2021 (CXS 33-1981; h*ttps://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?lnk=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B33-1981%252FCXS_033e.pdf*).
84. Codex Alimentarius Commission. Standard for named vegetable oils. Food and Agriculture Organization of the United Nations, World Health Organization; 2023 (*https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?lnk=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B210-1999%252FCXS_210e.pdf*).
85. Codex Alimentarius Commission. Standard for edible fats and oils not covered by individual standards. Food and Agriculture Organization of the United Nations, World Health Organization; 2023 (CX 19-1981; *https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?lnk=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B19-1981%252FCXS_019e.pdf*).
86. Food and Agriculture Organization of the United Nations, World Health Organization. Food control system assessment tool: dimension B – control functions. Rome: Food and Agriculture Organization of the United Nations, World Health Organization; 2019 (*https://iris.who.int/handle/10665/329868*). Licence: CC BY-NC-SA 3.0 IGO.

87. Code of practice for food premix operations. Washington, DC: Pan American Health Organization; 2005 (FCH/NU/66-16/04; ***https://iris.paho.org/handle/10665.2/37201***).
88. Institute of Medicine. Food chemicals codex, fifth edition. Washington, DC: The National Academies Press; 2003 (***https://doi.org/https://doi.org/10.17226/10731***).
89. Makhumula P, Dary O, Guamuch M, Tom C, Afidra R, Rambeloson Z. Legislative frameworks for corn flour and maize meal fortification. Ann NY Acad Sci. 2014;1312(1):91–104 (***https://doi.org/10.1111/nyas.12349***).
90. Blas E, Sivasankara Kurup A. Equity, social determinants and public health programmes. Geneva: World Health Organization; 2010 (***https://iris.who.int/handle/10665/44289***).
91. Jungjohann SM, Ara G, Pedro C, Friesen VM, Khanam M, Ahmed T et al. Vitamin A fortification quality is high for packaged and branded edible oil but low for oil sold in unbranded, loose form: findings from a market assessment in Bangladesh. Nutrients. 2021;13(3):794 (***https://doi.org/10.3390/nu13030794***).
92. Mkambula P, Mbuya MNN, Rowe LA, Sablah M, Friesen VM, Chadha M et al. The unfinished agenda for food fortification in low- and middle-income countries: quantifying progress, gaps and potential opportunities. Nutrients. 2020;12(2):354 (***https://doi.org/10.3390/nu12020354***).
93. Garcia-Casal MN, Peña-Rosas JP, De-Regil LM, Gwirtz JA, Pasricha S-R. Fortification of maize flour with iron for controlling anaemia and iron deficiency in populations. Cochrane Database Syst Rev. 2018;12(12):CD010187 (***https://doi.org/10.1002/14651858.CD010187.pub2***).
94. eCatalogue of indicators for micronutrient programmes [online database]. Geneva: World Health Organization; 2015 (***https://extranet.who.int/indcat/***).
95. Vitamin and mineral nutrition information system: micronutrients database [online database]. Geneva: World Health Organization; 2025 (***https://www.who.int/teams/nutrition-and-food-safety/databases/vitamin-and-mineral-nutrition-information-system***).
96. Global nutrition targets tracking tool [website]. World Health Organization; 2025 (***https://www.who.int/tools/global-targets-tracking-tool***).
97. Global database on the implementation of food and nutrition action (GIFNA) [online database]. Geneva: World Health Organization; 2024 (***https://gifna.who.int/about-us***).
98. WHO guideline development group meeting - fortification of wheat flour with vitamins and minerals in public health [website]. World Health Organization; 2025 (***https://www.who.int/news-room/events/detail/2020/09/16/default-calendar/whogdg-meeting-fortification-of-wheat-flour-with-vitamins-and-minerals-in-public-health***).
99. Diet and health subgroup [website]. World Health Organization; 2025 (***https://www.who.int/groups/nutrition-guidance-expert-advisory-group-(nugag)/diet-and-health***).
100. Call for experts – WHO guideline development group – food fortification with micronutrients [website]. World Health Organization; 2025 (***https://www.who.int/news-room/articles-detail/call-for-experts-who-guideline-development-group-food-fortification-with-micronutrients***).
101. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ et al. Cochrane handbook for systematic reviews of interventions, second edition. Chichester: John Wiley & Sons; 2019 (***https://training.cochrane.org/handbook***).
102. Guyatt G, Oxman AD, Akl EA, Kunz R, Vist G, Brozek J et al. GRADE guidelines: 1. Introduction—GRADE evidence profiles and summary of findings tables. J Clin Epidemiol. 2011;64(4):383–94 (***https://doi.org/10.1016/j.jclinepi.2010.04.026***).
103. Guyatt GH, Oxman AD, Vist GE, Kunz R, Falk-Ytter Y, Alonso-Coello P et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. BMJ. 2008;336(7650):924–6 (***https://doi.org/10.1136/bmj.39489.470347.AD***).
104. Declarations of interest [website]. World Health Organization; 2025 (***https://www.who.int/about/ethics/declarations-of-interest***).

Annex 1. GRADE summary of findings tables

Tables A1.1–A1.4 provide the GRADE summary of findings tables that informed this guideline.

Table A1.1. GRADE evidence profile for primary (critical) outcomes of vitamin A fortification of edible oils or fats compared with no vitamin A fortification of edible oils or fats

Certainty assessment							Number of participants		Effect		Certainty	Importance
Number of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Exposed to vitamin A-fortified oils or fats	Not exposed to vitamin A-fortified oils or fats	Relative (95% CI)	Absolute (95% CI)		
Serum retinol concentration (µmol/L) – randomized studies (follow-up: 6 months)												
2 *(1, 2)*	Randomized trials	Serious[a]	Serious[b]	Not serious	Not serious[c]	None	307	207	–	MD **0.35 µmol/L higher** (0.43 lower to 1.12 higher)	⊕⊕◯◯ Low	Critical
Serum retinol concentration (µmol/L) – non-randomized studies (follow-up: 2–5 months)												
2 *(3, 4)*	Controlled clinical trials	Serious[d]	Serious[b]	Not serious	Serious[e]	None	102	103	–	MD **0.31 µmol/L higher** (0.18 lower to 0.8 higher)	⊕◯◯◯ Very low	Critical
Subclinical vitamin A deficiency (serum retinol equal to or less than 0.70 µmol/L) – randomized studies (follow-up: 6 months)												
1 *(2)*	Randomized trials	Serious[f]	Not serious	Not serious	Serious[g]	None	0/268 (0.0%)	0/144 (0.0%)	Not estimable	–	⊕⊕◯◯ Low	Critical
Subclinical vitamin A deficiency (serum retinol equal to or less than 0.70 µmol/L) – non-randomized studies (follow-up: 2 months)												
1 *(4)*	Controlled clinical trials	Serious[f]	Not serious	Not serious	Very serious[h]	None	0/15 (0.0%)	2/16(12.5%)	RR 0.21 (0.01 to 4.10)	**99 fewer per 1 000** (from 124 fewer to 387 more)	⊕◯◯◯ Very low	Critical
Clinical vitamin A deficiency (xerophthalmia, night blindness) – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Critical
All-cause morbidity – randomized studies (follow-up: 6 months)												
2 *(1, 2)*	Randomized trials	Very serious[i]	Not serious	Not serious	Not serious	None	Out of two RCTs measuring morbidity, one (with 268 participants in the intervention group and 144 in the control group) reported results as frequency and duration of illness. This study reported no significant differences between study groups.				⊕⊕◯◯ Low	Critical
All-cause morbidity – non-randomized studies (follow-up: 2 months)												
1 *(4)*	Controlled clinical trials	Serious[f]	Not serious[j]	Not serious	Very serious[h]	None	One CCT reported morbidity scores (defined as frequency of illness multiplied by duration of illness) and described no significant differences between study groups.				⊕◯◯◯ Very low	Critical
All-cause mortality – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Critical
Adverse effects (hypervitaminosis, liver toxicity) – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Critical

CCT: controlled clinical trial; CI: confidence interval; MD: mean difference; RCT: randomized controlled trial; RR: risk ratio.

a Downgraded by one level for risk of bias (RoB) since both included studies were rated with some concerns for RoB.
b Downgraded for inconsistency as point estimates did vary widely, 95% CI did not overlap between studies, the direction of effect was not consistent and the magnitude of heterogeneity was high (I2 was 98%, P value for heterogeneity was < 0.0001). Subgroup analyses did not fully explain heterogeneity.
c Not downgraded for imprecision. Although only two studies were included, the magnitude of the median sample size was intermediate (*n* = 257), and the total sample size was larger than 400 (*n* = 514).
d Downgraded by one level for RoB since one out of two included studies was rated with a high RoB, and none of the included studies were rated with a low RoB.
e Downgraded by one level for imprecision since the number of included studies was small (*n* = 2), the magnitude of the median sample size was intermediate (*n* = 103), and the total sample size was smaller than 400 (*n* = 205).
f Downgraded by one level for RoB since the included study was rated with some concerns for RoB.
g Downgraded by one level for imprecision. There was only one study included, but the total sample size was larger than 400 (*n* = 412).
h Downgraded by two levels for imprecision since results are derived from one study, where total sample size was very low (*n* < 100).
i Downgraded by two levels for RoB, as results were not reported for one out of two studies, and additionally, because none of the included studies were rated with a low RoB.
j This is a single study so inconsistency cannot be judged.

Table A1.2. GRADE evidence profile for primary (critical) outcomes of vitamin D fortification of edible oils or fats compared with no vitamin D fortification of edible oils or fats

Certainty assessment							Number of participants		Effect		Certainty	Importance
Number of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Exposed to vitamin D-fortified oils or fats	Not exposed to vitamin D-fortified oils or fats	Relative (95% CI)	Absolute (95% CI)		
Serum 25(OH)D concentration (µmol/L) – randomized studies (follow-up: 12 weeks)												
1 *(5)*	Randomized trials	Not serious	Not serious[a]	Not serious	Very serious[b]	None	30	32	–	MD **6.59 nmol/L higher** (6.89 lower to 20.07 higher)	⊕⊕◯◯ Low	Critical
Vitamin D deficiency – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Critical
Osteomalacia in older persons – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Critical
Nutritional rickets in children – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Critical
Adverse effects (hypervitaminosis, hypercalcaemia, kidney stones) – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Critical
All-cause morbidity – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Critical
All-cause mortality – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Critical

CI: confidence interval; MD: mean difference; RR: risk ratio.

a This is a single study so inconsistency cannot be judged.
b Downgraded by two levels for imprecision since results are derived from one study, where total sample size was very low ($n < 100$).

Table A1.3. GRADE evidence profile for secondary outcomes of vitamin A fortification of edible oils or fats compared with no vitamin A fortification of edible oils or fats

Certainty assessment							Number of participants		Effect		Certainty	Importance
Number of studies	**Study design**	**Risk of bias**	**Inconsistency**	**Indirectness**	**Imprecision**	**Other considerations**	**Exposed to vitamin A-fortified oils or fats**	**Not exposed to vitamin A-fortified oils or fats**	**Relative (95% CI)**	**Absolute (95% CI)**		
Dietary vitamin A intake (µg RE/day) – randomized studies (follow-up: 6 months)												
1 *(2)*	Randomized trials	Serious[b]	Not serious[e]	Not serious	Serious[c]	None	268	144	–	MD **15.7 µg RE/day lower** (105.82 lower to 74.42 higher)	⊕⊕◯◯ Low	Important
Dietary vitamin A intake (µg RE/day) – non-randomized studies (follow-up: 2 months)												
1 *(4)*	Controlled clinical trials	Serious[b]	Not serious[e]	Not serious	Very serious[d]	None	15	16	–	MD **240.6 µg RE/day higher** (175.07 higher to 306.13 higher)	⊕◯◯◯ Very low	Important
Liver stores of vitamin A – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important
Retinol concentration in breast milk (µmol/L) – randomized studies (follow-up: 6 months)												
1 *(1)*	Randomized trials	Serious[b]	Not serious[e]	Not serious	Very serious[d]	None	37	26	–	MD **0.79 µmol/L higher** (0.72 higher to 0.86 higher)	⊕◯◯◯ Very low	Important
Mothers with low concentrations of retinol in breast milk (< 1.05 µmol/L) – randomized studies (follow-up: 6 months)												
1 *(1)*	Randomized trials	Serious[b]	Not serious[e]	Not serious	Very serious[d]	None	2/62 (3.2%)	39/39 (100%)	RR 0.04 (0.01 to 0.14)	**960 fewer per 1 000** (from 990 fewer to 860 fewer)	⊕◯◯◯ Very low	Important
Prevalence of biochemical vitamin A deficiency (modified relative dose response test) – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important
Retinol-binding protein concentration – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important
Beta-carotene concentration – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important
Serum ferritin concentration – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important
Anaemia – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important
Body mass index (kg/m2) – non-randomized studies (follow-up: 2 months)												
1 *(4)*	Controlled clinical trials	Serious[b]	Not serious[e]	Not serious	Very serious[d]	None	15	16	–	MD **0.69 kg/m2 higher** (0.09 higher to 1.29 higher)	⊕◯◯◯ Very low	–
Weight-for-height (z-score) – randomized studies (follow-up: 6 months)												
1 *(2)*	Randomized trials	Serious[b]	Not serious[e]	Not serious	Serious[f]	None	268	145	–	MD **0.02 z-scores lower** (0.21 lower to 0.17 higher)	⊕⊕◯◯ Low	–

Table A1.3. GRADE evidence profile for secondary outcomes of vitamin A fortification of edible oils or fats compared with no vitamin A fortification of edible oils or fats

Certainty assessment							Number of participants		Effect		Certainty	Importance
Number of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Exposed to vitamin A-fortified oils or fats	Not exposed to vitamin A-fortified oils or fats	Relative (95% CI)	Absolute (95% CI)		
IgG concentration (µmol/L) – non-randomized studies (follow-up: 2–5 months)												
2 *(3, 4)*	Controlled clinical trials	Serious[a]	Not serious	Not serious	Serious[g]	None	55	55	–	MD **1.24 µmol/L higher** (6.67 lower to 9.16 higher)	⊕⊕◯◯ Low	–
IgA concentration (µmol/L) – non-randomized studies (follow-up: 5 months)												
1 *(3)*	Controlled clinical trials	Very serious[h]	Not serious	Not serious	Serious[d]	None	40	40	–	MD **1.93 µmol/l higher** (0.56 higher to 3.3 higher)	⊕◯◯◯ Very low	–
IgM concentration (µmol/L) – non-randomized studies (follow-up: 5 months)												
1 *(3)*	Controlled clinical trials	Very serious[h]	Not serious	Not serious	Serious[d]	None	40	40	–	MD **0.03 µmol/L lower** (0.22 lower to 0.16 higher)	⊕◯◯◯ Very low	–
C-reactive protein concentration – not reported												
–	–	–	–	–	–	–	–	–	–	–	–	–
Type 2 diabetes mellitus												
1 *(6)*	Observational study	Extremely serious[i]	Not serious	Not serious	Not serious	None	1273/ 101 178 (1.3%)	1322/92 625 (1.4%)	RR 0.88 (0.82 to 0.95)	**2 fewer per 1 000** (from 3 fewer to 1 fewer)	⊕◯◯◯ Very low	–

CI: confidence interval; MD: mean difference; RE: retinol equivalents; RR: risk ratio.

a Downgraded by one level for RoB since one out of two included studies was rated with a high RoB, and none of the included studies were rated with a low RoB.
b Downgraded by one level for RoB since the included study was rated with some concerns for RoB.
c Downgraded by one level for imprecision. There was only one study included, but the total sample size was larger than 400 (*n* = 412).
d Downgraded by two levels for imprecision since results are derived from one study, where total sample size was very low (*n* < 100).
e This is a single study so inconsistency cannot be judged.
f Downgraded by one level for imprecision. There was only one study included, but the total sample size was larger than 400 (*n* = 413)
g Downgraded by one level for imprecision since results are derived from two small studies with a low total sample size (*n* = 110).
h Downgraded by two levels for RoB since the included study was rated with high RoB.
i Downgraded by three levels for RoB since the study included was a birth cohort study rated with serious concerns for RoB with the Risk of Bias in Non-randomised Studies of Interventions (ROBINS-I) tool.

Table A1.4. GRADE evidence profile for secondary outcomes of vitamin D fortification of edible oils or fats compared with no vitamin D fortification of edible oils or fats

Certainty assessment							Number of participants		Effect		Certainty	Importance
Number of studies	**Study design**	**Risk of bias**	**Inconsistency**	**Indirectness**	**Imprecision**	**Other considerations**	**Exposed to vitamin D-fortified oils or fats**	**Not exposed to vitamin D-fortified oils or fats**	**Relative (95% CI)**	**Absolute (95% CI)**		
Intake of dietary vitamin D (µg/day) (follow-up: 12 weeks)												
1 *(5)*	Randomized trials	Not serious	Not serious[a]	Not serious	Very serious[b]	None	30	32	–	MD **22.35 µg lower** (52.88 lower to 8.18 higher)	⊕⊕◯◯ Low	Important
Serum parathyroid hormone concentration (pmol/L) (follow-up: 12 weeks)												
1 *(5)*	Randomized trials	Not serious	Not serious[a]	Not serious	Very serious[b]	None	18	18	–	MD **0.1 pmol/L higher** (0.99 lower to 1.99 higher)	⊕⊕◯◯ Low	Important
Serum bone alkaline phosphatase levels (IU/L) (follow-up: 12 weeks)												
1 *(5)*	Randomized trials	Not serious	Not serious[a]	Not serious	Very serious[b]	None	18	18	–	MD **5.76 IU/L higher** (0.12 lower to 11.64 higher)	⊕⊕◯◯ Low	Important
Incidence of gestational diabetes mellitus												
1 *(7)*	Observational studies	Very serious[c]	Not serious[a]	Not serious	Serious[d]	None	297 out of 14 016 (2.1%)	361 out of 14 855 (2.4%)	RR 0.87 (0.75 to 1.01)	**3 fewer per 1 000** (from 6 fewer to 0 fewer)	⊕◯◯◯ Very low	Important
Weight gain during pregnancy – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important
Pre-eclampsia												
1 *(8)*	Observational studies	Very serious[c]	Not serious[a]	Not serious	Serious[d]	None	1 310 out of 35 124 (3.7%)	1 370 out of 38 113 (3.6%)	RR 1.04 (0.96 to 1.12)	**1 more per 1 000** (from 1 fewer to 4 more)	⊕◯◯◯ Very low	Important
Gestational hypertension												
1 *(8)*	Observational studies	Very serious[c]	Not serious[a]	Not serious	Serious[d]	None	399 out of 35 124 (1.1%)	396 out of 38 113 (1.0%)	RR 1.09 (0.95 to 1.26)	**1 more per 1 000** (from 1 fewer to 3 more)	⊕◯◯◯ Very low	Important
Eclampsia												
1 *(8)*	Observational studies	Very serious[c]	Not serious[a]	Not serious	Serious[d]	None	267 out of 35 124 (0.8%)	396 out of 38 113 (1.0%)	RR 0.98 (0.83 to 1.16)	**0 fewer per 1 000** (from 2 fewer to 2 more)	⊕◯◯◯ Very low	Important
Preterm birth – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important
Low birth weight (less than 2 500 g)												
1 *(9)*	Observational studies	Very seriousc	Not serious[a]	Not serious	Serious[d]	None	301 out of 4 836 (6.2%)	308 out of 5 716 (5.4%)	RR 1.16 (0.99 to 1.35)	**9 more per 1 000** (from 1 fewer to 19 more)	⊕◯◯◯ Very low	Important
Postpartum depression – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important
Postpartum haemorrhage – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important

Certainty assessment							Number of participants		Effect		Certainty	Importance
Number of studies	**Study design**	**Risk of bias**	**Inconsistency**	**Indirectness**	**Imprecision**	**Other considerations**	**Exposed to vitamin D-fortified oils or fats**	**Not exposed to vitamin D-fortified oils or fats**	**Relative (95% CI)**	**Absolute (95% CI)**		
Breast-milk vitamin D content of lactanting women - not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important
Body mass index												
1 *(10)*	Observational studies	Very serious[c]	Not serious[a]	Not serious	Not serious	None	5 029	5 803	–	MD **0.1 kg/m² lower** (0.17 lower to 0.03 lower)	⊕⊕◯◯ Low	Important
Overweight												
1 *(10)*	Observational studies	Very serious[c]	Not serious[a]	Not serious	Not serious	None	1 172 out of 5 029 (23.3%)	1 469 out of 5 803 (25.3%)	RR 0.92 (0.86 to 0.98)	**20 fewer per 1 000** (from 35 fewer to 5 fewer)	⊕⊕◯◯ Low	Important
Obesity												
1 *(10)*	Observational studies	Very serious[c]	Not serious[a]	Not serious	Not serious	None	523 out of 5 029 (10.4%)	708 out of 5 803 (12.2%)	RR 0.85 (0.77 to 0.95)	**18 fewer per 1 000** (from 28 fewer to 6 fewer)	⊕⊕◯◯ Low	Important
Type 1 diabetes mellitus												
1 *(11)*	Observational studies	Very serious[c]	Not serious[a]	Not serious	Not serious	None	298 out of 127 207 (0.2%)	404 out of 134 749 (0.3%)	RR 0.78 (0.67 to 0.91)	**1 fewer per 1 000** (from 1 fewer to 0 fewer)	⊕⊕◯◯ Low	Important
Type 2 diabetes mellitus – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important
Fracture events												
1 *(12)*	Observational studies	Very serious[c]	Not serious[a]	Not serious	Not serious	None	12 330 out of 10 4406 (11.8%)	16058 out of 11 3577 (14.1%)	RR 0.84 (0.82 to 0.85)	**23 fewer per 1 000** (from 25 fewer to 21 fewer)	⊕⊕◯◯ Low	Important
Childhood asthma												
1 *(13)*	Observational studies	Not serious	Not serious[a]	Not serious	Not serious	None	1 427 out of 106 347 (1.3%)	1 613 out of 115 900 (1.4%)	RR 0.96 (0.90 to 1.03)	**1 fewer per 1 000** (from 1 fewer to 0 fewer)	⊕⊕◯◯ Low	Important
Hypertension – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important
Cardiovascular disease – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important
Respiratory infections – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important
Menopause onset – not measured												
–	–	–	–	–	–	–	–	–	–	–	–	Important

Certainty assessment							Number of participants		Effect		Certainty	Importance
Number of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Exposed to vitamin D-fortified oils or fats	Not exposed to vitamin D-fortified oils or fats	Relative (95% CI)	Absolute (95% CI)		
Inflammatory bowel disease												
1 *(14)*	Observational studies	Very serious[c]	Not serious[a]	Not serious	Not serious	None	875 out of 98 856 (0.9%)	1 102 out of 108 044 (1.0%)	RR 0.87 (0.79 to 0.95)	**1 fewer per 1 000** (from 2 fewer to 1 fewer)	⊕⊕◯◯ Low	–
Total cholesterol concentration (mmol/L) (follow-up: 12 weeks)												
1 *(15)*	Randomized trials	Not serious	Not serious[a]	Not serious	Very serious[b]	None	30	32	–	MD **0.01 mmol/L higher** (0.38 lower to 0.4 higher)	⊕⊕◯◯ Low	–
Low-density lipoprotein cholesterol concentration (mmol/L) (follow-up: 12 weeks)												
1 *(15)*	Randomized trials	Not serious	Not serious[a]	Not serious	Very serious[b]	None	30	32	–	MD **0.02 mmol/L higher** (0.25 lower to 0.29 higher)	⊕⊕◯◯ Low	–
High-density lipoprotein cholesterol concentration (mmol/L) (follow-up: 12 weeks)												
1 *(15)*	Randomized trials	Not serious	Not serious[a]	Not serious	Very serious[b]	None	30	32	–	MD **0.03 mmol/L higher** (0.07 lower to 0.13 higher)	⊕⊕◯◯ Low	–
Triglyceride concentration (mmol/L) (follow-up: 12 weeks)												
1 *(15)*	Randomized trials	Not serious	Not serious[a]	Not serious	Very serious[b]	None	30	32	–	MD **0.02 mmol/L higher** (0.34 lower to 0.38 higher)	⊕⊕◯◯ Low	–
Energy intake (kcal) (follow-up: 12 weeks)												
1 *(5)*	Randomized trials	Not serious	Not serious[a]	Not serious	Very serious[b]	None	31	30	–	MD **280 kcal lower** (513.33 lower to 46.67 lower)	⊕⊕◯◯ Low	–
High birth weight (more than 4 000 g)												
1 *(9)*	Observational studies	Very serious[c]	Not serious[a]	Not serious	Not serious	None	407 out of 4 836 (8.4%)	583 out of 5 716 (10.2%)	RR 0.83 (0.73 to 0.93)	**17 fewer per 1 000** (from 28 fewer to 7 fewer)	⊕⊕◯◯ Low	–
Coeliac disease												
1 *(16)*	Observational studies	Very serious[c]	Not serious[a]	Not serious	Serious[d]	None	148 out of 98 856 (0.1%)	199 out of 108 044 (0.2%)	RR 0.81 (0.66 to 1.01)	**0 fewer per 1 000** (from 1 fewer to 0 fewer)	⊕◯◯◯ Very Low	–

CI: confidence interval; IU: international units; MD: mean difference; RR: risk ratio.

a This is a single study so inconsistency cannot be judged.
b Downgraded by two levels for imprecision since results are derived from one study, where total sample size was very low (n < 100).
c Downgraded by two levels for RoB since the study included was a birth cohort study rated with some concerns for RoB with the ROBINS-I tool.
d Downgraded by one level for imprecision since results are derived from one study, and the CI was crossing the line of no effect.

References

1. Atalhi N, El Hamdouchi A, Barkat A, Elkari K, Hamrani A, El Mzibri M et al. Combined consumption of a single high-dose vitamin A supplement with provision of vitamin A fortified oil to households maintains adequate milk retinol concentrations for 6 months in lactating Moroccan women. Appl Physiol Nutr Metab [Physiologie appliquée, nutrition et métabolisme]. 2020;45(3):275–82 (*https://doi.org/10.1139/apnm-2019-0116*).
2. Candelaria LV, Magsadia CR, Velasco RE, Pedro MRA, Barba CVC, Tanchoco CC. The effect of vitamin A-fortified coconut cooking oil on the serum retinol concentration of Filipino children 4–7 years old. Asia Pac J Clin Nutr. 2005;14(1):43–53 (*https://apjcn.qdu.edu.cn/14_1_12.pdf*).
3. Donglan W, Qingmin X, Yan H, Shutian LI, Weiqiang C, Yingyong C. Effects of vitamin A-fortified edible oil on improving the immune function of children. Acta Nutr Sin. 2006;28(1):40–6 (*https://search.bvsalud.org/gim/resource/es/wpr-558269*).
4. Marliyati SA, Martianto D, Andarwulan N, Fauzi S. Efficacy of non-branded cooking oil fortified with carotene from RPO on blood retinol and IgG of children aged 7–9 years. PJN. 2016;15(5):419–26 (*https://doi.org/10.3923/pjn.2016.419.426*).
5. Ghasemifard N, Hassanzadeh-Rostami Z, Abbasi A, Naghavi AM, Faghih S. Effects of vitamin D-fortified oil intake versus vitamin D supplementation on vitamin D status and bone turnover factors: A double blind randomized clinical trial. Clin Nutr ESPEN. 2022;47:28–35 (*https://doi.org/10.1016/j.clnesp.2021.12.025*).
6. Keller A, Ängquist L, Jacobsen R, Vaag A, Heitmann BL. A retrospective analysis of a societal experiment among the Danish population suggests that exposure to extra doses of vitamin A during fetal development may lower type 2 diabetes mellitus (T2DM) risk later in life. Br J Nutr. 2017;117(5) (*https://doi.org/10.1017/S000711451700037X*).
7. Keller A, Stougård M, Frederiksen P, Thorsteinsdottir F, Vaag A, Damm P et al. In utero exposure to extra vitamin D from food fortification and the risk of subsequent development of gestational diabetes: the D-tect study. Nutr J. 2018;17:1–9 (*https://doi.org/10.1186/s12937-018-0403-5*).
8. Stougaard M, Damm P, Frederiksen P, Jacobsen R, Heitmann BL. Extra vitamin D from fortification and the risk of preeclampsia: the D-tect study. PLoS ONE. 2018;13(1):e0191288 (*https://doi.org/10.1371/journal.pone.0191288*).
9. Jensen CB, Berentzen TL, Gamborg M, Sørensen TIA, Heitmann BL. Does prenatal exposure to vitamin D-fortified margarine and milk alter birth weight? A societal experiment. Br J Nutr. 2014;112(5):785–93 (*https://doi.org/10.1017/S0007114514001330*).
10. Jensen CB, Gamborg M, Berentzen TL, Sørensen TIA, Heitmann BL. Prenatal exposure to vitamin-D from fortified margarine and milk and body size at age 7 years. Eur J Clin Nutr. 2015;69(10):1169–75 (*https://doi.org/10.1038/ejcn.2015.55*).
11. Jacobsen R, Hypponen E, Sørensen TIA, Vaag AA, Heitmann BL. Gestational and early infancy exposure to margarine fortified with vitamin D through a national Danish programme and the risk of type 1 diabetes: The D-tect study. PLoS ONE. 2015;10(6):e0128631 (*https://doi.org/10.1371/journal.pone.0128631*).
12. Händel MN, Frederiksen P, Osmond C, Cooper C, Abrahamsen B, Heitmann BL. Prenatal exposure to vitamin D from fortified margarine and risk of fractures in late childhood: period and cohort results from 222 000 subjects in the D-tect observational study. Br J Nutr. 2017;117(6):872–81 (*https://doi.org/10.1017/S000711451700071X*).
13. Thorsteinsdottir F, Maslova E, Jacobsen R, Frederiksen P, Keller A, Backer V et al. Exposure to vitamin D fortification policy in prenatal life and the risk of childhood asthma: results from the D-Tect study. Nutrients. 2019;11(4):924 (*https://doi.org/10.3390/nu11040924*).
14. Duus KS, Moos C, Frederiksen P, Andersen V, Heitmann BL. Prenatal and early life exposure to the Danish mandatory vitamin D fortification policy might prevent inflammatory bowel disease later in life: a societal experiment. Nutrients. 2021;13(4):1367 (*https://doi.org/10.3390/nu13041367*).
15. Ghasemifard N, Nasimi N, Hassanzadeh-Rostami Z, Abbasi A, Faghih S. Effects of vitamin D fortified canola oil on vitamin D and lipid profiles in healthy adults: a double-blind randomized trial. Iran J Nutr Sci Food Technol. 2020;15(2):fa1–9 (*https://www.cabidigitallibrary.org/doi/full/10.5555/20203305091*).
16. Moos C, Duus KS, Frederiksen P, Heitmann BL, Andersen V. Exposure to the Danish mandatory vitamin D fortification policy in prenatal life and the risk of developing coeliac disease – the importance of season: a semi ecological study. Nutrients. 2020;12(5):1243 (*https://doi.org/10.3390/nu12051243*).

Annex 2. Summary of guideline development group considerations to determine the direction and strength of the recommendations

Table A2.1 provides a summary of the guideline development group considerations to determine the direction and strength of the recommendations.

Table A2.1. Summary of guideline development group considerations

Criterion	Guideline development group considerations
Certainty of evidence	The most common GRADE ratings for critical outcomes for the evidence related to vitamin A were low and very low certainty. The recommendation for vitamin A fortification was considered strong, although the certainty of evidence was low or very low for the priority critical outcomes. The most common GRADE rating for critical outcomes for the evidence related to vitamin D was low certainty. The recommendation for vitamin D fortification was considered conditional; the certainty of evidence was very low for the priority critical outcomes. Additional information showed that not all fortification vehicles are adequately fortified, and that, in some other cases, the doses used may exceed daily nutrient requirements leading to a risk of toxicity in some contexts. The group noted that fortification of sugar has increased vitamin A intake, serum retinol concentration and mean total body vitamin A stores in Guatemala. Fortification of milk *(1)* has improved children's growth in different countries, but the effect of vitamin A cannot be isolated. Fortification of other foods with vitamin D suggests fortification of food with vitamin D might increase serum 25(OH)D concentration to varying degrees depending on the fortification vehicle, fortification dose and characteristics of the population. Fortification of food with vitamin D may have a positive impact on bone turnover. Fortification of food with vitamin D may have only a limited effect on anthropometric parameters and may improve some lipid parameters..
Priority of the problem	Vitamin A deficiency is a public health problem in more than half of all countries, especially those in Africa and South-East Asia. The most severe effects of vitamin A deficiency are seen in young children and pregnant women in low-income countries. The group concluded that vitamin A deficiency is an important public health problem and health care providers, policy-makers and communities in all settings are likely to place a high value on food fortification to combat vitamin A deficiency. The global prevalence of vitamin D deficiency is unknown. People with limited access to foods rich in vitamin D including adolescents, adults, pregnant and lactating women, older adults, people with obesity, people living at extreme latitudes and people living in regions with insufficient sunshine or with sun avoidance behaviours that prevent or impede dermal synthesis, are particularly at risk of vitamin D deficiency. The group agreed that vitamin D deficiency has a negative impact on bone health and that there is increasing interest in vitamin D and its possible positive effects on health. However, the group acknowledged that there is uncertainty in estimating the global prevalence of vitamin D deficiency.

Criterion	Guideline development group considerations
Balance of benefits and harms	Evidence from the included studies suggests that oils fortified with vitamins A and D have little or no effect on health. However, more studies are needed as the sample sizes of the included studies were very small and, as such, the studies may not have been able to detect small effects that might be meaningful at the population level. No studies included in the systematic review reported on adverse effects. Evidence from a pre- and post-survey in Indonesia (2011–2012) showed a large increase in retinol concentration in breast milk, and in the serum of young (lactating and non-lactating) women, suggesting that infants aged 6–11 months benefit from increased maternal vitamin A intake during pregnancy and/or breastfeeding. Changes in serum retinol concentrations were larger in areas with the lowest baseline retinol levels. The Malawi Micronutrient Survey (2015–2016) showed elevated vitamin A concentrations in preschool- and school-aged children, raising concerns about potential unintended negative effects when vitamin A is delivered through multiple interventions. The Oman National Nutrition Survey (2016–2017) reported a relatively low prevalence of vitamin A deficiency among women (0.2%) and children (9.5%), which was attributed to consumption of vitamin A-rich foods and fortified oils and fats. At the time of the survey, the food fortification programme in Oman mandated that all cooking oils and margarine be fortified. The risk and severity of excessive vitamin A intake are a function of amount of intake from different dietary sources including all interventions and the duration of intake. Edible oil fortification is generally considered as safe. Given the nature of edible oils and their intended use, it is practically impossible for consumers to exceed safety limits. The group discussed case studies from Guatemala *(2)*, India *(3)*, Northern Cape (South Africa) *(4)* and Zambia *(5)*. As of 2023, more than 80 countries implement universal vitamin A supplementation programmes targeted to children aged 6–59 months through semi-annual national campaigns. The Global Alliance for Vitamin A framework provides a useful framework for reviewing policies, overlapping interventions and associated biomarkers to inform programme design. Vitamin D toxicity *(6)* does not arise from vitamin D synthesized in the skin. It can be an effect of prolonged intake of excessive supplemental doses, and in rare cases has been caused by excessive food fortification. Vitamin D toxicity *(7)* results in hypercalcaemia and/or hypercalciuria, which may be accompanied by impaired renal function and increased intestinal absorption of calcium and bone resorption. Case studies *(8)* have shown that 50 000–300 000 IU/day of vitamin D supplementation can cause hypercalcaemia within a few weeks. The United States Institute of Medicine, currently named the National Academy of Medicine, is the only major consensus group to have formally designated a threshold for serum 25(OH)D concentrations above which there were concerns of adverse effects (125 nmol/L). A published review of vitamin D safety concluded that serum 25(OH)D concentrations less than 220 nmol/L poses no known risk of harm; furthermore, available data indicate that some individuals attain serum 25(OH)D concentrations greater than 125 nmol/L naturally from ultraviolet B exposure *(9)*. The group noted that although a combined fortification strategy using multiple fortification vehicles appears to be an effective option for reaching all segments of the population, monitoring of multiple interventions (e.g. supplementation) in the same population group is critical to prevent excessive intakes. The group recommended that fortification of edible oils and fats should be integrated in, and monitored as part of, national programmes for the prevention and control of micronutrient deficiencies and insufficiencies and prevention of diet-related noncommunicable diseases.

Criterion	Guideline development group considerations
Equity, gender equality and human rights	**Equity** Equity was discussed from the perspective of coverage, potential to reach the most vulnerable population groups with the fortification vehicle (oils and fats), potential to reach the most vulnerable population groups with the intervention (fortification), and affordability. The group agreed that high population coverage is needed to have a public health impact. One of the major challenges faced in the fortification of oil is reaching the unorganized market where oil is sold in loose, unbranded packs. Proper regulations and a normative framework for fortification are necessary to ensure the sustainable reach of the fortification programme to the most vulnerable population groups. There is often a difference in access to fortified oils and fats between urban and rural areas, between certain socioeconomic groups and between population groups residing in border areas and those in other areas, emphasizing the importance of regulatory monitoring. The group discussed country experiences from Bangladesh *(10, 11)*, Guatemala, Nigeria *(12)* and Pakistan *(13)*. The group recommended that equitable access to and consumption of fortified oils and fats must be evaluated with regard to each country's context. In general, effective nutrition interventions are more likely to decrease health inequities if they are accompanied by concurrent interventions that address the root cause of the problem. Strategies for fortification therefore need to be aligned with relevant programmes, especially poverty reduction and other social interventions. **Gender equality** The most severe effects of vitamin A deficiency are seen in young children and pregnant women in low-income countries. The prevalence of vitamin D deficiency is reported to be higher among adolescents, adults, pregnant women and people wearing concealing clothing compared with the rest of the population. Most studies included in the systematic review of vitamin A fortification were conducted in children (preschool-aged children and school-aged children aged 5–12 years); one was conducted in non-pregnant women aged 15–49 years. Studies included in the systematic review of vitamin D fortification were focused on children and adults. Gender and life stage considerations are generally incorporated as part of intervention design because different estimated average requirements, recommended dietary allowances and upper limits for a micronutrient are defined for different age, gender and physiological status groups (e.g. different values for pregnant or lactating women). Because adult males tend to have the highest consumption rates of staple foods, and thus the highest micronutrient intakes if a staple food were to be mass fortified and the greatest risk of excessive micronutrient intakes, this group is used to determine safety limits. There are multiple opportunities to ensure gender equality is considered across the fortification value chain. These include working with technical institutions to increase supply chain actors' awareness of the benefits of food fortification for women and children and their recognition of women as both caregivers and consumers of fortified foods. They also include building men's awareness of nutrition topics, communicating with vulnerable families and marginalized groups, and engaging with women in programme design and implementation. The group acknowledged that gender inequality is a cross-cutting determinant of health that intersects with other determinants. The group concluded that fortification of edible oils and fats with vitamins A and D could increase gender equity. The group recommended that gender equality must be integrated into all stages of programme design and contextualized to the needs of different settings.

Criterion	Guideline development group considerations
Equity, gender equality and human rights	**Human rights** Nutrition is one of many underlying socioeconomic determinants of the right to the highest attainable standard of health. The group reflected on the different human rights instruments that relate to nutrition. The group recommended that programmes for fortification of edible oils and fats should be designed using a human rights-based approach. Recognizing that access to a diverse diet may not be possible in all contexts at all times, food fortification can help to fill gaps in nutrient intake. A human rights-based approach to fortification could relate to the physical availability of fortified foods; economic access to fortified foods; food habits and cultural norms; links with gender equality and equity; the regulatory environment for fortification; and the role of different stakeholders in the fortification value chain. Strategies for fortification need to be aligned with relevant programmes, especially poverty reduction and other social interventions. The group also acknowledged that proper regulations and a normative framework for fortification are necessary to ensure the sustainable reach of the fortification programme to the most vulnerable population groups. Competent authorities should determine whether fortification of edible oils and/or fats with vitamins A and/or D should be mandatory or voluntary, depending on the severity and extent of the public health problem. Input from all involved sectors, including oil processors and producers, industry and trade organizations, public institutions, academia, research organizations, and consumer associations and protection groups, when developing regulations and standards will help to ensure a realistic approach.
Costs, acceptability and feasibility	**Costs** The group discussed examples of costs and cost–effectiveness evaluations for fortification of oil with vitamin A in Bangladesh *(14)*, Burkina Faso *(15)*, Ethiopia *(16)* and the United Republic of Tanzania *(17)* and. The group noted that fortification of edible oils with vitamin A, supported by appropriate legislation and enforcement of regulations, can be a cost-effective strategy to address vitamin A deficiency. Little evidence is available regarding the cost–effectiveness of vitamin D-fortified oils or fats. Based on the systematic review and available implementation information, in countries with low exposure to sunlight or seasonal variations in sunlight exposure and where dietary sources of vitamin D are not consumed regularly, co-fortification of oils and fats with both vitamin A and D can be considered as a cost-effective strategy to help increase intakes of vitamin D in the population. **Acceptability** The group discussed acceptability through the perspective of different stakeholders in the fortification value chain, including consumers, industry and government. The group noted that 35 countries have legislation for mandatory fortification of edible oils and eight countries have legislation for voluntary fortification. Of these 43 countries, 39 countries have standards that include vitamin A, 13 countries have standards that include vitamin D and two countries have standards that include vitamin E. The amount of vitamin A permitted to be added in the 39 countries with mandatory and voluntary standards for vitamin A ranges from 3.5 to 41.25 mg/kg oil. For the 13 countries with mandatory and voluntary standards for vitamin D, the amount permitted to be added ranges from 0.01 to 0.75 mg/kg oil. The group noted that, for fats such as margarine, fortification was widely practised and accepted by consumers, especially in European countries. In some countries such as Belgium, Poland and Sweden, fortification of margarine, fat spreads and related products is mandatory. The Netherlands has a national covenant to fortify margarine and other spreadable fats with vitamin D. Most countries allow voluntary fortification, including Finland, Germany and Norway.

Criterion	Guideline development group considerations
Costs, acceptability and feasibility	**Feasibility** The group noted that 176 countries have data on availability of oil for human consumption, which ranges from 0 to 71.81 g per capita per day. Thirty-three countries have legislation for mandatory fortification of edible oils and seven countries have legislation for voluntary fortification. Of these 40 countries, 39 countries have standards that include vitamin A, 13 countries have standards that include vitamin D and two countries have standards that include vitamin E. In most countries, edible oils and fats are processed centrally by medium- and large-scale producers, which makes it easier to implement and monitor fortification processes. However, the group noted the following limitations: • limited data are available on processing status; • in some countries, not all oil is industrially processed *(18)* – for example, only 52% is in Bangladesh, 85% is in Pakistan and 60% is in the United Republic of Tanzania, compared with 100% in Canada being commercially manufactured goods; and • not all oil is sold in packaging (some is sold loose or in bulk). The group concluded that fortification of edible oils with vitamins A and D can be considered a feasible intervention. However, implementation by centralized oil producers and regular monitoring by governments are critical, preferably alongside other agricultural, health, education and social intervention programmes that promote the consumption of adequate quantities of good quality nutritious foods among the nutritionally vulnerable. The feasibility of a fortification programme should be assessed in each context before rollout to assess the stability of the nutrient under prevailing storage, transportation, packing and cooking conditions; the variations in per capita consumption of oils and fats; and the feasibility of determining a dosage level that would provide meaningful quantities of nutrients to those at the lower end of the consumption spectrum while not providing unacceptable levels for those at the higher end.
Values and preferences	The group discussed values and preferences regarding the health outcomes related to vitamin A and D and regarding fortification of edible oils and fats with vitamins A and D as an intervention. Regarding health outcomes, the group's discussion was centred on end users, that is, those who buy fortified oils and fats. The group considered that micronutrient deficiencies are important public health problems and that health care providers, policy-makers and family members in all settings are likely to place a high value on food fortification to combat micronutrient deficiencies. The group concluded there is no important variability in how much the general population value the main health outcomes related to vitamin A and D deficiency. Regarding fortification as an intervention, the group's discussion was again centred on end users, that is, those who buy fortified oils and fats. The group noted that fortification requires less behaviour change than other interventions, meaning it is likely to be acceptable and feasible. However, the group also recognized that some groups oppose adding any substance to foods. The group also noted there might be reservations from industry for various reasons including insufficient business incentives; limited awareness, internal capacity and technical knowledge; varying costs of premix and equipment; and varying demands of consumers. The group concluded there is possibly important variability in how much the general population values fortification of edible oils and fats as a way to receive vitamins A and D in their diets.

References

1. The contribution of school milk programmes to the nutrition of children worldwide. Brussels: International Dairy Federation; 2020 (*https://shop.fil-idf.org/products/school-milk-programmes-2020*).
2. Bielderman I, Vossenaar M, Melse-Boonstra A, Solomons NW. The potential double-burden of vitamin A malnutrition: under- and overconsumption of fortified table sugar in the Guatemalan highlands. Eur J Clin Nutr. 2016;70(8):947–53 (*https://doi.org/10.1038/ejcn.2016.36*).
3. Reddy GB, Pullakhandam R, Ghosh S, Boiroju NK, Tattari S, Laxmaiah A et al. Vitamin A deficiency among children younger than 5 y in India: an analysis of national data sets to reflect on the need for vitamin A supplementation. Am J Clin Nutr. 2021;113(4):939–47 (*https://doi.org/10.1093/ajcn/nqaa314*).
4. van Stuijvenberg ME, Schoeman SE, Nel J, le Roux M, Dhansay MA. Liver is widely eaten by preschool children in the Northern Cape province of South Africa: Implications for routine vitamin A supplementation. Matern Child Nutr. 2020;16(3):e12931 (*https://doi.org/10.1111/mcn.12931*).
5. Tanumihardjo SA, Gannon BM, Kaliwile C, Chileshe J. Hypercarotenodermia in Zambia: which children turned orange during mango season? Eur J Clin Nutr. 2015;69(12):1346–9 (*https://doi.org/10.1038/ejcn.2015.143*).
6. Roth DE, Abrams SA, Aloia J, Bergeron G, Bourassa MW, Brown KH et al. Global prevalence and disease burden of vitamin D deficiency: a roadmap for action in low- and middle-income countries. Ann N Y Acad Sci. 2018;1430(1):44–79 (*https://doi.org/10.1111/nyas.13968*).
7. Cranney A, Horsley T, O'Donnell S, Weiler H, Puil L, Ooi D et al. Effectiveness and safety of vitamin D in relation to bone health. Evid Rep Technol Assess (Full Rep). 2007(158):1–235 (*https://pmc.ncbi.nlm.nih.gov/articles/PMC4781354/*).
8. Schwartzman MS, Franck WA. Vitamin D toxicity complicating the treatment of senile, postmenopausal, and glucocorticoid-induced osteoporosis. Four case reports and a critical commentary on the use of vitamin D in these disorders. Am J Med. 1987;82(2):224–30 (*https://doi.org/10.1016/0002-9343(87)90060-x*).
9. Hathcock JN, Shao A, Vieth R, Heaney R. Risk assessment for vitamin D. Am J Clin Nutr. 2007;85(1):6–18 (*https://doi.org/10.1093/ajcn/85.1.6*).
10. Jungjohann S, Ara G, Pedro C, Khanam M, Friesen V, Ahmed T et al. Fortification quality is high for packaged and branded edible oil but low for oil sold in unbranded, loose form: findings from a market assessment in Bangladesh (P24-047-19). Curr Dev Nutr. 2019;3:2037 (*https://doi.org/https://doi.org/10.1093/cdn/nzz044.P24-047-19*).
11. Raghavan R, Aaron GJ, Nahar B, Knowles J, Neufeld LM, Rahman S et al. Household coverage of vitamin A fortification of edible oil in Bangladesh. PLoS One. 2019;14(4):e0212257 (*https://doi.org/10.1371/journal.pone.0212257*).
12. Large scale food fortification compliance in Nigeria: state of the nation report, 2022. Jabi: Global Alliance for Improved Nutrition; 2022 (*https://www.gainhealth.org/resources/reports-and-publications/state-nation-report-large-scale-food-fortification-lsff-nigeria*).
13. Evaluation of the Supporting Nutrition in Pakistan Food Fortification Programme: endline evaluation report. Oxford: e-Pact; 2021 (*https://iati.fcdo.gov.uk/iati_documents/90000152.pdf*).
14. Fiedler JL, Lividini K, Guyondet C, Bermudez OI. Assessing alternative industrial fortification portfolios: a Bangladesh case study. Food Nutr Bull. 2015;31(1):57–74 (*https://doi.org/10.1177/156482651503600106*).
15. Food systems community of practice [website]. World Health Organization; (*https://www.whofoodsystems.org/*).
16. Vosti S, Engle-Stone R, Woldegebreal D, Luo H, Asfaw E, Kagin J et al. Estimated effectiveness and cost-effectiveness of a fortified edible oils program in Ethiopia. Curr Dev Nutr. 2020;4 (*https://doi.org/10.1093/cdn/nzaa064_025*).
17. Walters D, Ndau E, Saleh N, Mosha T, Horton S. Cost-effectiveness of sunflower oil fortification with vitamin A in Tanzania by scale. Matern Child Nutr. 2019;15(S3):e12720 (*https://doi.org/10.1111/mcn.12720*).
18. Total quantity and proportion of industrially processed food vehicles [website]. Global Fortification Data Exchange; 2025 (*https://fortificationdata.org/chart-quantity-and-proportion-of-industrially-processed-food-vehicles/*).

Annex 3. Logic model for fortification of maize flour and corn meal with vitamins and minerals for public health

Fig. A3.1. Logic model for fortification of maize flour and corn meal with vitamins and minerals for public health

Annex 4. WHO steering committee for food fortification

Dr Francesco Branca
Director, Department of Nutrition and Food Safety

Dr Elaine Borghi
Unit Head, Monitoring Nutritional Status and Food Safety Events Unit
Department of Nutrition and Food Safety

Dr Luz Maria De-Regil
Unit Head, Multisectoral Actions in Food Systems Unit
Department of Nutrition and Food Safety

Dr Maria Nieves Garcia-Casal
Scientist, Food and Nutrition Actions in Health Systems Unit
Department of Nutrition and Food Safety

Dr Bente Mikkelsen
Director, Department of Noncommunicable Diseases

Dr Jason Montez
Scientist (Nutrition, Obesity and Diet-related NCDs), Standards and Scientific Advice on Food Nutrition Unit
Department of Nutrition and Food Safety

Dr Juan Pablo Peña-Rosas
Head, Global Health Initiatives
Department of Nutrition and Food Safety

Dr Teri Reynolds
Unit Head, Clinical Services and Systems Unit
Department of Integrated Health Services

Dr Lisa Rogers
Technical Officer, Food and Nutrition Actions in Health Systems Unit
Department of Nutrition and Food Safety

Annex 5. WHO guideline development group on food fortification with micronutrients

Note: The areas of expertise of each guideline development group member are given in italics.

Dr Salima Ali Rashid Almamary
Department of Nutrition
Ministry of Health
Oman
Public health nutrition

Dr Ayed Amr
Department of Nutrition and Food Technology
University of Jordan
Jordan
Food science, oil chemistry

Dr Maria Elena del Socorro Jefferds
Nutrition Branch, Division of Nutrition, Physical Activity and Obesity
U.S. Centers for Disease Control and Prevention
United States of America
Behaviour science, programme evaluation

Dr Rukhsana Haider
Training and Assistance for Health and Nutrition Foundation
Bangladesh
Breastfeeding, capacity-building on counselling and nutrition

Mr Phillip Joseph Makhumula-Nkhoma
Lifesciences Consulting
Malawi
Programme implementation, fortification standards, quality assurance and quality control, capacity assessment and strengthening

Ms Andrea Papan
Institute of Development Studies
Sweden
Human rights-based approaches to gender equality and sustainable development

Professor Dalip Ragoobirsingh
Medical Biochemistry and Diabetology, and Diabetes Education Programme
University of the West Indies
Jamaica
Diabetes, noncommunicable diseases

Ms Rusidah Selamat
Nutrition Division, Department of Public Health
Ministry of Health
Malaysia
Public health nutrition

Professor C Murray Skeaff
Department of Human Nutrition
University of Otago
New Zealand
Fats and fatty acids, biomarkers, diet and health, human nutrition

Ms Ayodele Elizabeth Tella
Country Program Manager
TechnoServe
Nigeria
Programme implementation

Annex 6. Systematic review team

Ms Ildikó Csölle
Cochrane Hungary, Clinical Center of the University of Pécs, Medical School, and
National Laboratory on Human Reproduction, and
Doctoral School of Health Sciences, Faculty of Health Sciences
University of Pécs
Hungary

Ms Regina Felső
Cochrane Hungary, Clinical Center of the University of Pécs, Medical School
National Laboratory on Human Reproduction, and
Hungarian Centre for Genomics and Bioinformatics, Szentágothai Research Center
University of Pécs
Hungary

Dr Tamás Ferenci
Physiological Controls Research Center
Obuda University
Hungary, and
Department of Statistics
Corvinus University of Budapest
Hungary

Ms Kazahyet Ibrahim
Department of Public Health Medicine, Medical School
University of Pécs
Hungary

Dr Daniela Kuellenberg de Gaudry
Cochrane Hungary, Clinical Center of the University of Pécs, Medical School
University of Pécs
Hungary

Dr Szimonetta Lohner
Cochrane Hungary, Clinical Center of the University of Pécs, Medical School, and
Department of Public Health Medicine, Medical School
University of Pécs
Hungary

Ms Maria-Inti Metzendorf
Institute of General Practice, Medical Faculty
Heinrich Heine University Düsseldorf
Germany

Mr Patrick Nyamemba Nyakundi
Doctoral School of Health Sciences, Faculty of Health Sciences, and
Department of Public Health Medicine, Medical School
University of Pécs
Hungary

Ms Éva Szabó
Cochrane Hungary, Clinical Center of the University of Pécs, Medical School, and
Department of Biochemistry and Medical Chemistry, Medical School
University of Pécs
Hungary

Note: The technical lead of the systematic review team participated in technical presentations and in discussions related to those presentations, providing information, feedback and clarification when required. They did not participate in formulating, or influence, the recommendations.

Annex 7. Peer-reviewers

Note: The names and affiliations of peer-reviewers are provided here as an acknowledgement and by no means indicate their endorsement of the recommendations in this guideline. The acknowledgement of the peer-reviewers does not necessarily represent the views, decisions or policies of the institutions with which they are affiliated.

Dr Andreas Bluethner
Founding Director - Dr. Blüthner & Partner
Germany

Dr Karen Codling
Independent consultant – Food fortification
Director of the AfriCat Foundation
AfriCat and Okonjima Namibia
Namibia

Dr Reina E Stone
Associate Professor
UC Davis Department of Nutrition
United States of America

Mr Penjani Mkambula
Deputy Director – Programmes
Global Alliance for Improved Nutrition (GAIN)
United Kingdom of Great Britain and Northern Ireland

Dr N. Sreekumaran Nair
Professor & Head
Department of Biostatistics
Jawaharlal Institute of Postgraduate Medical Education and Research (JIPMER)
India

Dr Sirimavo Nair
Professor
The Maharaja Sayajirao University of Baroda
India

Annex 8. Questions in PICO format

Tables A8.1–A8.3 provide the guideline questions in PICO format.

1. Should edible oils and fats be fortified with vitamin A to:
   - improve vitamin A status (as defined by trialists) and reduce the prevalence of vitamin A deficiency in populations; and
   - improve dietary intake of vitamin A in populations?

If so, what fortificant(s) should be used and in what amounts?

Table A8.1. Population, intervention, control, outcomes and setting for guideline question on fortification of edible oils and fats with vitamin A

Element	Details
Population	• General (healthy) population • Subpopulations — By age group: infants (0–6 months), young children (6–23 months), preschool-aged children (24–59 months), school-aged children (5–12 years), adolescents (10–19 years), non-pregnant women of reproductive age (15–49 years), pregnant and lactating women, adults (50–59 years), older persons (60 years and older) — By prevalence of vitamin A deficiency[a]: countries with a public health problem, countries with no public health problem — By risk of inadequate vitamin A intake in the population (as defined by trialists) — By exposure to additional vitamin A through other programmes: supplementation, fortification, biofortification, micronutrient powder, no programme — By number of additional vitamin A programmes: 0, 1, 2, 3, 4
Intervention[b]	• Edible oils and fats fortified with vitamin A alone, or in combination with other micronutrients • Subgroup analyses — By type: oils, solid fats (as classified by Codex standards of identity) *(1, 2)* — By processing method: trans-esterification, hydrogenation, fractionation, other, no modification — By amount of vitamin A added to the oil or fat — By type of vitamin A compound: acetate, palmitate — By consumption pattern: high consumption of edible oils and fats, low consumption of edible oils and fats (as defined by trialists based on dietary guidelines) — By method of cooking: deep frying, stir or shallow frying, reuse of oil for frying, no frying — By method to stabilize vitamin A: quality of oil or fat *(1, 2)*, packaging (coloured, bulk, retail, other), storage conditions, addition of antioxidants, nitrogen flushing, other — By type of vitamin A fortification intervention: mandatory, market driven (voluntary) — By delivery platform: retail, social safety net, institutional, other
Control	• Same oils or fats without added vitamin A, or no intervention
Outcomes	• Primary (critical) outcomes — Vitamin A deficiency (xerophthalmia, night blindness) — Subclinical vitamin A deficiency[c] (plasma or serum retinol concentration) — Adverse effects (hypervitaminosis, liver toxicity) — Morbidity (all causes) — Mortality (all causes) • Secondary outcomes — Vitamin A status[c] (as defined by trialists – liver stores of vitamin A, concentration of retinol in breast milk, MRDR test results, retinol-binding protein concentration, beta-carotene concentration) — Intake of dietary vitamin A — Iron status[c] (ferritin), anaemia

Element	Details
Setting	• All countries

MRDR: modified relative dose response.

a Refer to WHO cut-off values for public health significance *(3, 4)*.
b Co-interventions were permitted when common to the intervention and control.
c Adjusted for inflammation or non-adjusted.

2. Should edible oils and fats be fortified with vitamin D to:
   - improve vitamin D status (as defined by trialists) and reduce the prevalence of vitamin D deficiency in populations; and
   - improve dietary intake of vitamin D in populations?

If so, what fortificant(s) should be used and in what amounts?

Table A8.2. Population, intervention, control, outcomes and setting for guideline question on fortification of edible oils and fats with vitamin D

Element	Details
Population	• General (healthy) population • Subpopulations — By age group: infants (0–6 months), young children (6–23 months), preschool-aged children (24–59 months), school-aged children (5–12 years), adolescents (10–19 years), non-pregnant women of reproductive age (15–49 years), pregnant and lactating women, adults (50–60 years), older people (more than 60 years) — By prevalence of vitamin D deficiency: countries with a public health problem, countries with no public health problem — By risk of inadequate vitamin D intake in the population (as defined by trialists) — By exposure to additional vitamin D through other programmes: supplementation, fortification, biofortification, micronutrient powder, no programme — By number of additional vitamin D programmes: 0,1, 2, 3, 4 — By skin pigmentation based on Fitzpatrick skin tone chart (5): 3 or less, 4 or more, mixed or unknown — By latitude: between the Tropic of Cancer and the Tropic of Capricorn, north of the Tropic of Cancer, south of the Tropic of Capricorn — By sun exposure and sun protection behaviour — By exposure to environmental pollutants[a] (as defined by trialists) — By BMI classification (as defined by trialists)
Intervention[b]	• Edible oils and fats fortified with vitamin D alone, or in combination with other micronutrients • Subgroup analyses — By type: oils, solid fats (as classified by Codex standards of identity) *(1, 2)* — By type of processing: trans-esterification, hydrogenation, fractionation, other, no modification — By amount of vitamin D added to the oil or fat — By type of vitamin D compound: vitamin D2 (ergocalciferol), vitamin D3 (cholecalciferol) — By consumption pattern: high consumption of edible oils and fats, low consumption of edible oils and fats (as defined by trialists based on dietary guidelines) — By method of cooking: deep frying, stir or shallow frying, reuse of oil for frying, no frying — By method to stabilize vitamin D: quality of oil or fat *(1, 2)*, packaging (coloured, bulk, retail, other), storage conditions, addition of antioxidants, nitrogen flushing, other — By type of vitamin D fortification intervention: mandatory, market driven (voluntary) — By delivery platform: retail, social safety net, institutional, other
Control	• Same oils or fats without added vitamin D, or no intervention

Element	Details
Outcomes	• Primary (critical) outcomes — Vitamin D deficiency (serum 25(OH)D concentration; as defined by trialists) — Vitamin D concentration (serum 25(OH)D concentration) — Osteomalacia in older persons — Nutritional rickets in children — Adverse effects (hypervitaminosis, hypercalcaemia, kidney stones) — Morbidity (all causes) — Mortality (all causes) • Secondary outcomes — Intake of dietary vitamin D — Maternal and infant outcomes (preterm birth, low birth weight and small for gestational age, gestational diabetes, weight gain during pregnancy, pre-eclampsia, postpartum depression, postpartum haemorrhage) — Concentration of vitamin D in breast milk — Growth, BMI, weight change (overweight and obesity) — Parathyroid hormone concentration (serum concentration) — Alkaline phosphatase concentration (serum concentration) — Longer-term outcomes (bone health, diabetes, hypertension, other cardiovascular diseases, respiratory infections, asthma, menopause onset (as defined by trialists))
Setting	• All countries

BMI: body mass index.

a Air pollution includes particulate matter and gases *(6)*.
b Co-interventions were permitted when common to the intervention and control.

3. Should edible oils and fats be fortified with vitamins A and D to:
   - improve vitamin A and D status (as defined by trialists) and reduce the prevalence of vitamin A and D deficiency in populations; and
   - improve dietary intake of vitamin A and D in populations?

If so, what fortificant(s) should be used and in what amounts?

Table A8.3. Population, intervention, control, outcomes and setting for guideline question on fortification of edible oils and fats with vitamins A and D

Element	Details
Population	• General (healthy) population • Subpopulations — By age group: infants (0–6 months), young children (6–23 months), preschool-aged children (24–59 months), school-aged children (5–12 years), adolescents (10–19 years), non-pregnant women of reproductive age (15–49 years), pregnant and lactating women, adults (50–60 years), older people (more than 60 years) — By prevalence of vitamin A[a] and D deficiency: countries with a public health problem with vitamin A deficiency, countries with a public health problem with vitamin D deficiency, countries with a public health problem with both vitamin A and D deficiency, countries with no public health problem — By risk of inadequate vitamin A and D intake in the population (as defined by trialists) — By exposure to additional vitamin A and D through other programmes: supplementation, fortification, biofortification, micronutrient powder, no programme — By number of additional vitamin A and D programmes: 0, 1, 2, 3, 4 — By skin pigmentation based on Fitzpatrick skin tone chart *(5)*: 3 or less, 4 or more, mixed or unknown — By latitude: between the Tropic of Cancer and the Tropic of Capricorn, north of the Tropic of Cancer, south of the Tropic of Capricorn — By sun exposure and sun protection behaviour — By exposure to environmental pollutants[b] (as defined by trialists) — By BMI classification (as defined by trialists)

Element	Details
Intervention[c]	• Edible oils and fats fortified with vitamins A and D alone, or in combination with other micronutrients • Subgroup analyses — By type: oils, solid fats (as classified by Codex standards of identity) *(1, 2)* — By processing method: trans-esterification, hydrogenation, fractionation, others, no modification — By amount of vitamin A and/or D added to the oil or fat — By type of vitamin A or D compound: acetate, palmitate, vitamin D2 (ergocalciferol), vitamin D3 (cholecalciferol), others — By consumption pattern: high consumption of edible oils and fats, low consumption of edible oils and fats (as defined by trialists based on dietary guidelines) — By method of cooking: deep frying, stir or shallow frying, reuse of oil for frying, no frying — By method to stabilize vitamin A and D: quality of oil/fat *(1, 2)*, packaging (coloured, bulk, retail, other), storage conditions, addition of antioxidants, nitrogen flushing, other — By type of vitamin A and D fortification intervention: mandatory, market driven (voluntary) — By delivery platform: retail, social safety net, institutional, other
Control	• Same oils or fats without added vitamin A and D, or no intervention
Outcomes	• Primary (critical) outcomes — Vitamin A deficiency (xerophthalmia, night blindness) — Subclinical vitamin A deficiencyd (plasma or serum retinol concentration) — Vitamin D deficiency (serum 25(OH)D concentration; as defined by trialists) — Osteomalacia in older persons — Vitamin D concentration (serum 25(OH)D concentration) — Nutritional rickets in children — Adverse effects (hypervitaminosis, liver toxicity, hypercalcaemia, kidney stones) — Morbidity (all causes) — Mortality (all causes) • Secondary outcomes — Vitamin A status[d] (as defined by trialists – liver stores of vitamin A, concentration of retinol in breast milk, MRDR test results, retinol-binding protein concentration, beta-carotene concentration) — Intake of dietary vitamin A and D — Iron statusd (ferritin), anaemia — Maternal and infant outcomes (preterm birth, low birth weight and small for gestational age, gestational diabetes, weight gain during pregnancy, pre-eclampsia, postpartum depression, postpartum haemorrhage) — Concentration of vitamin D in breast milk — Growth, BMI, weight change (overweight and obesity) — Parathyroid hormone concentration (serum) — Alkaline phosphatase concentration (serum) — Longer-term outcomes (bone health, diabetes, hypertension, other cardiovascular diseases, respiratory infections, asthma, menopause onset (as defined by trialists))
Setting	• All countries

BMI: body mass index; MRDR: modified relative dose response.

a Refer to WHO cut-off values for public health significance *(3, 4)*.
b Air pollution includes particulate matter and gases *(6)*.
c Co-interventions were permitted when common to the intervention and control.
d Adjusted for inflammation or non-adjusted.

References

1. Codex Alimentarius Commission. Standard for edible fats and oils not covered by individual standards. Food and Agriculture Organization of the United Nations, World Health Organization; 2023 (CX 19-1981; *https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?lnk=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B19-1981%252FCXS_019e.pdf*).
2. Codex Alimentarius Commission. Standard for named vegetable oils. Food and Agriculture Organization of the United Nations, World Health Organization; 2023 (*https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?lnk=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B210-1999%252FCXS_210e.pdf*).
3. Vitamin A deficiency [website]. World Health Organization; 2024 (*https://www.who.int/data/nutrition/nlis/info/vitamin-a-deficiency*).
4. Xerophthalmia and night blindness for the assessment of clinical vitamin A deficiency in individuals and populations. Geneva: World Health Organization; 2014 (*https://iris.who.int/handle/10665/133705*).
5. Fitzpatrick TB. The validity and practicality of sun-reactive skin types I through VI. Arch Dermatol. 1988;124(6):869–71 (*https://doi.org/10.1001/archderm.124.6.869*).
6. WHO global air quality guidelines: particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. Geneva: World Health Organization; 2021 (*https://iris.who.int/handle/10665/345329*). Licence: CC BY-NC-SA 3.0 IGO.

Annex 9. WHO Secretariat

Dr Luz Maria De-Regil
Unit Head, Multisectoral Actions in Food Systems Unit
Department of Nutrition and Food Safety

Dr Maria Nieves Garcia-Casal
Scientist, Food and Nutrition Actions in Health Systems Unit
Department of Nutrition and Food Safety

Ms Kaia Engesveen
Technical Officer, Multisectoral Actions in Food Systems Unit
Department of Nutrition and Food Safety

World Health Organization

Department of of nutrition and food safety

20 Avenue Appia
1211 Geneva 27
Switzerland